

Biochemistry

World of Revision

Marrow

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Contents

Biochemistry

Enzymes	1
Basics of enzymes • Classification of enzymes • Enzyme mechanism of action • Enzyme kinetics • Enzyme inhibition • Regulation of enzyme activity • Clinical enzymology • Enzyme markers • Serine proteases	
Carbohydrates : Introduction	12
Chemistry of carbohydrates • Dietary fibres • Glycosaminoglycans (GAGs)/Mucopolysaccharides • Mucopolysaccharidoses (MPS) • Glucose transporters	
Metabolism of Carbohydrates : Part 1	18
Fed & fasting state metabolism • Glycolysis : Steps & energetics • Regulation of glycolysis • Metabolism in cancer cells • Pyruvate dehydrogenase (PDH)	
Metabolism of Carbohydrates : Part 2	24
Glycogen synthesis • Glycogenolysis • Regulation of glycogen metabolism • Glycogen storage disorders • Gluconeogenesis • Galactose & fructose metabolism • Hexose monophosphate (HMP shunt) pathway • Uronic acid pathway	
Amino Acids : Part 1	33
Classification of amino acids • Properties of amino acids • Proteins • Collagen • Elastin, keratin & fibrillin • Amino acid metabolism • Urea cycle • Urea cycle disorders	
Amino Acids : Part 2	44
Phenylalanine & tyrosine : Metabolism • Phenylalanine & tyrosine : Disorders • Derivatives of tyrosine • Tryptophan : Metabolism & derivatives • Tryptophan metabolism : Associated disorders • Sulphur containing amino acids : Metabolism • Sulphur containing amino acids : Disorders & derivatives • Glycine & serine • Basic amino acids • Branched chain amino acids • Overview	
Metabolism of Lipids	57
Classification of fatty acids (FA) • Phospholipids (PL) & glycolipids • Sphingolipidoses • Metabolism in fasting state • Beta oxidation • Other oxidation pathways & disorders of fatty acid oxidation • Ketone body synthesis • Metabolism in fed state • Lipoproteins • Disorders of lipoproteins	
Molecular Biology	74
DNA • Transcription • RNA • Mutations and epigenetics • Genomic analysis • Recombinant DNA technology • PCR : Genome editing and DNA sequencing • Overview of techniques • Nucleotides	
Vitamins and Minerals	91
Vitamin A vs. Vitamin D • Vitamin E vs. Vitamin K • Vitamin B ₁ vs. B ₂ • Vitamin B ₃ , B ₅ , B ₆ , B ₇ • Folic acid vs. Vitamin B ₁₂ • Vitamin C (Ascorbic acid) • Minerals • Miscellaneous topics on nutrition	
Miscellaneous Topics in Biochemistry	100
TCA cycle • Electron transport chain (ETC) • Heme metabolism	

ENZYMES

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Basics of Enzymes

00:00:20

Definition :

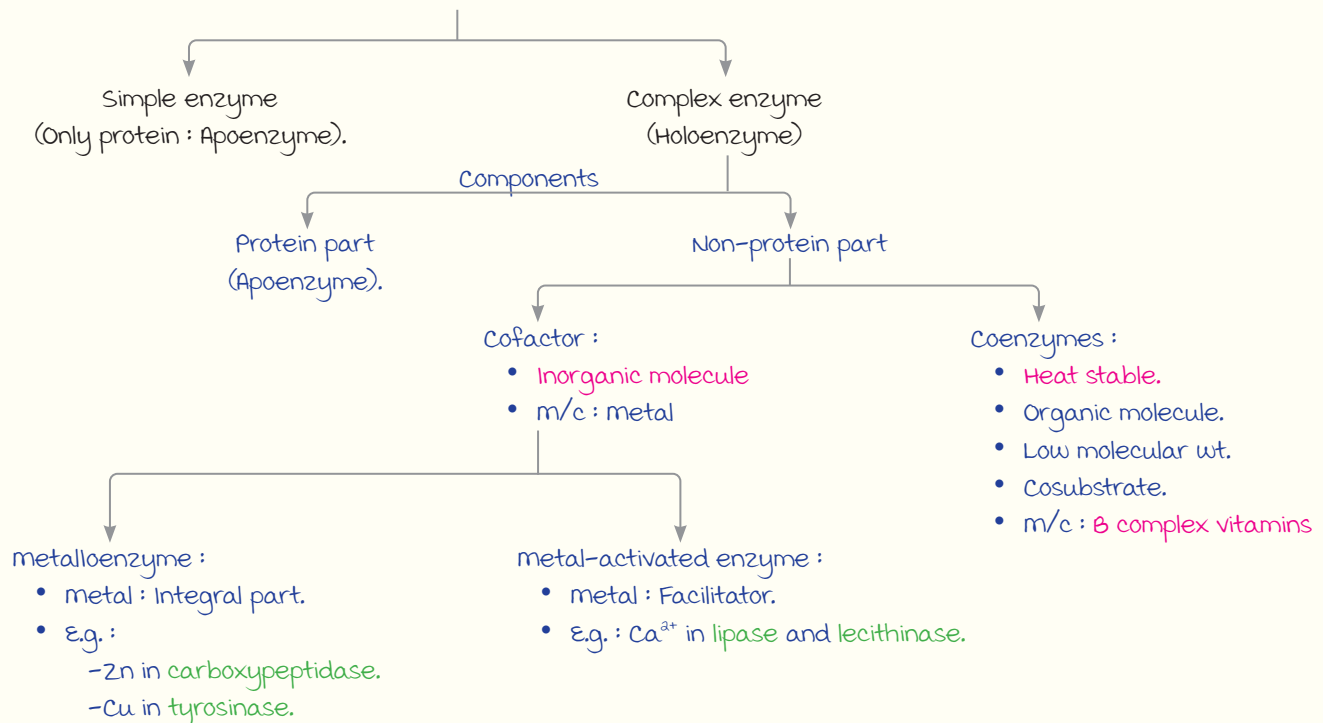
- Specialized proteins that act as **biological catalyst**.
- Exception : **Ribozymes** (RNA).

Ribozyme	Location	Function
Peptidyl transferase	28 Sr RNA	Peptide bond synthesis
Sn RNA	Spliceosome	RNA splicing
Group II introns (Self splicing)	-	
Ribonuclease P	-	Post-transcriptional modification of tRNA

Properties of Enzymes :

- 16% by weight : Nitrogen.
- Heat labile**.
- Precipitated by protein precipitating agents.

Types of Enzymes :



Prosthetic group : Co-enzyme/Co-factor tightly integrated into apoenzyme.

----- Active space ----- Coenzymes :

Vitamin	Active form	Reaction involved
Thiamine	Thiamine di/pyrophosphate (TDP/TPP)	- Oxidative decarboxylation : BCKD, α-KGDH, PDH - Transketolase
Riboflavin	FAD, FMN	Dehydrogenase (Succinate, acyl CoA)
Niacin	NAD ⁺ , NADP ⁺	most dehydrogenases
Pantothenic acid	CoA, Acyl carrier protein	All reactions with acetyl CoA, succinyl CoA & fatty acid synthase complex
Pyridoxine	Pyridoxal phosphate	- Transamination - Transsulfuration (Cystathionine-β synthase & cystathionase) - Decarboxylation - ALA synthase : Heme synthesis - Glycogen phosphorylase : Carbohydrate metabolism Amino acid metabolism
Folic acid	THFA	All 1 Carbon reactions
Cobalamin	- methyl B₁₂ - Adenosyl B₁₂	- methionine synthase - methylyl malonyl CoA mutase
Lipoate	Lipoamide	Oxidative decarboxylation
Ascorbic acid	-	Hydroxylation (Prolyl & lysyl hydroxylase) : Collagen synthesis

Cofactors  :

metal	Reaction catalysed		
Zinc	- Carbonic anhydrase - Carboxypeptidase A & B - Alcohol dehydrogenase - Cytosolic SOD (Free radical scavenger)		
magnesium	ATP/PO ₄ group involved : <table border="0"> <tr> <td> - Kinase - Phosphatase </td><td> - mutase - Enolase </td></tr> </table>	- Kinase - Phosphatase	- mutase - Enolase
- Kinase - Phosphatase	- mutase - Enolase		
Iron	Heme iron : <table border="0"> <tr> <td> - Hemoglobin, myoglobin - Complex III & IV of ETC (Cytochrome) - Nitric oxide synthase </td><td> - Peroxidase - Catalase - Tryptophan pyrrolase </td></tr> </table> Non-heme iron : Complex I, II and III of ETC (Fe-S complex)	- Hemoglobin, myoglobin - Complex III & IV of ETC (Cytochrome) - Nitric oxide synthase	- Peroxidase - Catalase - Tryptophan pyrrolase
- Hemoglobin, myoglobin - Complex III & IV of ETC (Cytochrome) - Nitric oxide synthase	- Peroxidase - Catalase - Tryptophan pyrrolase		

Zn deficiency features

↓ Retinol dehydrogenase



Visual disturbances

no deficiency

- ↓ Xanthine oxidase
- Hypouricemia

Cu deficiency

- Depigmentation/hypopigmentation
- X-ray : Similar to scurvy
- Menke's disease : Lysyl oxidase defect (ATP 7A gene)

metal	Reaction catalysed
manganese	- Kinase - Phosphatase - mitochondrial SOD - Arginase - Ribonucleotide reductase
molybdenum	Xanthine oxidase
Potassium	$\text{Na}^+ - \text{K}^+$ ATPase - Pyruvate Kinase
Copper	- Tyrosinase (melanin production) - Complex IV of ETC (Cytochrome C oxidase) - Lysyl oxidase : - Covalent cross linking of collagen (Quarter staggered arrangement) - Oxidative deamination of lysyl and hydroxyllysyl residues.
Nickel	Urease
Calcium	- Lecithinase - Lipase

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Note :

vit. C deficiency leads to defective collagen D/t ↓ lysyl hydroxylase and prolyl hydroxylase.

Classification of Enzymes

00:20:48

Class	Class name	Details & examples
I : Oxidoreductases	Dehydrogenase	Transfer hydrogen elements to an acceptor : $\text{NAD}^+ \rightarrow \text{NADH}$ $\text{FAD} \rightarrow \text{FADH}_2$: - SDH - Acyl CoA DH $\text{NADP}^+ \rightarrow \text{NADPH}$: - GL6PD 6-Phosphogluconate DH - Cytoplasmic ICDH } HMP shunt
	Oxygenase	monooxygenase : Phenylalanine/tyrosine/tryptophan hydroxylase (Require BH_4), cytochromes
	Oxidase	Forms H_2O or $\text{H}_2\text{O}_2 \rightarrow$ Source of ROS
	Peroxidase	- Glutathione peroxidase ($\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O}$) - Free radical scavenger
	Catalase	$\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O}$ - Enzyme marker of peroxisome - Free radical scavenger
	Reductase	Reductive biosynthesis : - NADPH utilizer (generated by HMP shunt) - Fatty acid, cholesterol & steroid synthesis
II	Transferase	- Transfers functional group - E.g. : - Kinases (Hexo/glucokinase) - Phosphorylases (Glycogen phosphorylase)
III	Hydrolase	- Breaks covalent bonds by adding H_2O - E.g. : - All digestive enzymes - Phosphatase - Arginase

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Class	Class name	Details & examples
IV	Lyase	• Breaks covalent bond without H_2O addition • Form double bonds by atom elimination or adding groups • E.g. : - Aldolase - Fumarase - Aconitase - Enolase - Simple decarboxylase (require PLP)
V	Isomerase	E.g. : - Isomerase (Produces isomers) - Racemase (D & L isomers) - mutase (Intramolecular PO_4 transfer)
VI	Ligase	• Coupling of molecules with breakdown of ATP • E.g. : - Synthetase - Carboxylase (Requires biotin)
VII	Translocase	• Transfer of ions/molecules across membrane • E.g. : H^+ pump/Ca^{2+} channel

Carboxylation & decarboxylation reactions :

Types	Reactions & cofactors	Applied biochemistry
Simple decarboxylation : Class IV	Synthesis of : • GABA • Serotonin • Epinephrine • Norepinephrine } Requires PLP (vit B_6)	Vit B_6 deficiency : Resistant seizures in neonate
Oxidative decarboxylation : Class I	• multienzyme complexes : - E1 : PDH/α-KGDH/BCKD - E2 : Dihydrolipoyl transacylase - E3 : Dihydrolipoamide DH • 5 coenzymes : - TPP : vit B_1 - CoA : vit B_5 - Lipoamide - FAD : vit B_a - NAD^+ : vit B_3	• Chronic alcoholism : vit. B_1 def. $\rightarrow$ $\downarrow$ PDH, α-KGDH $\rightarrow$ $\downarrow$ ATP (Fatigue) • Beri-beri : vit. B_1 def. $\rightarrow$ - Peripheral edema - Ptois - Confabulation • Enzyme assay : erythrocyte transketolase
B_7 dependent carboxylation	Requires : • ATP • Biotin • CO_a	Avidin (Raw egg) $\xrightarrow{\ominus}$ Biotin dependent carboxylases
B_7 independent carboxylation	• AIR carboxylase • Gamma carboxylase : Needs vit. K • CPS : I and II	-

Enzyme Mechanism of Action

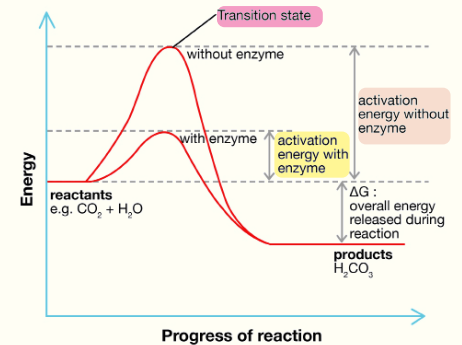
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- Substrate binding : **Active site**.
- Site for regulator/modifier : **Allosteric site**.

Free energy change (ΔG)  :

- Free energy change =
 $\Delta G = \text{Energy of product} - \text{Energy of substrate}$.
- Enzyme $\rightarrow$ $\downarrow$ Activation energy.
 $(\epsilon_{\text{act}}^{\text{catalysed}} < \epsilon_{\text{act}}^{\text{uncatalysed}})$
 $\rightarrow$ No change in ΔG .



ΔG interpretation

- +ve : Unfavourable/endergonic reaction
- ve : Favourable/exergonic reaction

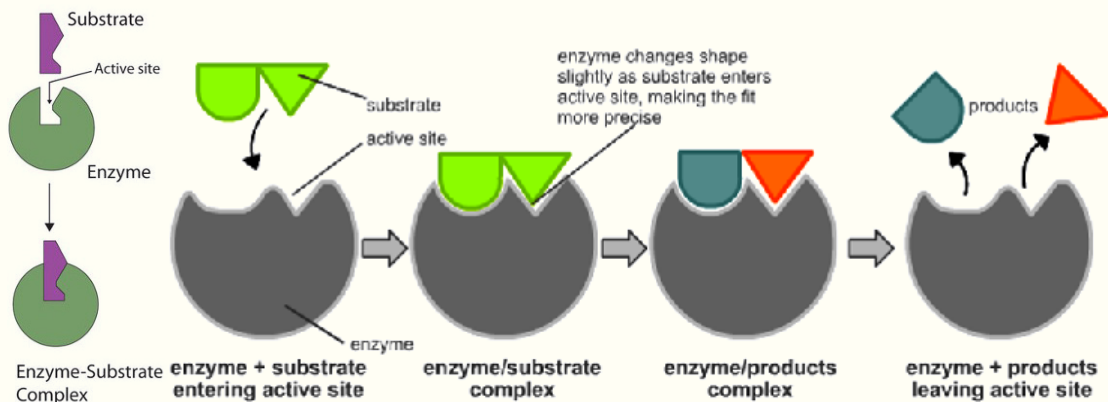
Enzyme-Substrate Complex :

Emil-Fischer's template theory :

- Lock and key mechanism.
- Could not explain dynamic changes.

Koshland's induced fit theory :

Conformational change in active site induced by substrate during binding.



mechanisms that Lower Activation Energy (ϵ_{act}) :

mechanism	Details	Examples
Catalysis by proximity	Reactants in bond-forming distance	-
Acid-base catalysis	Amino acid at catalytic site Proton donors Proton acceptors	Aspartate proteases (Pepsin)
Covalent catalysis	Transient covalent bond b/w : Enzyme + substrate	Serine proteases
metal-ion catalysis	metal ions facilitate reaction	Zn in Carboxy peptidase
Catalysis by strain	Enzyme strains & breaks substrate	Lytic reaction $\rightarrow$ Lyase Hydrolyase

----- Active space -----

Enzyme Kinetics

01:01:20

Equilibrium constant (K_{eq}):

- Independent of enzyme action.
- $K_{eq} = \frac{[Products]}{[Substrates]}$
- $K_{eq} = 1$: Reversible reaction at equilibrium.

Factors Affecting Rate of Reaction :

1. Substrate concentration:

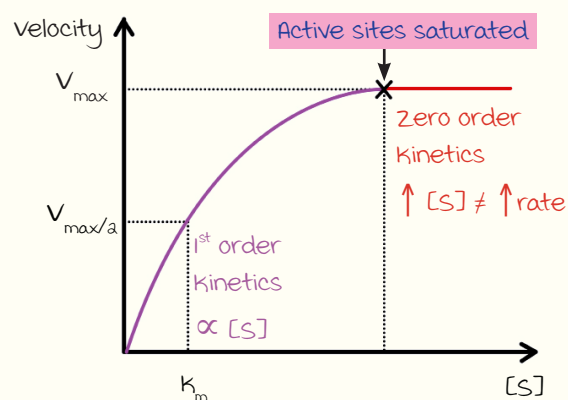
- Hyperbolic curve:

Michaelis-Menten equation:

$$V_i = \frac{V_{max} \times [S]}{K_m + [S]}$$

 K_m value (Michaelis constant)

- $[S]$ at $V_{max}/2$.
- $\propto \frac{1}{\text{enzyme affinity to substrate}}$
- If $[S] = 1000 K_m$: Zero order kinetics
- Signature of E-S complex
- Ideal substrate: Lowest K_m

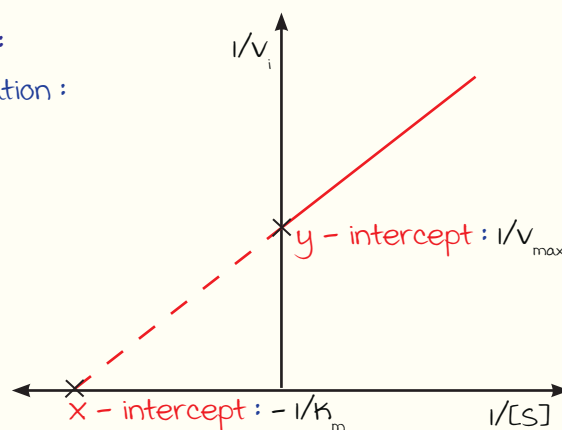


- Lineweaver-Burk plot (Straight line):

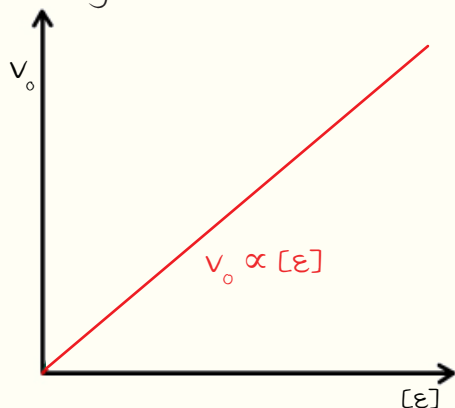
Reciprocal of Michaelis-Menten equation:

$$\frac{1}{V_i} = \frac{K_m}{V_{max}} \times \frac{1}{[S]} + \frac{1}{V_{max}}$$

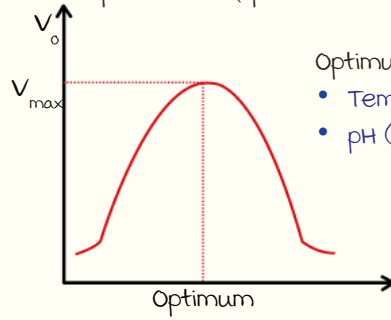
$\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
 y a $\times$ x $+ b$



2. Enzyme concentration:



3. Temperature & pH :



Optimum :

- Temperature (35 to 40 °C).
- pH (5 to 9).

$Q_{10} = 2$
$10^\circ\text{C} \uparrow \rightarrow 2 \times \text{rate of reaction}$

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Catalytic Constant :

- AKA turnover number.

$$\text{Catalytic efficiency} = \frac{k_{cat}}{K_m} = \frac{y}{x}$$

$$k_{cat} = \frac{[V_{max}]}{[E_t]}$$

E_t = Total enzyme concentration = Free enzyme + ES complex.

Enzyme Inhibition

01:10:32

	Competitive inhibition	Non-competitive inhibition	Uncompetitive inhibition
Features	• Inhibitor : Structural analogue of substrate • Compete for same site as substrate • Reversible	• Inhibitor : Not a structural analogue • Distinct binding site • Usually irreversible	Inhibitor binds to enzyme-substrate complex.
Effect on V_{max} & K_m	• V_{max} : Constant • K_m : $\uparrow$ $K_m' > K_m$	• V_{max} : $\downarrow$ • K_m : Constant $V_{max}' < V_{max}$	• V_{max} : $\downarrow$ • K_m : $\downarrow$ $V_{max}' < V_{max}$ $K_m' < K_m$
Line-weaver Burk plot	$\frac{-1}{K_m}$ $\frac{-1}{K_m'}$	$\frac{-1}{K_m'} = \frac{-1}{K_m}$	$\frac{1}{V_{max}'} \frac{1}{V_{max}}$

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	Competitive inhibition		Non-competitive inhibition		uncompetitive inhibition
Examples	Inhibitor	Enzyme	Inhibitor	Enzyme	Phenylalanine inhibits placental alkaline phosphatase
	methotrexate	Dihydrofolate reductase	Cyanide	Cytochrome C oxidase	
			CO		
	Statins	HMG CoA reductase	Fluoride	Enolase	
	Dicumarol	vit K epoxide	Iodoacetate	Glyceraldehyde 3-P-DH	
	Ethanol	Alcohol DH	Fluoroacetate	Aconitase	
malonate (Poison)	Succinate DH				

Phenylalanine inhibits placental alkaline phosphatase

Applied biochemistry	
1. Patient on :	
- methotrexate	• Bell shaped curve : michaelis menten equation • Straight line : Lineweaver-Burk plot
- Statins	
2. Dicumarol : Anticoagulant	
3. Hooch tragedy : methanol poisoning	Antidote → Ethanol (Competitive inhibition)
4. Consumption of KCN tablets :	Non-competitive inhibition
5. Fluoride oxalate used in	estimation of blood glucose
6. Fluoroacetate	Inhibits → Krebs cycle

Suicide Inhibition :

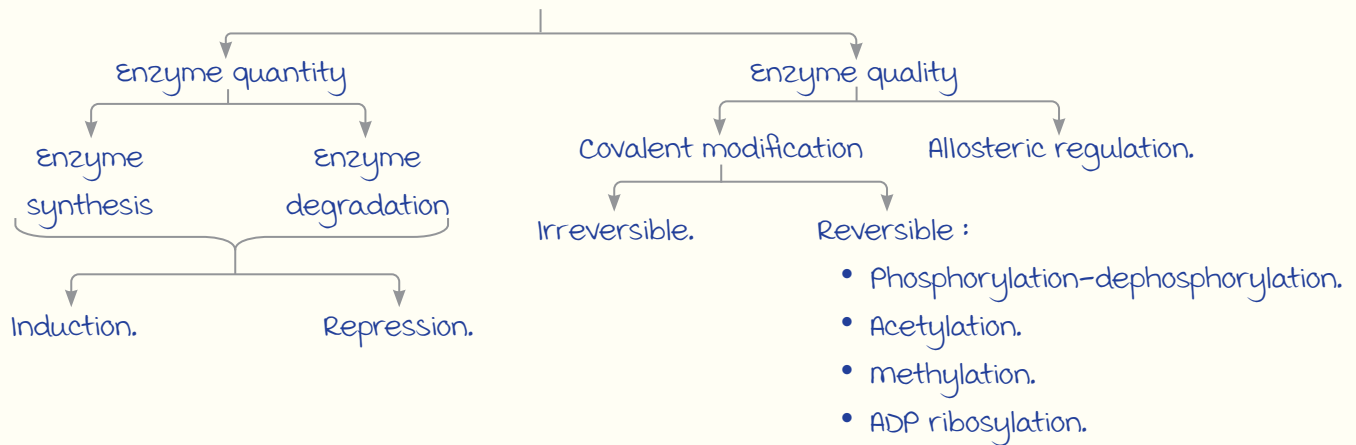
unreactive inhibitor $\xrightarrow{\text{Binds to enzyme}}$ Reactive inhibitor (Irreversible).

Suicide inhibitor	Enzyme
Allopurinol	Xanthine oxidase
DiFluoromethyl ornithine	Ornithine decarboxylase
Aspirin	Cyclooxygenase

Regulation of Enzyme Activity

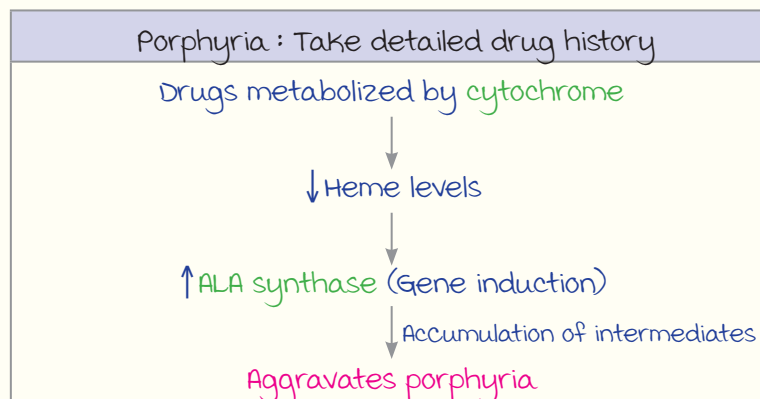
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Enzyme Quantity :

- Heme $\xrightarrow{\text{Represses}}$ ALA synthase gene.
- Dietary cholesterol $\xrightarrow{\text{Represses}}$ HMG CoA reductase gene.



Covalent modification (Hormonal Regulation) :

1. Zymogen activation (Irreversible) :

- Gastrointestinal enzyme (E.g. : Trypsinogen $\rightarrow$ Trypsin).
- Clotting factors.

2. Phosphorylation-dephosphorylation (Reversible) :

	Phosphorylated state	Dephosphorylated state
Insulin : Glucagon ratio	Low	High
Fed/fasting	Fasting	Fed
Enzymes activated	Inhibited by insulin : • Fructose -1, 6- biphosphatase • Glycogen phosphorylase • Hormone sensitive lipase	Inhibited by glucagon : • Phosphofructokinase • Glycogen synthase • Pyruvate DH • Acetyl CoA carboxylase • HMG CoA reductase

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Clinical Enzymology

01:29:15

Isoenzymes :

Physically distinct form of same enzyme and catalyse same reaction.

Lactate DH :

Isoenzyme	Subunits	Electrophoretogram	Tissue localization	% in serum
LDH-1	H ₄	Fastest	Heart	30
LDH-2	H ₃ M ₁	Fast	RBC	35
LDH-3	H ₂ M ₂	Intermediate	- Brain - Lung - Spleen - platelets	20
LDH-4	H ₁ M ₃	Slow	Liver & skeletal muscles	10
LDH-5	M ₄	Slowest		5

myocardial infarction

- Normal : LDH-2 > LDH-1
- MI : LDH-1 > LDH-2 (Flipped pattern).

Creatine kinase (CK) :

Isoenzyme	Subunits	Electrophoretogram	Tissue localization	% in serum	Clinical significance
CK-1	BB	Fastest	Brain	1	CVA
CK-2	mB	Intermediate	Heart	5	MI
CK-3	mm	Slowest	Skeletal muscle	80	muscle injury, myopathies

Alkaline phosphatase :

Isoenzymes	Location	Clinical significance
α -1-ALP	membrane epithelium of biliary canaliculi	marker of cholestasis
α -2-ALP (Heat labile)	Hepatic sinusoidal cells	marker of hepatic injury
α -2-ALP (Heat stable)	Placenta	most stable
Pre- β -ALP	Osteoblast	marker of bone formation : ↑ in Paget's & Vit D deficiency
Gamma ALP	Intestinal cells	Ulcerative colitis
Leukocyte ALP	Leukocytes	↓ : Acute lymphoblastic leukemia

Enzyme Markers

01:36:47

Cardiac Biomarkers :

markers for MI :

Name	Starts to rise	Peak	Return to baseline
CK-MB (Earliest enzyme marker)	4 - 8 h	24 h	48 - 72 h
- Troponin T - Troponin I } most specific	4 - 6 h	24 h	7 - 10 d
myoglobin (First to rise)	-	-	-

marker for circulatory overload : NT pro-BNP.

Liver :

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markers of hepatic injury :

- S. ALT.
- S. AST.

markers of cholestasis :

- S. ALP (m/c).
- 5' nucleotidase (most specific).
- S. GGT. : Chronic alcoholism.

Prostate :

- Acid phosphatase : Tartrate labile.
- Prostate specific antigen (PSA) : Specific.

Pancreas :

- Serum amylase (Also from salivary gland).
- Serum lipase : Specific.

Bone Disease :

Bone formation (From osteoblast) :

- Pre- β ALP.
- Osteocalcin.
- Propeptide of type I collagen.

Bone resorption (From osteoclast) :

- N-telopeptide of type I collagen.
- C-telopeptide of type I collagen.
- urine free deoxypyridinoline.

Acute Kidney Injury (AKI) :

- | | |
|---|---|
| <ul style="list-style-type: none"> • Kidney injury molecule 1 (Kim 1). • Neutrophil gelatin associated lipocalin (NGAL). • IL-8. • ALT. • Glutathione-S-transferase. | <ul style="list-style-type: none"> • GGT. • microalbumin. • Osteopontin. • Liver fatty acid binding protein. • Sodium hydrogen exchange isoform. • Exosomal fetuin. |
|---|---|

Serine Proteases

01:43:55

Serine Proteases :

Serine present in active site.

Enzymes :

- | | | |
|--|---|------------------------------------|
| <ol style="list-style-type: none"> 1. Chymotrypsin : Bulky amino acids 2. Trypsin : Basic amino acids 3. Elastase : Small amino acids | } | Cleaves peptide bond at this site. |
| <ol style="list-style-type: none"> 4. Clotting factors. 5. Complements. 6. PSA. | | |

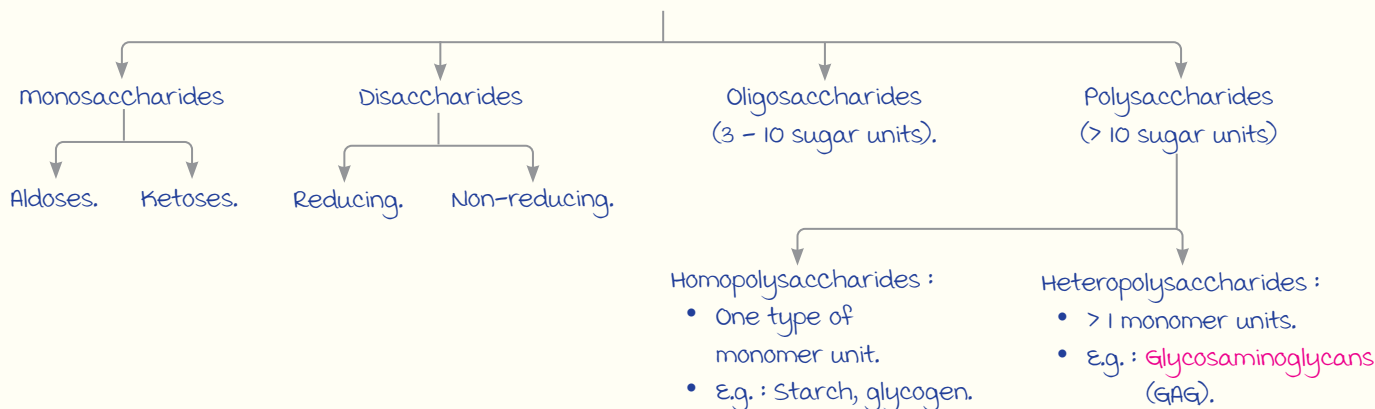
CARBOHYDRATES : INTRODUCTION

Chemistry of Carbohydrates

00:01:00

Carbohydrate : Aldose or keto derivative of polyhydroxy alcohol.

Classification :



monosaccharides :

No. of carbon atoms	Aldoses	Ketoses
Triose (Simplest carbohydrates)	Glyceraldehyde	Dihydroxy acetone
Tetrose	Erythrose	Erythrulose
Pentose	Ribose $\xrightleftharpoons{\text{Epimer}}$ Xylose	Ribulose $\xrightleftharpoons{\text{Epimer}}$ Xylulose
Hexose	Glucose, galactose, mannose	Fructose

Disaccharides :

Reducing disaccharides :

Name	Monomer units	Linkage
Maltose	Glucose + glucose	α -1, 4
Isomaltose	Glucose + glucose	α -1, 6
Lactose	Galactose + glucose	β -1, 4
Lactulose	Galactose + fructose	α -1, β -4

Non-reducing disaccharides :

Name	Monomer units	Linkage
Trehalose	Glucose + glucose	α -1, 1
Sucrose	Glucose + fructose	α -1, β -2

Dietary Fibres

00:10:30

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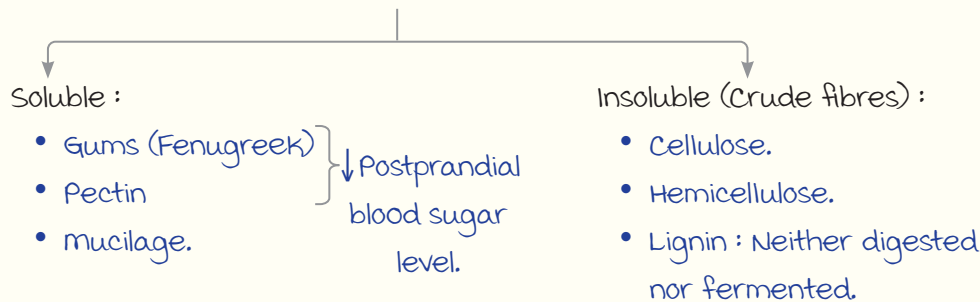
RDA : 40 g/2000 kcal.

Energy released per gram : 2 kcal/g.

Properties :

- Resistant to digestion & absorption by small intestine (Lacks cellulase to hydrolyse beta 1,4 linkage).
- undergoes complete/partial fermentation in large intestine.

Classification :



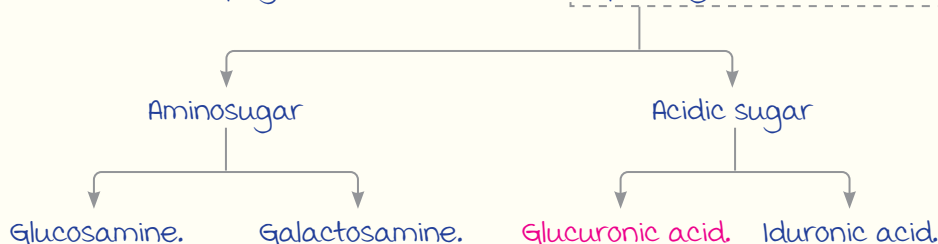
Uses :

- Adds bulk to the stool.
- Regulates bowel movements.
- Improves satiety.
- Prebiotic (Fibre) : Promote colonisation of probiotic bacteria.
- Sequesters bile salts → ↓ Cholesterol.
- Improves glucose tolerance.

Glycosaminoglycans (GAGs)/Mucopolysaccharides

00:16:30

Unbranched heteropolysaccharides made of repeating disaccharide units



Properties :

1. Negatively charged : Confers consistency of mucus & enables mobility at joints.
2. Absorbs water : Provides cushioning effect at weight bearing joints.

----- Active space -----

Significance :

GAG	Significance
Chondroitin sulphate	• most abundant • Found in bones & cartilage
Keratan sulphate (KS)	• Types : - KS-1 : Cornea → maintain corneal transparency - KS-2 : Loose connective tissue • Contains galactose instead of uronic acid
Heparin	• Only intracellular GAG • Anticoagulant • IV heparin : used for estimation of lipoprotein lipase (LpL)
Heparan sulphate	• Responsible for charge selectiveness of GBM • Present in synaptic vesicles • Anchors LpL to vascular endothelium • Acts as plasma membrane receptors
Dermatan sulphate	• Widely distributed GAG found in the dermis • maintains structure of sclera • Atherogenic GAG
Hyaluronic acid	• Important role in cell migration • Not covalently bound to proteins • Sulphate group : Absent

Mucopolysaccharidoses (MPS)

00:28:12

Disorders a/w degradation of GAG → Belong to **lysosomal storage disorders**.General Features :

Short stature

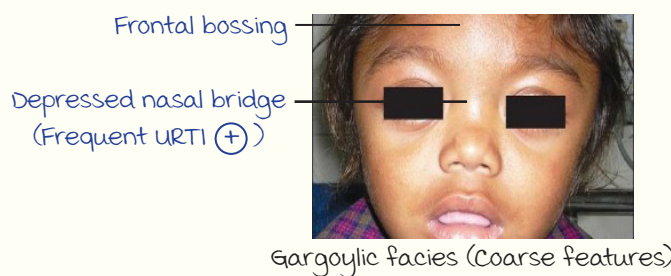


Corneal clouding

Umbilical
herniaProtuberant abdomen
(visceromegaly)



Claw hand



----- Active space -----

X-ray Features :

Dysostosis multiplex

Bullet shaped
middle phalanx

Beaking of vertebrae

Types :

	Enzyme defect	Specific features	GAG accumulated	Treatment
Hurler's disease (MPS-I H)	α -L-iduronidase	- Intellectual disability - mesenchymal defects	- Heparan sulphate - Dermatan sulphate - Keratan sulphate	Aldurazyme (Laronidase) MPS-IH : Stem cell therapy
Scheie disease (MPS-I S)	α -L-iduronidase (Partial defect)	- Hirsutism - Normal intelligence	Dermatan sulphate	
Hunter's disease (MPS II)	α -L-Iduronate sulphatase	- Clear vision - Affects males (X-linked recessive)	- Heparan sulphate - Dermatan sulphate - Keratan sulphate	Elaprase (Idursulfase)
Sanfilippo disease (MPS III : m/c)	Enzyme that degrades heparan sulphate	- mental retardation - No short stature and corneal clouding	Heparan sulphate	-
Morquio disease (MPS IV)	- Type A : N-acetyl galactosamine-6- sulphatase - Type B : β-galactosidase	- Normal intelligence - No visceromegaly & Reilly body inclusions	- Keratan sulphate - Chondroitin sulphate	Type A : Elosulfase (Vimizim)
Maroteaux Lamy disease (MPS VI)	N-acetyl galactosamine -4-sulphatase (Aryl sulphatase B)	Normal intelligence	Dermatan sulphate	Naglazyme (Galsulfase)
Natowicz syndrome (MPS IX)	Hyaluronidase	- Joint pain - Short stature	-	-

----- Active space -----

Note :

I (Inclusion) cell disease : Lysosomal targeting disorder.



Glucose Transporters

00:47:12

SGLT  :

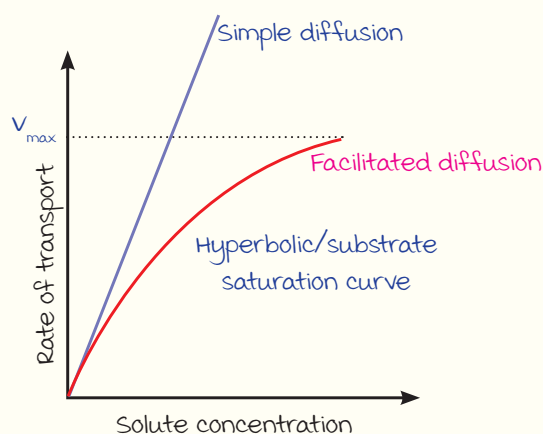
- Sodium dependent.
- Sodium-glucose symport.
- Unidirectional.
- Against concentration gradient.
- Secondary active transport.

Types :

Type	Location	Function	Applied biochemistry
SGLT-1	Luminal side of intestine	Absorption of glucose	ORS contains Na^+ & glucose
	PCT		-
SGLT-2	PCT	Reabsorption of glucose	• SGLT-2 defect : $\downarrow$ Renal threshold $\rightarrow$ Renal glycosuria • SGLT-2 inhibitor/Gliflozins : - OHAs - Glycosuria $\rightarrow$ UTIs (s/e)

GLUTs  :

- Sodium independent.
- Bidirectional.
- Along concentration gradient.
- Ping pong mechanism.
- Facilitated carrier mediated process (Passive).

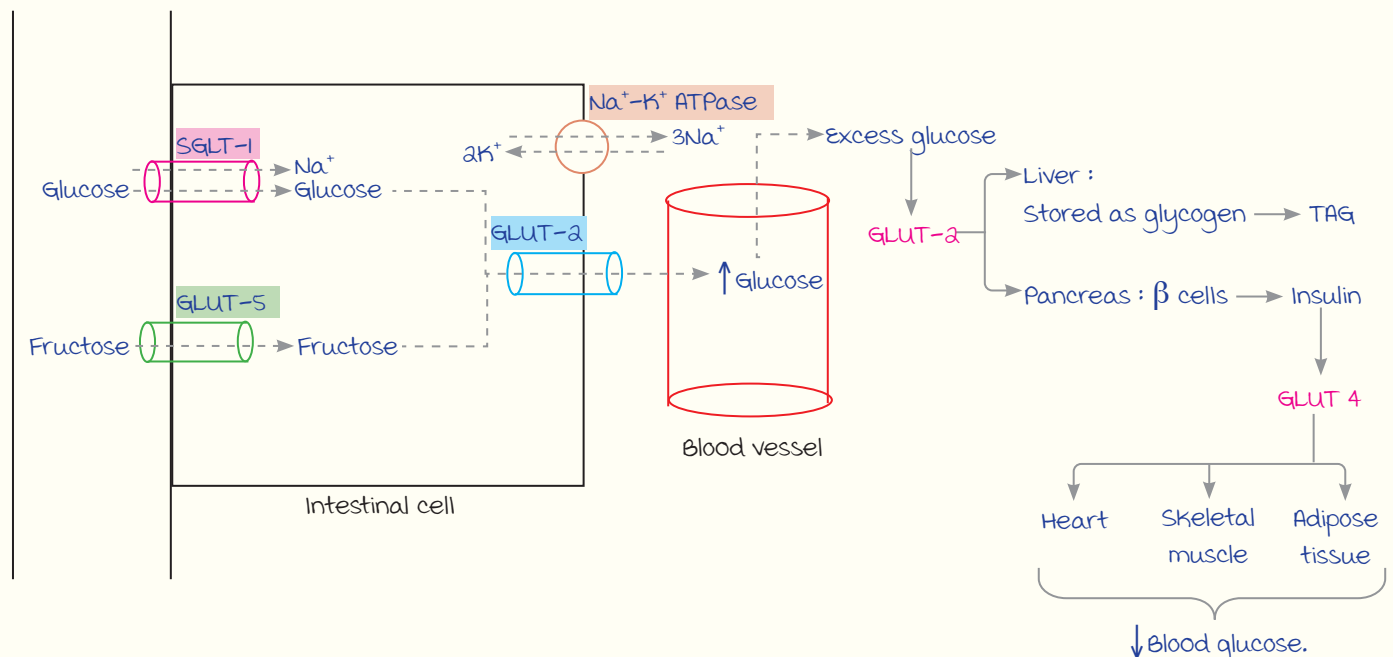


Types :

----- Active space -----

Types	Location	Important points
GLUT-1	- Brain - Placenta - Kidney - RBC - Retina - Colon	- Widely distributed - Low K_m (High affinity)
GLUT-2	β cells of pancreas : ATP mediated insulin release. - Sinusoidal cells of liver : Excess glucose stored as : Glycogen $\rightarrow$ TAG - Serosal side of intestine. - Proximal renal tubules.	- High K_m (Low affinity) - Active in the fed state
GLUT-3	- Neurons - Placenta - Kidney	Lowest K_m (Highest affinity)
GLUT-4	- Heart - Skeletal muscle - Adipose tissue	- Insulin dependent - GLUT 4 levels : - Overnight fast : $\downarrow$ - Exercise ($\uparrow$ Insulin sensitivity) : $\uparrow$ - Diabetes mellitus : $\downarrow$
GLUT-5	- Luminal side of intestine - Testes & sperm	Fructose transporter
GLUT-6	- Spleen - Leukocyte	-
GLUT-7	Liver SER	Transports glucose-6- PO_4 into SER of liver

Overview :



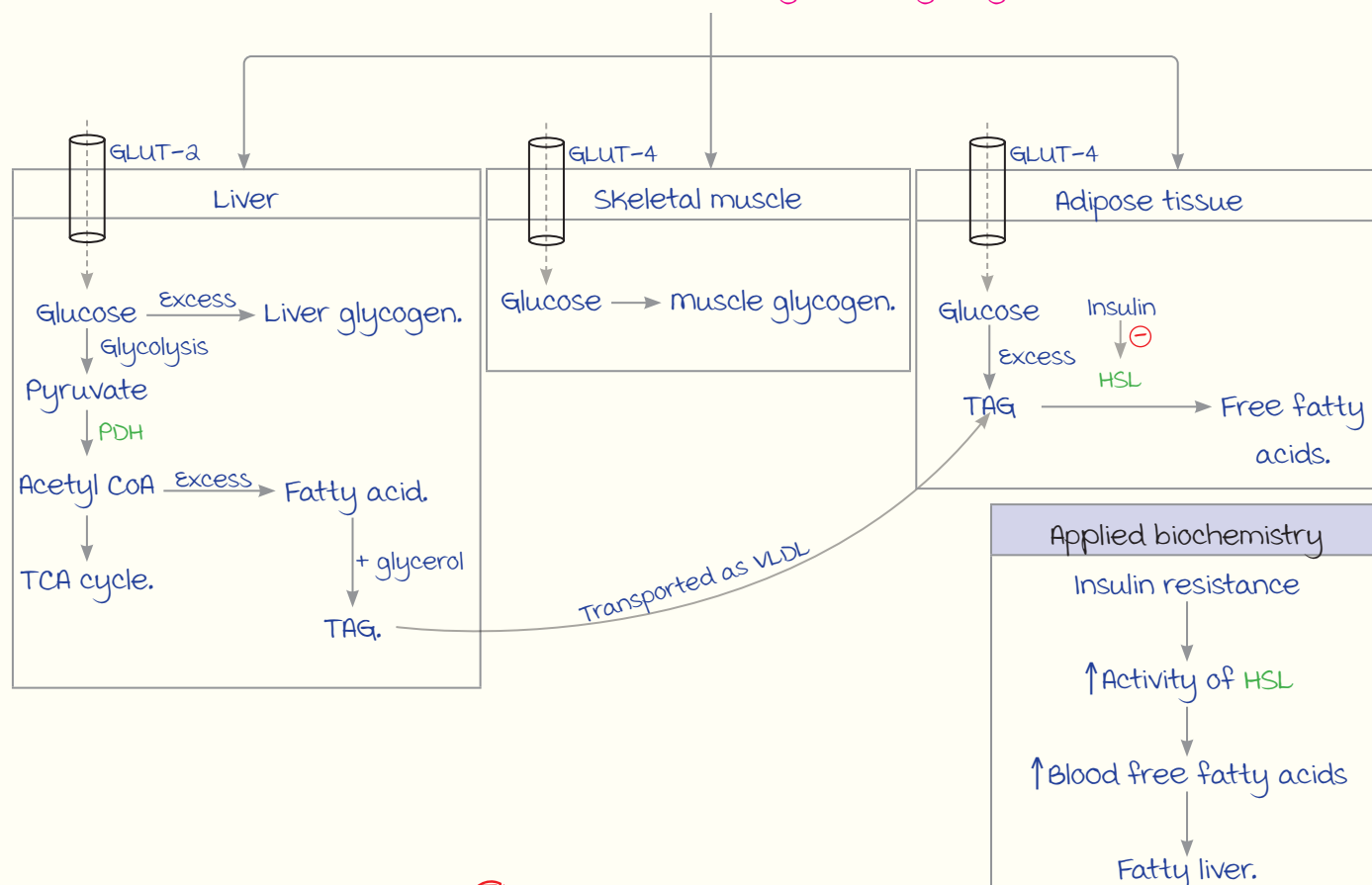
METABOLISM OF CARBOHYDRATES : PART 1

Fed & Fasting State Metabolism

00:00:10

Well Fed State :

Within 1 - 4 hours of food intake → High insulin/glucagon ratio.



Fasting State :

Stage	Duration post food intake	metabolic fuel	Source of energy
Early fasting	4 - 16 hours	Glucose	Hepatic glycogenolysis
Fasting	16 - 48 hours		Gluconeogenesis (Source of ATP : Fatty acid oxidation)
Prolonged fasting (Starvation)	2 - 5 days	Ketone bodies	Ketone body synthesis
Prolonged starvation	> 5 days	muscle proteins	muscle proteolysis (Cachexia)

Glycolysis : Steps & Energetics

00:08:36

----- Active space -----

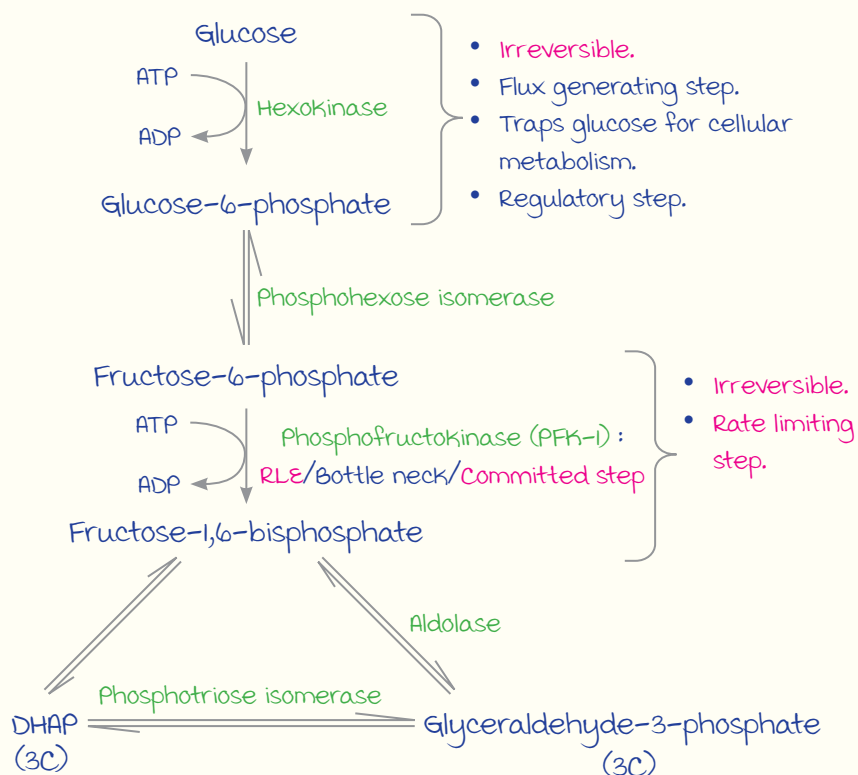
Features :

- Site : All organs (In cytoplasm).
- Only pathway that can operate **aerobically** & **anaerobically**.

Condition		Reason
Defect in glycolytic enzymes ↓ Hemolysis		RBCs rely exclusively on anaerobic glycolysis for ATP in fed/fasting state (Lack mitochondria)
Tolerance to hypoxia	Heart muscles : Low	Low glycolytic capacity
	Skeletal muscles : High	Enormous glycolytic capacity

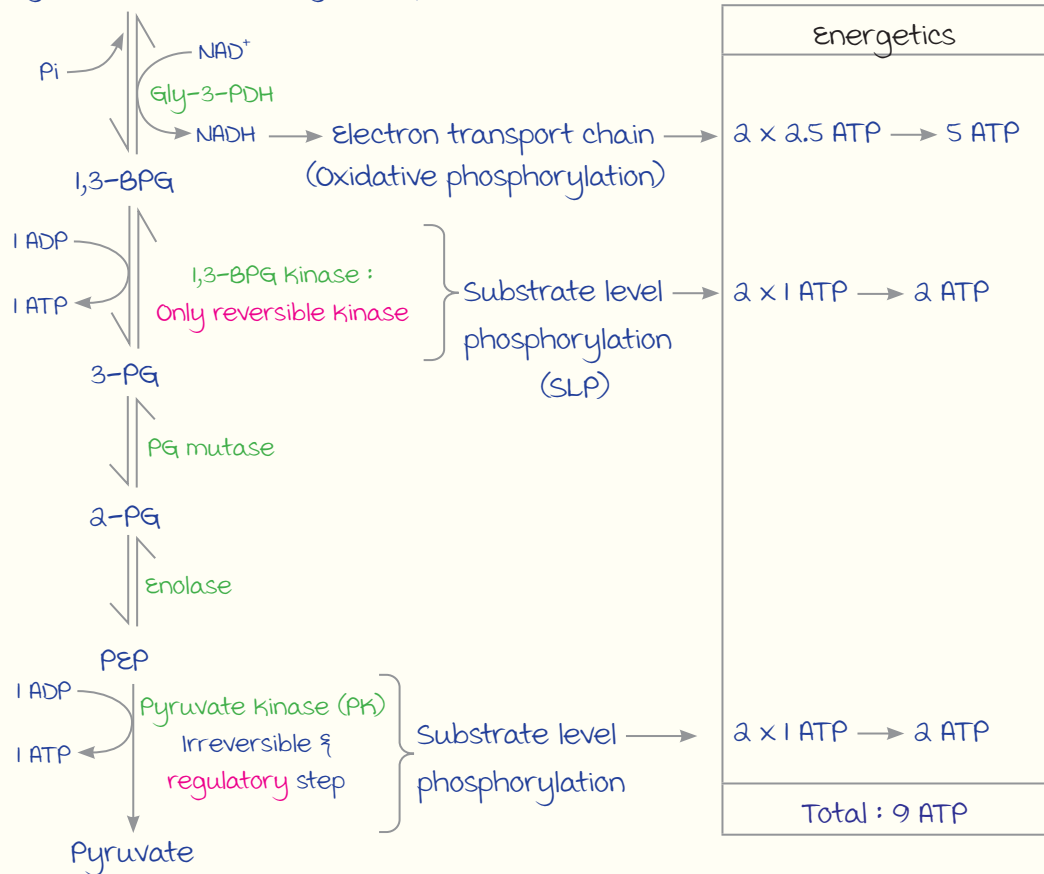
Aerobic Glycolysis :

Preparatory phase : Stage of ATP utilization (2 ATPs used).



----- Active space ----- Pay off phase :

2 x Glyceraldehyde-3-phosphate (Gly-3-PO₄)



Net ATP : 9 ATP - 2 ATP (utilised in preparatory phase) = 7 ATP.

Applied biochemistry :

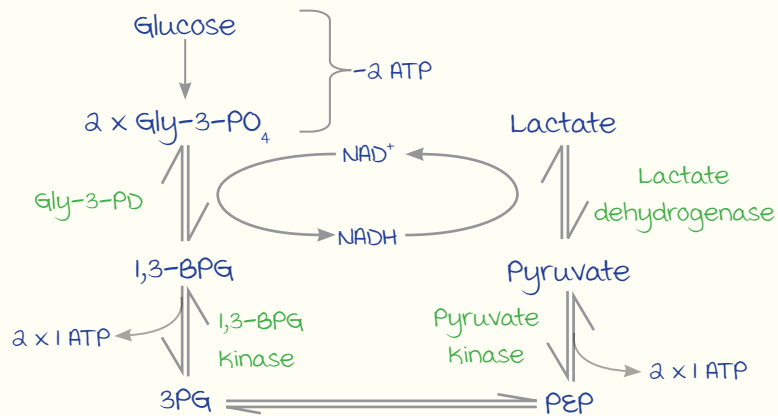
Compound	Enzyme inhibited	Reason/Application
Fluoride	Enolase	Blood glucose estimation : Gray vacutainer containing NaF & oxalate used
Arsenate	Gly-3-PDH	Arsenate-phosphoglycerate : No ATP formed
Iodoacetate		Inhibits -SH containing enzymes

Hexokinase vs. Glucokinase :

	Hexokinase	Glucokinase
Isoforms	I, II, III	Hexokinase IV
K _m values	Low	High (Active in fed state)
Affinity	High	Low
Inducible/Constitutive	Constitutive (Housekeeping enzyme)	Inducible
Inhibited by G-6-P	⊕	⊖

Anaerobic Glycolysis :

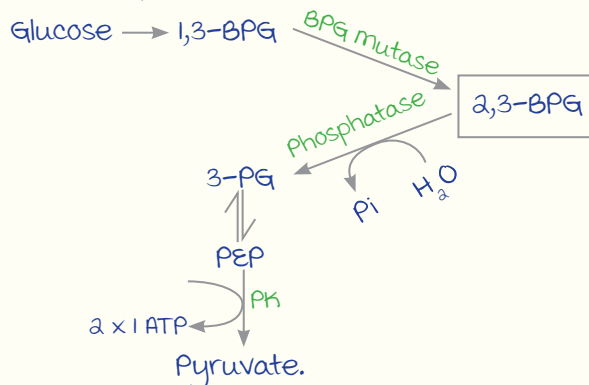
----- Active space -----



Energetics	
• No net generation of NADH : utilized by LDH	} 4 ATP
• 1,3-BPG : 2 x 1 ATP	
• Pyruvate Kinase : 2 x 1 ATP	
• Utilization : 2 ATP	
Total : 4 ATP - 2 ATP = 2 ATP	

Rapaport Leubering Cycle/2,3-BPG Shunt :

Site : Takes place in RBCs.



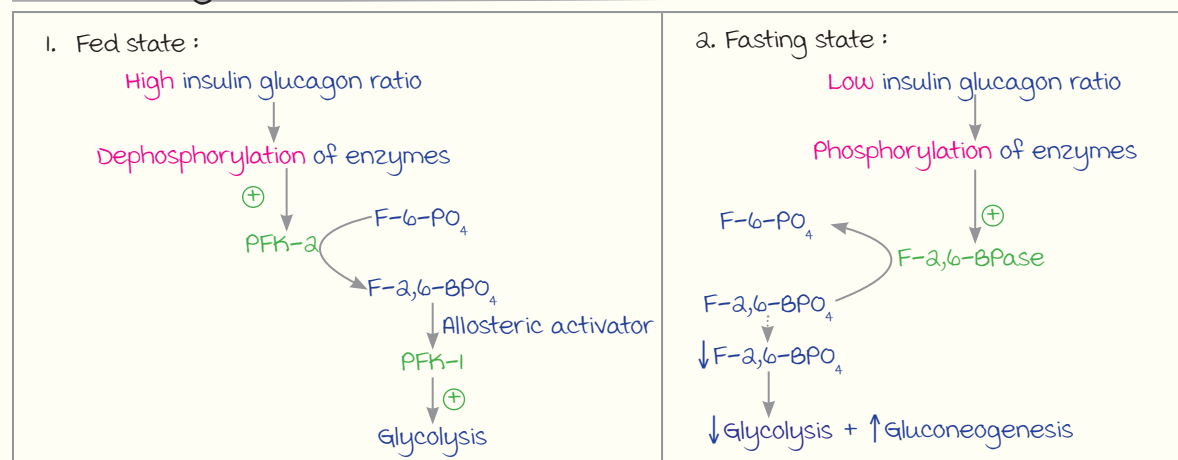
Energetics	
• Pyruvate kinase : +2 ATP • Hexokinase : -1 ATP • PFK : -1 ATP	
Net ATP : 0	

Significance of 2,3-BPG :

- maintains taut state (T-state) of hemoglobin.
- Responsible for unloading of O₂ at tissue site → Shifts oxygen dissociation curve to the right.

Regulation of Glycolysis

00:36:44

Hormonal Regulation/Covalent modification :

----- Active space -----

Allosteric Regulation :

Substrates : Activators.

Products : Inhibitors.

Enzyme	Allosteric activator	Allosteric inhibitor
Hexokinase	-	ATP
PFK-1	- F-6-P - F-2,6-BP 5' AMP	- ATP - Citrate - Low pH
Pyruvate kinase	-	ATP

Metabolism in Cancer Cells

00:43:23

Warburg hypothesis :

1. Cancer cells have $\uparrow$ levels of glucose uptake.
2. Glucose in cancer cells $\xrightarrow{\text{Aerobic fermentation}}$ Lactate.

metabolic reprogramming :

	Normal cells	Cancer cells
Glycolytic pathway	Glucose ↓ PEP PKM1 (Pyruvate Kinase) : • Tetramer • High catalytic activity ↓ Pyruvate ↓ Acetyl CoA → TCA cycle	uses glucose via aerobic glycolysis. Glucose ↓ PEP PKM2 : • Dimer • Low catalytic activity ↓ Pyruvate ↓ O_a Lactate
Energetics	Glucose ↓ 7 ATP Pyruvate ↓ 2 NADH = 5 ATP Acetyl CoA ↓ TCA cycle: $2 \times 10 \text{ ATP} = 20 \text{ ATP}$ Aerobic oxidation : 1 Glucose → 32 ATP.	$\uparrow$ Affinity for glucose. Glucose ↓ Pyruvate ↓ Lactate Aerobic glycolysis : 2 ATP. Application : usage of fluoro-deoxy-glucose in PET scan ↓ $\uparrow$ uptake by cancer cells Functional screening

Pyruvate Dehydrogenase (PDH)

00:51:28

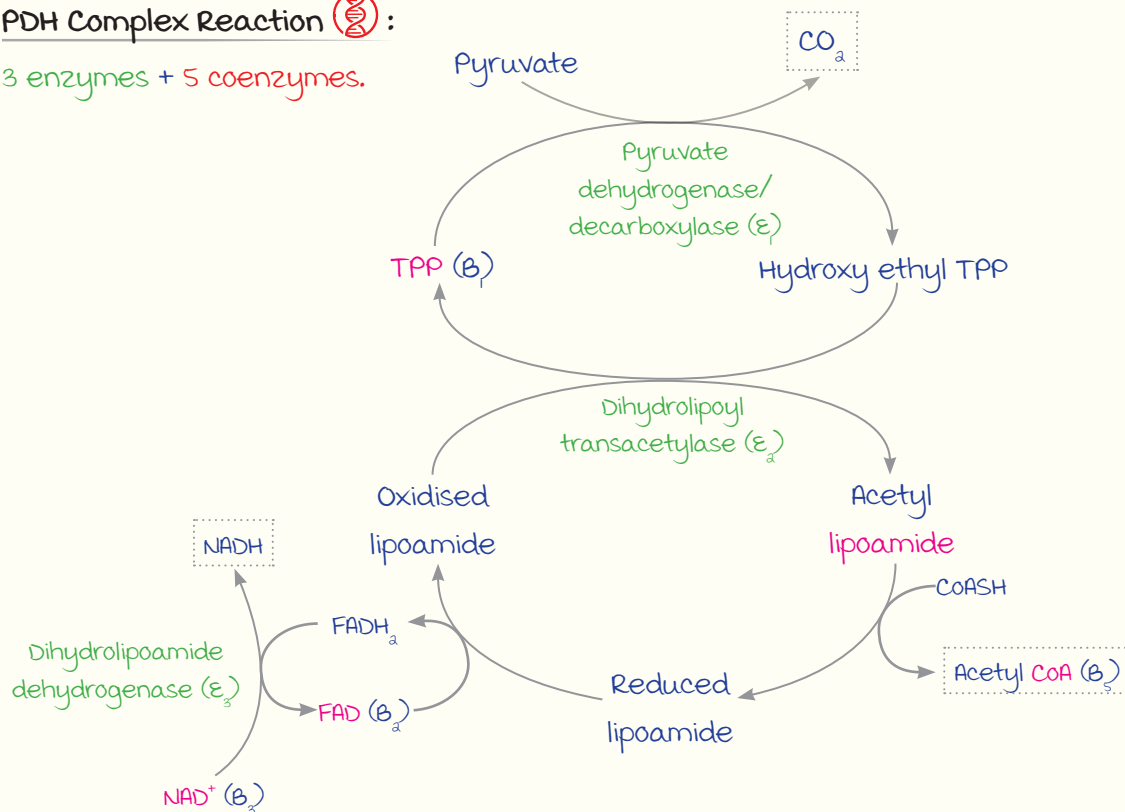
----- Active space -----

Link reaction :

- Links glycolysis to TCA cycle.
- Site : mitochondria.
- Net reaction : Pyruvate (3C) $\xrightarrow[\text{CO}_2]{\text{NAD}^+ \rightarrow \text{NADH}}$ Acetyl CoA (2C).

PDH Complex Reaction :

3 enzymes + 5 coenzymes.



Significance :

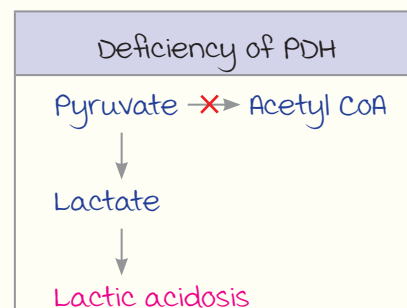
- Irreversible & cannot be circumvented by another enzyme.
- Fat : **Never converted** to glucose $\rightarrow$ Exceptions :
 - Glycerol.
 - Odd chain fatty acid (Propionyl CoA).

Clinical aspect :

Excess carbohydrate $\rightarrow$ Pyruvate $\rightarrow$ Acetyl CoA
 (Exclusive carbohydrate diet)

Sufficient energy stores $\oplus$

Acetyl CoA $\rightarrow$ Fatty acid $\rightarrow$ TAG $\rightarrow$ $\uparrow$ VLDL



METABOLISM OF CARBOHYDRATES : PART 2

Glycogen Synthesis

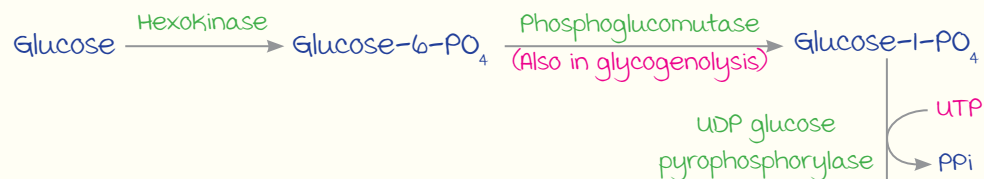
00:00:14

Occurs in the well-fed state; high insulin - glucagon ratio.

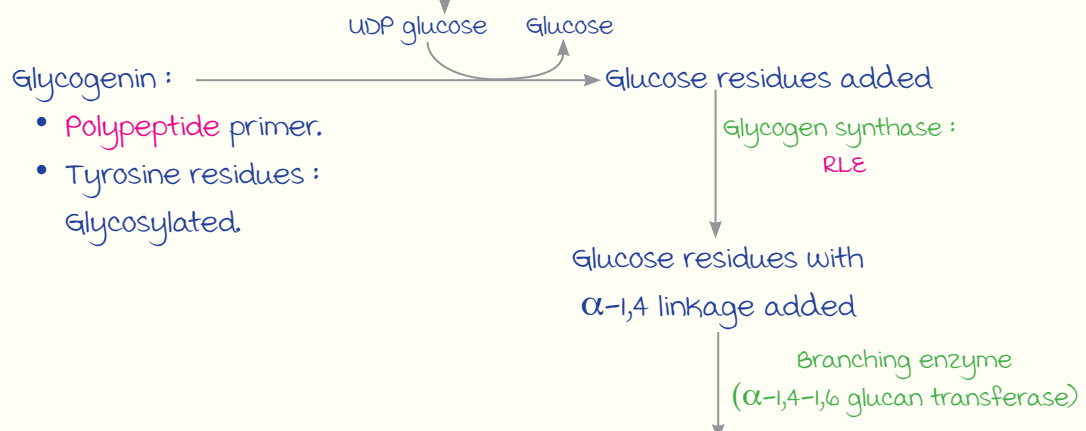
Site : Cytoplasm of liver & skeletal muscle.

Stages :

1. Synthesis of UDP glucose :



2. Synthesis of linear polymer :



3. Branching of linear polymer : Adds branches at α -1,6 linkage.

Glycogenolysis

00:05:27

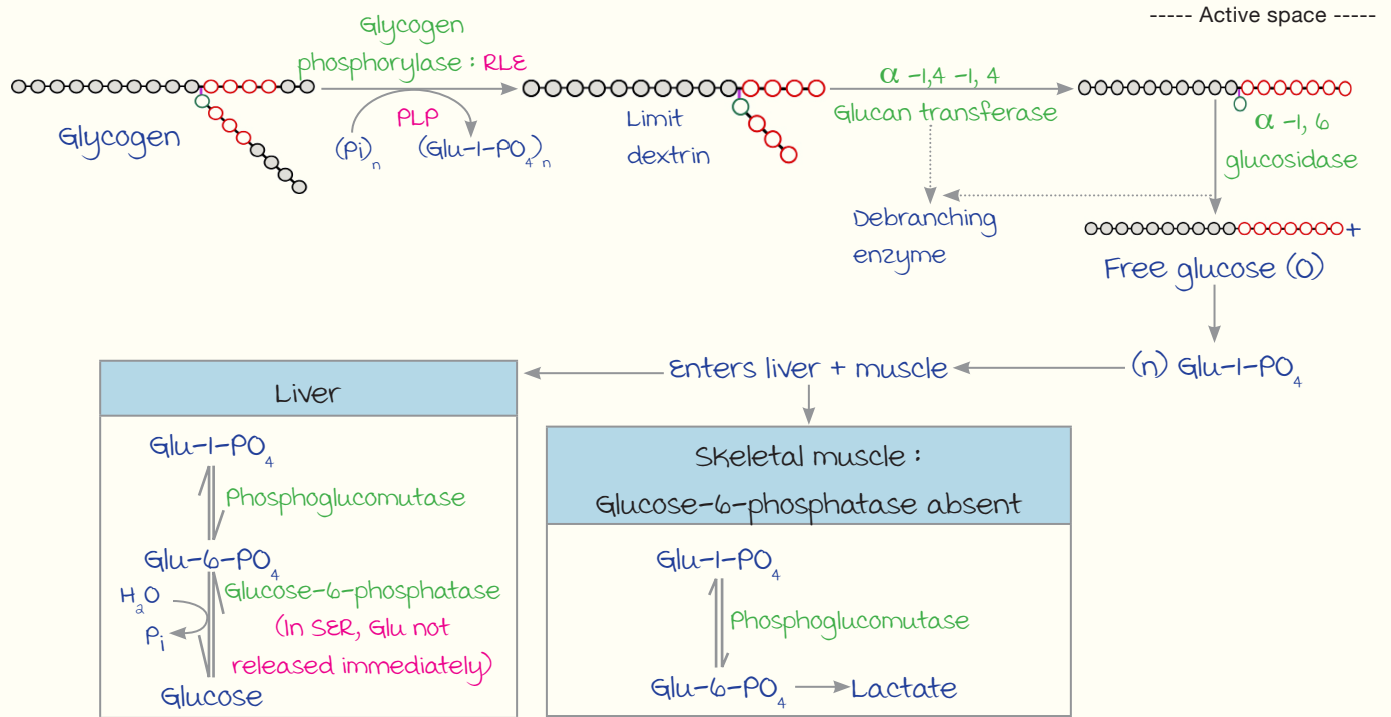
- Occurs in the fasting state (4 - 16 hours without food).
- Hepatic glycogenolysis : Early fasting.

Site :

- Cytoplasm of liver & skeletal muscle.
- Lysosomes (minor role).
- SER

Stages :

1. Release of terminal glucose by glycogen phosphorylase.
2. Removal of branches.
3. Conversion of Glu-1- PO_4 to free glucose.

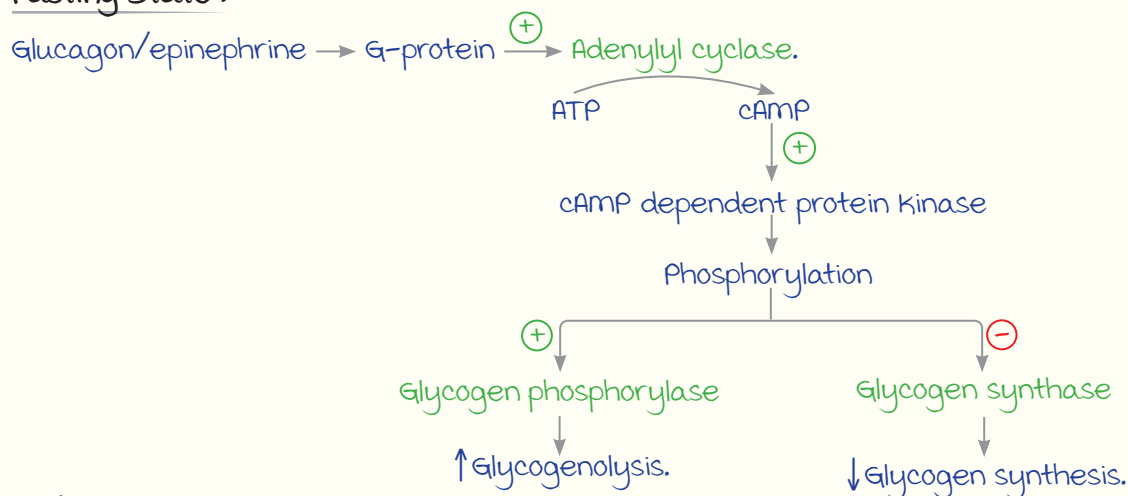


Energetics in muscle : 1 $\text{Glu-6-PO}_4 \rightarrow 3 \text{ ATP}$ (No hexokinase step).

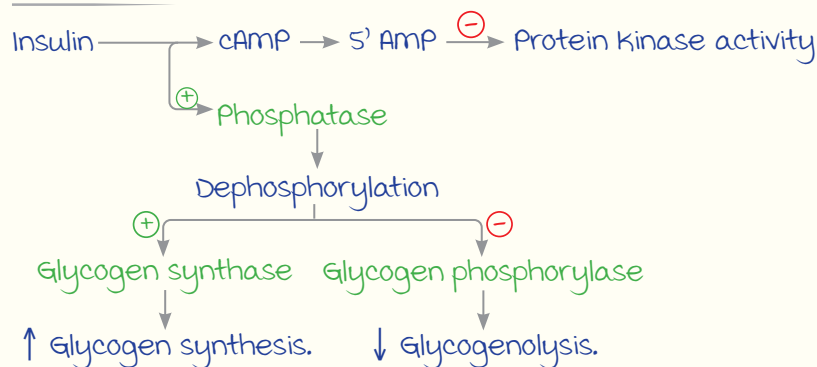
Regulation of Glycogen Metabolism

00:15:22

Fasting State :



Fed State :



----- Active space -----

In the muscle :

Regulation by :

1. cAMP dependent pathway.
2. cAMP independent **calcium calmodulin** dependent pathway $\xrightarrow{+}$ **glycogen phosphorylase**.
3. **5' AMP** :
 - Allosteric activator of **glycogen phosphorylase**.
 - Activated in **extreme state of anoxia**.

Allosteric Regulation :

Inhibitors of glycogenolysis :

- Glu-6-PO₄.
- Glucose (Only in liver).
- ATP.

Activators of glycogen synthesis : Glucose-6-PO₄.Glycogen Storage Disorders

00:22:23

Diseases	Defect in	Features	Investigations
Type Ia : Von Gierke's disease (m/c liver GSD)	Glucose -6-phosphatase	• Protruding abdomen • Doll-like facies • Thin extremities • Convulsions, coma on fasting • O/e : massive hepatomegaly 	• S. glucose ↓ : Severe Fasting hypoglycemia • S. uric acid ↑↑ : Hyperuricemia • S. Lactate ↑↑ : Lactic acidosis • AST & ALT : Normal • Liver biopsy : Accumulation of normal glycogen. • Ketosis : Rothera's test Positive • Hyperlipidemia
Type Ib :	Glu-6-PO ₄ transporter defect (in SER)	All features of Ia + recurrent bacterial infections	Neutropenia
Type 2 : Pompe's disease (Lysosomal storage disorder)	Acid α-1,4 glucosidase/acid maltase	• Floppy infant • macroglossia • Generalized hypotonia • Cardiac failure • Hepatomegaly ⊕/⊖	 Chest x-ray : massive cardiomegaly

----- Active space -----

Diseases	Defect in	Features	Investigations
Type III : Cori's disease/Limit dextrinosis	Debranching enzyme $(\alpha-1,4 - 1,4$ $\text{glucantrans-ferase})$	- Early morning hypoglycemia - Liver cirrhosis (Non fatal) 	S. glucose ↓ - Rothera's test : +/- S. uric acid } Normal S. lactate } Normal - AST & ALT : ↑ - Liver biopsy : Accumulation of limit dextrin
Type IV : Anderson's disease/Amylopectinosis	Branching enzyme $(\alpha-1,4-1,6$ $\text{glucan transferase})$	- Fasting hypoglycemia - Portal hypertension - Cirrhosis (Fatal) Progression → Liver failure 	S. glucose ↓ - Rothera's test : Negative S. uric acid } Normal S. lactate } Normal - AST & ALT : ↑↑ - Liver biopsy : Accumulation of amylopectin (Abnormal glycogen)
Type VI : Her's disease	Hepatic glycogen phosphorylase	Hypoglycemia (Less severe)	-
Type V : McArdle's disease (m/c muscle GSD)	muscle glycogen phosphorylase	- Adolescent male - Pain in calf muscle on exercise - No hemolysis - Second wind phenomena : Exercise intolerance - Pain in calf during exercise → Rest → Resume activity with more ease - Rhabdomyolysis → myoglobinuria →	S. Glucose : ↓ during symptoms S. Lactate : ↓ - AST & ALT : Normal - Creatine Kinase : ↑↑ - Burgundy coloured urine
Type VII : Tarui's disease	muscle & erythrocyte PFK	- Exercise intolerance (No 2nd wind phenomena) - myoglobinuria - Hemolysis	S. Glucose : ↓ during exercise S. Lactate : ↓ - Creatine Kinase : ↑
Fanconi Bickel syndrome	GLUT-2 defect	-	No glycogen accumulation in liver
Type 0 GSD	Glycogen synthase	- No hepatomegaly - Fasting hypoglycemia	No glycogen accumulation

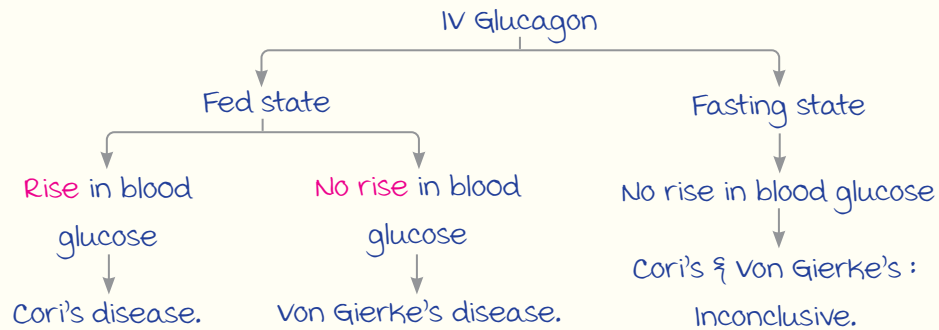
Liver GSD

muscle GSD

----- Active space -----

IV glucagon challenge test :

Differentiates b/w von Gierke's & Cori's disease.



Gluconeogenesis

00:51:40

- 16 - 48 hours of fasting.
- Synthesis of glucose from non-carbohydrate sources.

Site :

Cytoplasm + mitochondria of :

- Liver
- Kidney

Note :

Pathways occurring in both cytoplasm + mitochondria :

mnemonic : PUBG

- Pyrimidine synthesis.
- Urea cycle.
- Blood : Heme synthesis.
- Gluconeogenesis.

Substrates Forming Glucose :

- Glucogenic amino acids : Alanine (Glucose-Alanine/Cahill cycle).
- Lactate (Glucose-Lactate/Cori's cycle).
- Propionyl CoA (Odd chain fatty acid).
- Glycerol.

Enzymes :

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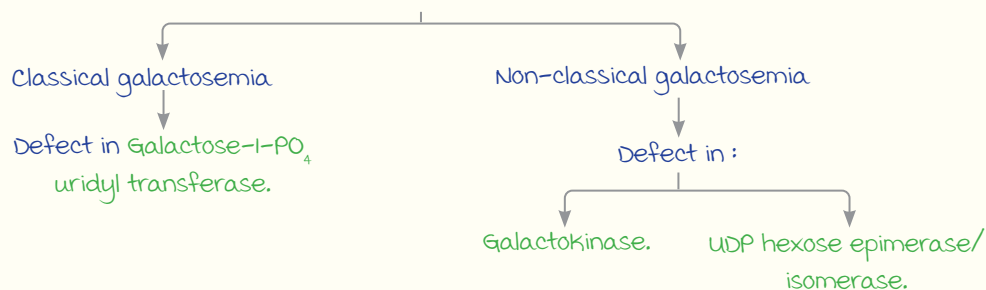
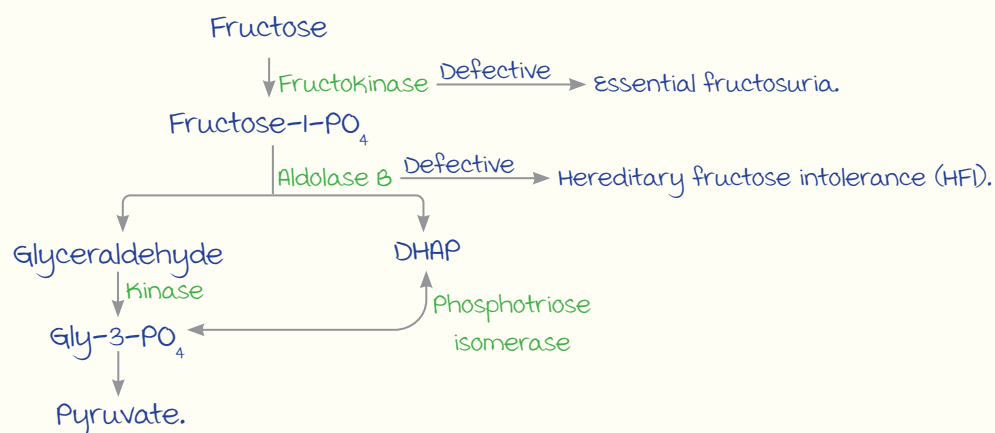
Irreversible steps of glycolysis	key enzymes of gluconeogenesis
Hexokinase	$\text{Glu-6-PO}_4 \xrightarrow[\text{H}_2\text{O}]{\text{Glu-6-phosphatase}} \text{Glucose} + \text{P}_i$
PFK	$\text{F-1,6-BP} \xrightarrow[\text{H}_2\text{O}]{\text{Fructose-1,6-bisphosphatase : RLE}} \text{F-6-PO}_4 + \text{P}_i$
Pyruvate kinase (PK)	a. Pyruvate $\xrightarrow[\text{ATP} + \text{Biotin} + \text{CO}_a]{\text{Pyruvate carboxylase : RLE}} \text{OAA}$ - Acetyl CoA : Allosteric activator of pyruvate carboxylase - mitochondrial step - OAA : Transported to cytoplasm via malate aspartate shuttle b. OAA $\xrightarrow[\text{CO}_a]{\text{PEP carboxykinase (PEPCK)}} \text{PEP} + \text{GTP} \rightarrow \text{GDP}$

Galactose & Fructose Metabolism

01:00:30

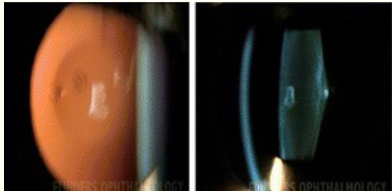
Galactose metabolism :

Types of disorders :

Fructose metabolism :

----- Active space -----

Classical Galactosemia vs. HFI :

		Classical galactosemia	HFI
Enzyme defect		Galactose-1-PO ₄ uridylyl transferase	Aldolase-B
Age of onset		1 st two weeks of life	4 - 6 months
Trigger		Breastfeeding	Fruit juices (Sucrose/fructose)
c/f		- Intellectual disability - Fasting hypoglycemia, Convulsions, Vomiting, Jaundice - Hepatomegaly, liver failure - Feeding difficulty, poor weight gain	
Cataract		⊕ : Oil drop cataract 	Fructose → Low renal threshold ↓ Fructosuria ↓ No cataract
Accumulated compound		Galactose-1-PO ₄ : - Hepatotoxic & neurotoxic ↓ Inorganic PO₄ ↓ Activity of glycogen phosphorylase - Eye : Converted to dulcitol/galactitol ↓ Oil drop cataract	Fructose-1-PO ₄ : - Hepatotoxic & neurotoxic ↓ Inorganic PO₄ ↓ Activity of glycogen phosphorylase
Lab diagnosis	Urine Benedict's test	Positive	Positive
	Glucose oxidase test	Negative	Negative
	Enzyme studies & genetic mutation test		
	Specific test	mucic acid test : Positive	Rapid furfural test/Seliwanoff's test : Positive
Rx		Lactose free diet	Fructose free diet

Hexose Monophosphate (HMP Shunt) Pathway

01:11:28

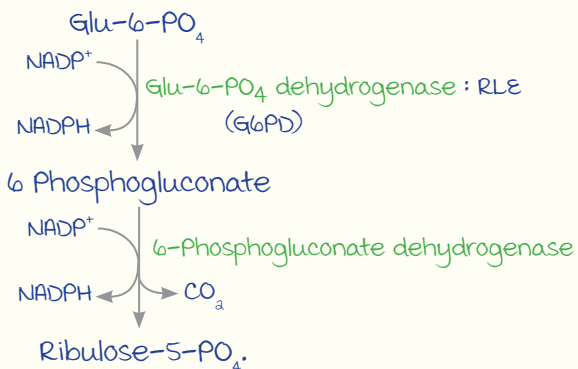
----- Active space -----

- AKA Pentose phosphate pathway.
- Does not generate ATP.

Site : Cytoplasm

Phases :

Oxidative phase :

- Irreversible.
- Generate NADPH.
- Steps :
 

- Functions of NADPH :

1. Free radical scavenging :



Sites :

- RBC : Prevents hemolysis (maintains membrane integrity).
- Lens : Prevents cataract (maintains transparency of lens).
- 2. maintains iron in Hb in the reduced state → Prevent formation of methemoglobin.
- 3. Reductive biosynthesis of fatty acids & steroids in :
 - Liver.
 - Adipose tissue.
 - Lactating mammary gland.
 - Gonads.
 - Adrenal cortex.

Non-oxidative phase :

- Reversible.
- Generate pentoses.
- Enzymes :
 - Transketolase : Requires vit. B₁
 - Transaldolase.
- Site (Rapidly proliferating cells) : Bone marrow, skin, mucosa.

----- Active space -----

G6PD Deficiency 🧬:

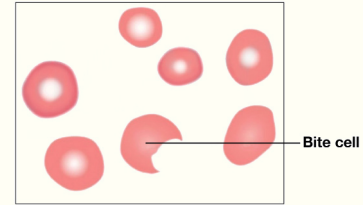
- m/c enzyme deficiency.
- x-linked recessive.

c/f: Defect in G6PD → ↓ NADPH →

- Hemolytic anemia.
- Jaundice (↑ Indirect bilirubin).
- methemoglobinemia → Cyanosis.

Precipitating factors :

- Drugs :
 - Antimalarials (Primaquine/Chloroquine).
 - Griseofulvin.
 - Dapsone.
- Fava beans (Favism).
- Infections.



Peripheral smear

Peripheral smear :

- Bite cells/Blister cells/Degmacytes.
- Heinz bodies (Denatured Hb).

Protective against : *Plasmodium falciparum*Uronic Acid Pathway

01:20:58

Oxidative pathway for glucose.

Site :

Liver.

Organelle :

Cytoplasm.

Functions 🧬:

- Produces uronic acid : Glucuronic acid → Conjugation of bilirubin.
- Synthesis of pentoses → DNA & RNA synthesis.
- Synthesis of ascorbic acid : Absent in humans d/t lack of L-gulonolactone oxidase.

Essential Pentosuria :

- Benign condition.
- Benedict test : Positive.
- Bial's test : Positive.
- Defect : xylitol dehydrogenase/xylulose reductase → Excretion of L-xylulose.
- Part of Garrod's tetrad : CAAP
 - Cystinuria.
 - Albinism.
 - Alkaptonuria.
 - Pentosuria.

AMINO ACIDS : PART 1

----- Active space -----

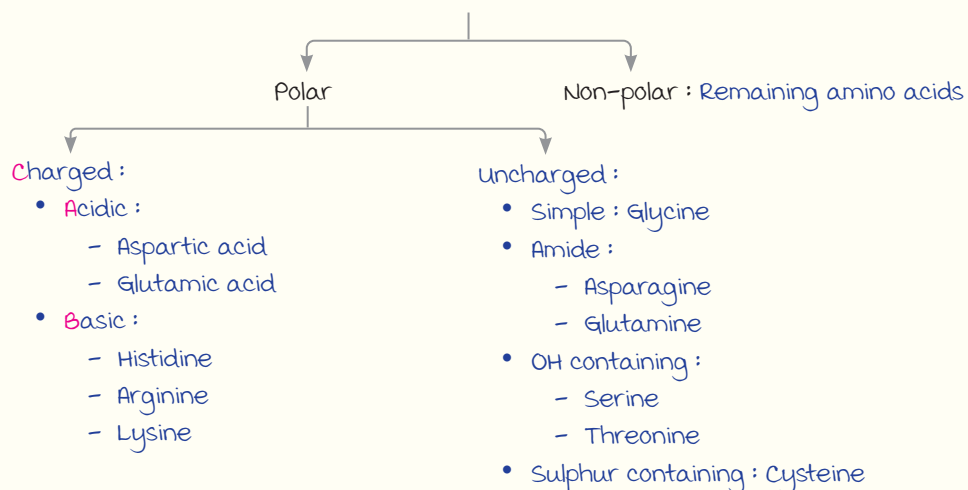
Classification of Amino Acids

00:00:35

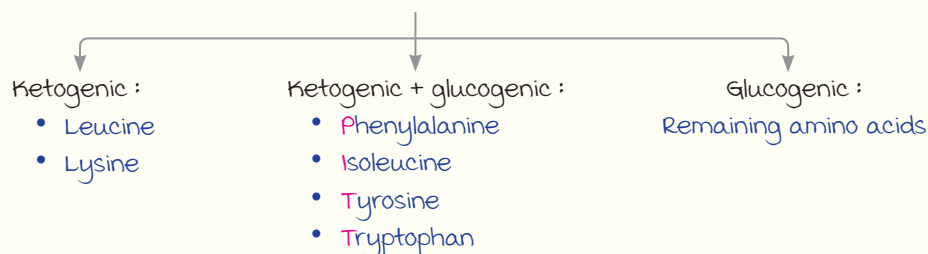
Based on Side Chain :

Group		Amino acids	Group	Amino acids
Aliphatic	Simple	- Glycine - Alanine	Amides	- Asparagine - Glutamine
	Branched chain	- Leucine - Isoleucine - Valine	Basic	- Histidine - Arginine - Lysine
	OH group containing	- Serine - Threonine	Aromatic	- Phenylalanine : Benzene ring - Tyrosine : Phenol ring - Tryptophan : Indole ring - Histidine : Imidazole ring
	Sulphur containing	- Cysteine - methionine		Imino acid
Acidic		- Aspartic acid (Aspartate) - Glutamic acid (Glutamate)		

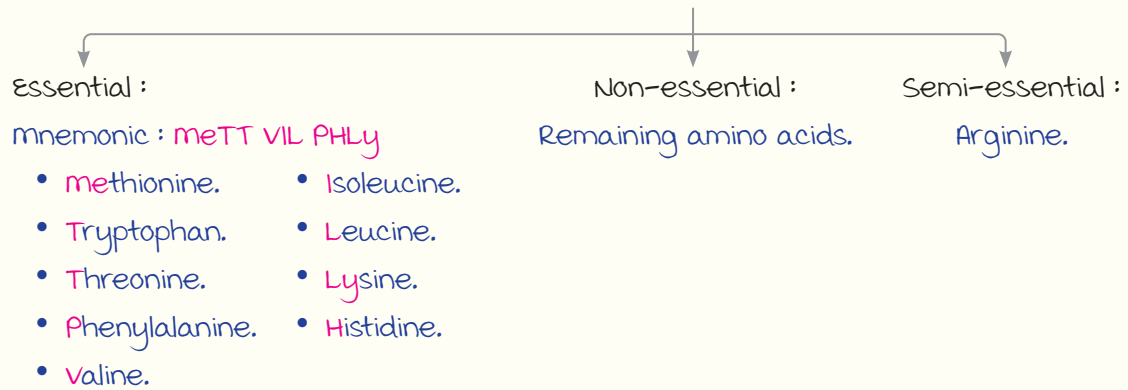
Based on Side Chain Characteristics :



Based on Metabolic Fate :



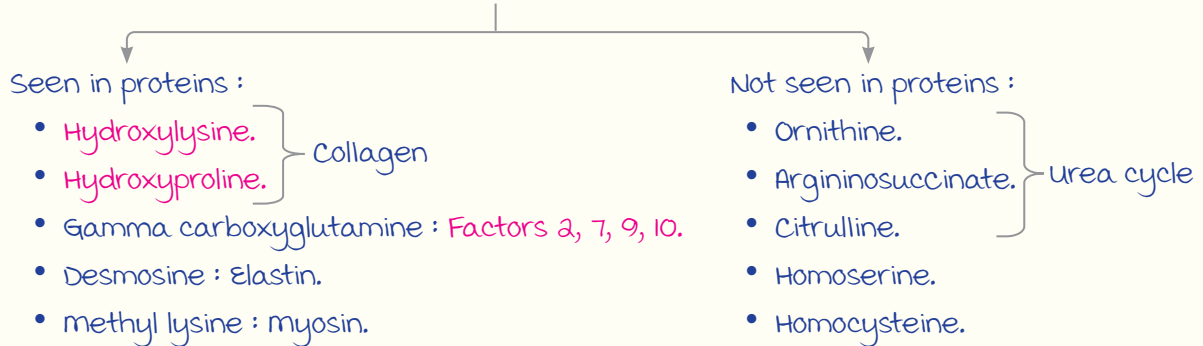
----- Active space -----

Based on Nutritional Requirement :Derived Amino Acids :

Properties :

- No codons.
- Formation :
 - Post translational modification.
 - Intermediate of metabolic pathway.

Classification :

Standard Amino Acids :

Selenocysteine :

- Coded by stop codon : UGA.
- Precursor amino acid : Serine.
- 21st protein forming amino acid.
- Constituent of :
 - Glutathione peroxidase.
 - Thioredoxin reductase.
 - Deiodinase.
 - Selenoprotein P.

Pyrrolysine :

- 22nd protein forming amino acid.
- Codon : UAG.
- Absent in humans.

Properties of Amino Acids

00:08:15

----- Active space -----

1. Isomerism :

- D & L isomerism.
- Exception : **Glycine** (Optically inactive).

2. Absorption of light :

- Colorless : Do not absorb visible light.
- UV light absorption :
 - 250 - 290 nm.
 - **Tryptophan** (280 nm $\rightarrow$ max. absorption) > Phenylalanine & tyrosine.

3. Buffering :

- maximum at ionization constant ($pK_a = pH$).
- Imidazole group of **histidine** : maximum buffering capacity ($pK_a = 6.5 - 7.4$).

4. Exist in charged state :

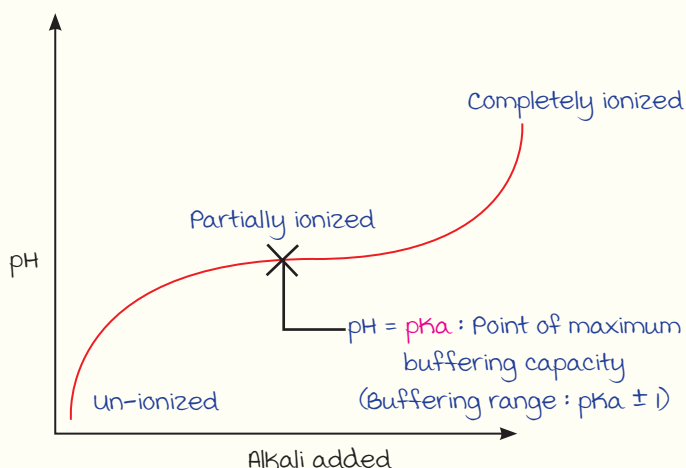
Ampholytes/zwitter ions : Compounds with isoelectric point (pI) $\rightarrow$ No net charge.

- No mobility in electric field.
- **maximum precipitability.**
- minimum solubility & buffering.

Titration Curve :

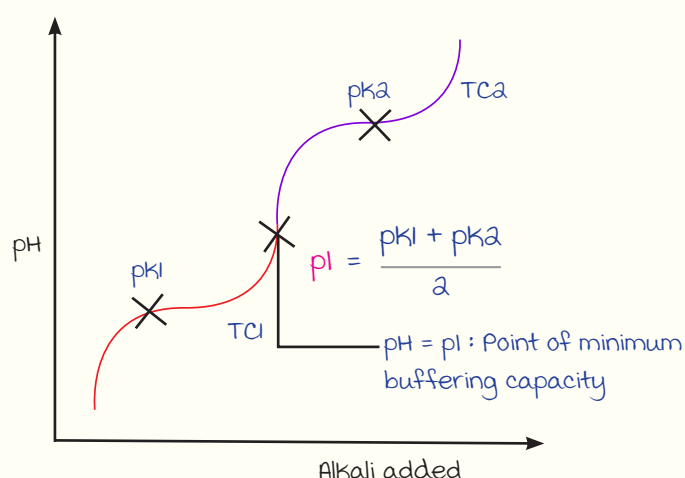
Compounds with single ionizable group :

E.g. : Acetic acid.



Compounds with multiple ionizable groups :

E.g. : Amino acids.



----- Active space -----

Proteins

00:18:10

Peptide bond :

- Between 2 amino acids.
- Uncharged & polar.
- Partial double bond.
- In trans configuration.

Structure of Proteins :

Primary :

- Linear sequence of amino acids.
- Peptide (Covalent) bond.



Primary structure

Secondary : Stabilized by non-covalent bonds.

Alpha helix :

- m/c 2° structure.
- most stable : Intrachain hydrogen bonds
- Disrupted by proline
- Glycine : Induces bends

Beta sheet :

- 2nd m/c 2° structure
- Stabilized by interchain hydrogen bonds



Secondary structure

Tertiary :

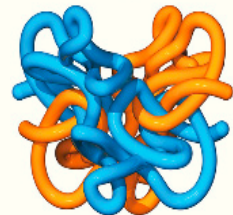
- 3D structure that can perform a function.
- Non-covalent forces.
- E.g. : Domain.
 - Immunoglobulin fold.
 - Rossmann fold.



Tertiary structure

Quaternary :

- > 1 polypeptide interact via subunits.
 - Non-covalent forces :
 - Hydrophobic interaction.
 - van der Waals forces.
 - Subunit interaction.
- Also in tertiary structures.



Quaternary structure

Protein Folding :

- Conversion of 1° structure protein → Higher order structure.
- mediated by molecular chaperones (E.g. : Heat shock proteins).

Protein misfolding diseases :

----- Active space -----

- Prion diseases.
- Amyloidosis.
- Prion related protein diseases :
 - Alzheimer's disease.
 - Parkinson's disease.
 - β thalassemias.
 - Cystic fibrosis.
 - Huntington's disease.
 - Fronto-temporal dementia (FTD).
 - Amyotrophic lateral sclerosis (ALS).
 - Dementia with lewy body (DLB).

Protein Degradation :

	Proteasomal degradation	Lysosomal degradation
ATP dependence	ATP dependent	ATP independent
Target proteins	• Short lived proteins • misfolded cellular proteins • membrane proteins	• Long lived proteins • Extracellular proteins • membrane proteins
Signal molecule	Ubiquitin : Requires PeST sequence for binding : • Proline • Glutamate • Serine • Threonine	-

Collagen

00:26:20

Features  :

- most abundant protein.
- Fibrous protein in ECM.
- Glycine : most abundant amino acid (33%).

Structure :

1. Poly proline α chain :

- Glycine X-Y repeat (Hydroxy proline & hydroxy lysine).
- Left handed turn.
- Composed of 1000 amino acids.

3. Quarter staggered arrangement :

- Lateral arrangement of triple helix.
- 2nd layer: 1/4th distance away from 1st layer.
- Cu^{2+} & lysyl oxidase $\rightarrow$ Covalent cross-links $\oplus$.

2. Triple helix 3 α chain :

- 3 together in $\textcircled{R}$ direction.
- vit C $\xrightarrow{+}$ Hydroxyproline & hydroxylysine $\rightarrow$ Facilitates hydrogen bonding.

----- Active space -----

Synthesis :

	Intracellular	Extracellular
Site	RER of fibroblast	ECM
Product	Procollagen	Tropocollagen
Events	1. Hydroxylation : vit C 2. Glycosylation : Hydroxylysine residue 3. Triple helix	1. Cleavage 2. Quarter staggered : Lysyl oxidase & Cu^{2+} 3. Covalent cross-links

Type :

Type	Diseases associated
Type I : • most abundant type in ECM • Present in bone • Ubiquitous in hard & soft tissue	• Osteogenesis imperfecta • Osteoporosis • Ehlers-Danlos type VII
Type 2 : Present in cartilage	• Severe chondrodysplasias • Osteoarthritis
Type 3	Ehlers-Danlos type IV
Type 4 : Present in GBM	Alport syndrome
Type 6 : Ubiquitous in microfibrils	Bethlem myopathy
Type 7 : Seen in anchoring fibrils	Epidermolysis bullosa (Dystrophic)
Type 10	Schmid metaphyseal dysplasia

Elastin, Keratin & Fibrillin

00:34:05

Elastin :

Elastin vs. Collagen :

	Collagen	Elastin
Types	many types	Only 1 type
Triple helix	⊕	⊖
Gly-X-Y	⊕	⊖
Hydroxylysine	⊕	⊖
Glycosylation	⊕	⊖
Cross-links	Covalent cross-links	Desmosine cross-link
Extension peptide	N & C terminal	C terminal

Disorders a/w elastin :

1. Williams Beuren syndrome.
2. Cutis laxa.

Keratin :

----- Active space -----

- Component of outer layer of skin, nails & hair.
- Rich in **cysteine** : Confers hardness to nails.
- Associated disorder : Defect in keratin 5/14 → **Epidermolysis bullosa** (Classical type).

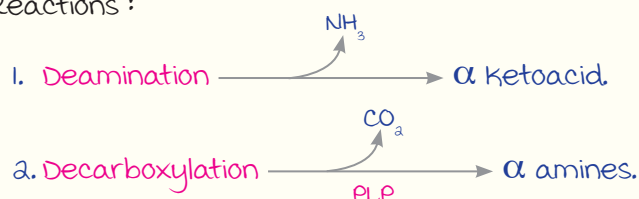
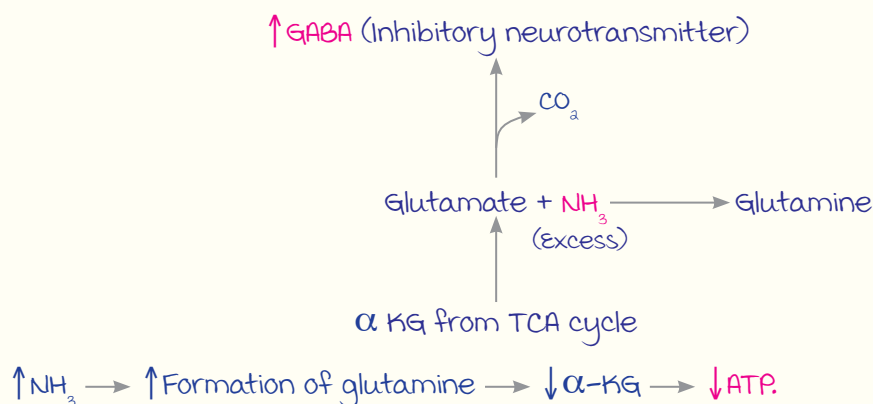
Fibrillin :

- Glycoprotein in myofibrils.
- Found in scaffolding of elastin.
- **Fibrillin I** mutation → **marfan's syndrome**.

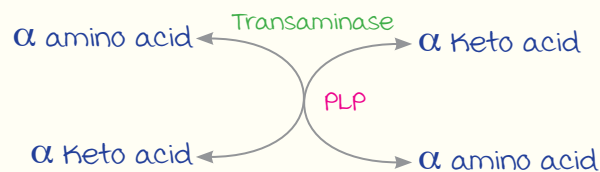
Amino Acid Metabolism

00:36:32

Reactions :

Toxic nature of ammonia  :metabolism of Ammonia :**Transdeamination** = Transamination + Oxidative deamination.1. Transamination  :

- most abundant source of NH_3 : **α amino group** of amino acid.
- Reaction :



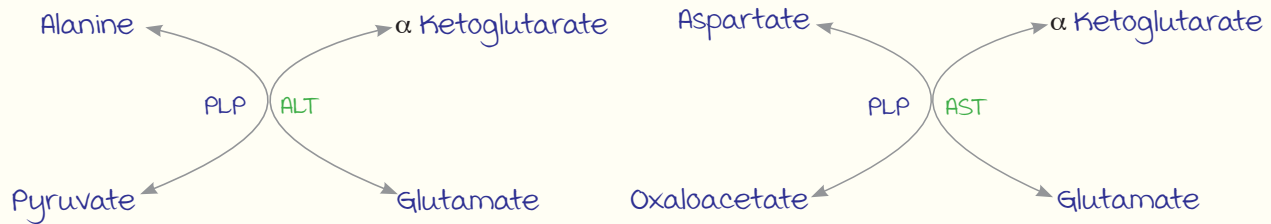
- Site : Cytoplasm of all organs.

----- Active space -----

- Features :

- Reversible.
- Toxic amino group $\rightarrow$ Non-toxic glutamates.
- Biosynthesis of non-essential amino acids.
- Ping pong mechanism.

- E.g. :



- Transamination of non-alpha amino acids : δ ornithine aminotransferase.

- Exceptions to transamination :

- Proline.
- Hydroxyproline.
- Lysine.
- Threonine.

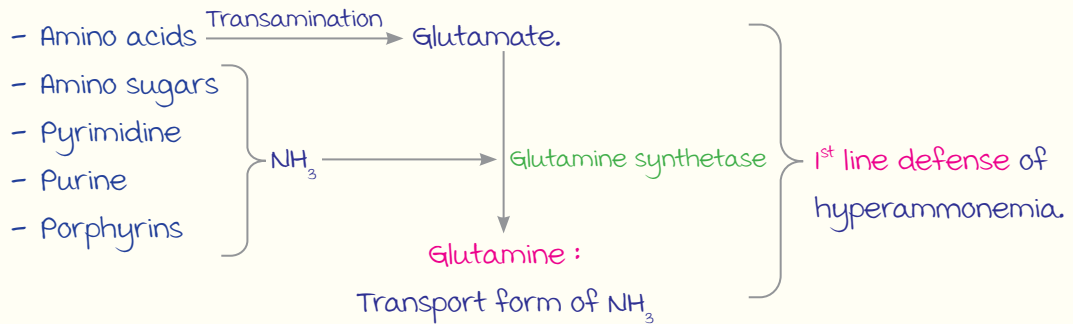
Applied biochemistry

Gyrate atrophy of retina & choroid :

- Defect in δ ornithine aminotransferase.
- Treatment :
 - Restrict ornithine & arginine.
 - Supplement PLP (B_6).

2. Transport of ammonia (DNA icon) :

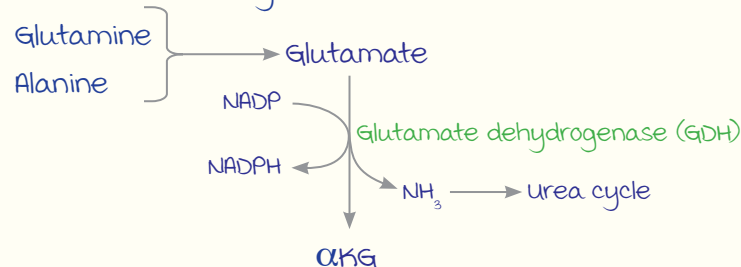
- Sources of ammonia :



- Transport form of NH_3 from skeletal muscle : Alanine.

3. Oxidative deamination :

Sites : Liver > Kidney.



Urea Cycle

00:50:45

----- Active space -----

- AKA Kreb's Henseleit/Ornithine cycle.

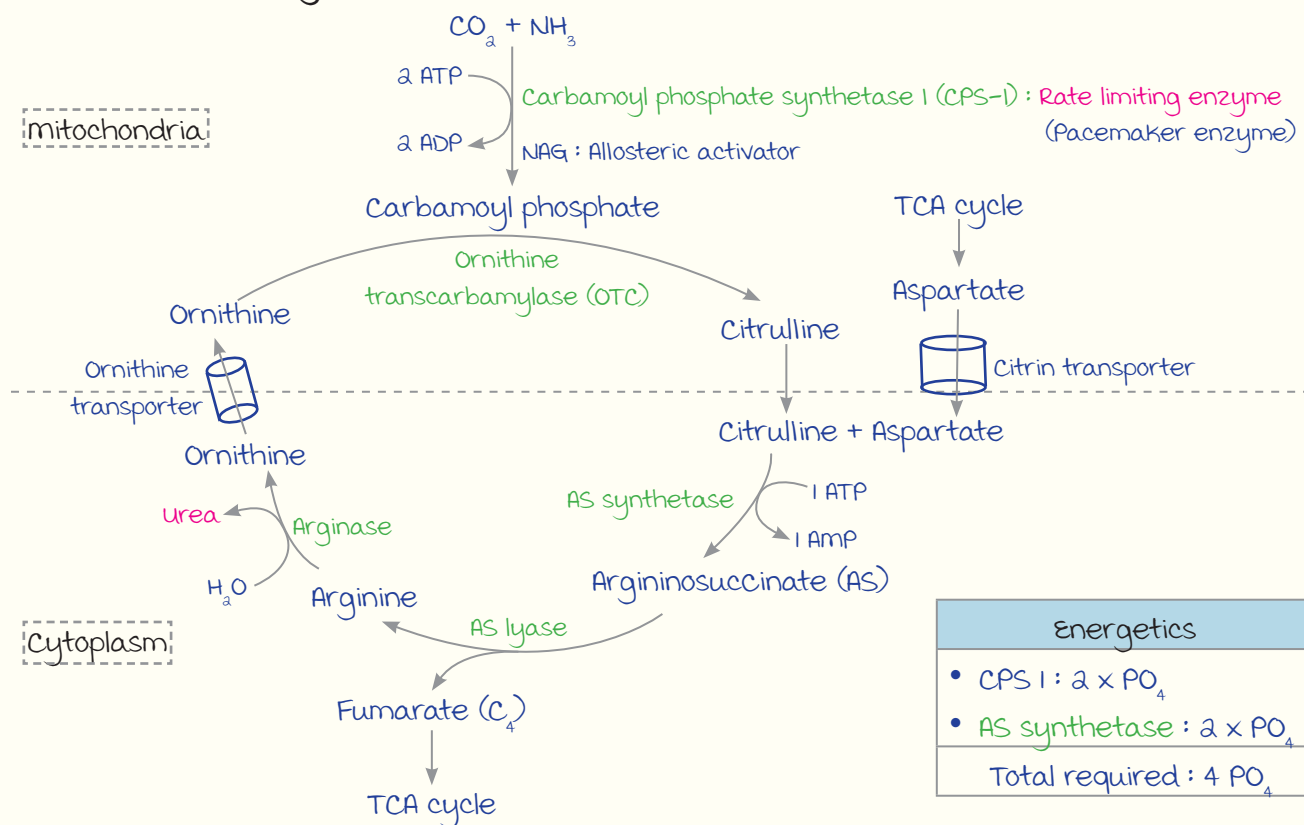
Site : Exclusively in liver.

Organelle : Cytoplasm + mitochondria.

Contributions to Structure :

- 1st nitrogen : Ammonia.
- 2nd nitrogen : Aspartate.
- Carbon atom : Respiratory CO_2 .

Reactions of Urea Cycle :



Urea Cycle Disorders

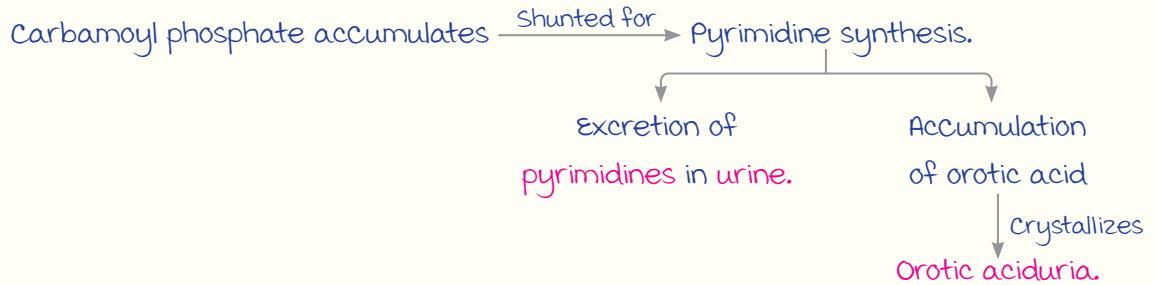
00:54:42

Defect	Disorder
CPS I	Hyperammonemia type I
OTC	Hyperammonemia type II
AS synthetase	Citrullinemia type I
AS Lyase	Argininosuccinic aciduria
Arginase	Argininemia
Citrin transporter	Citrullinemia type 2
Ornithine transporter	HHH syndrome

----- Active space -----

Hyperammonemia type II :

- m/c urea cycle disorder.
- X-linked recessive.
- Defect : OTC.

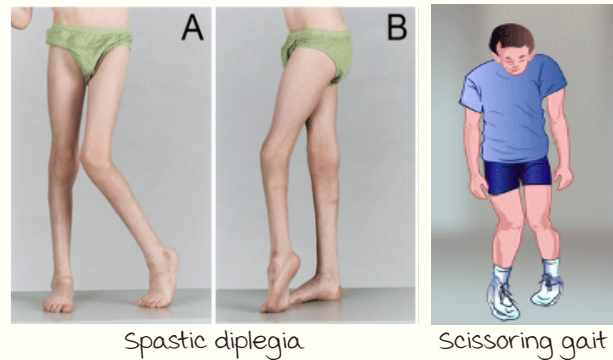


HHH syndrome :

- Defect : Ornithine transporter $\longrightarrow$ Hyperornithinemia.
- Carbamoyl phosphate + lysine $\longrightarrow$ Homocitrulline $\longrightarrow$ Homocitrullinemia.
- Hyperammonemia.

Argininemia :

- Least hyperammonemia.
- Spastic diplegia + scissoring of lower limbs.

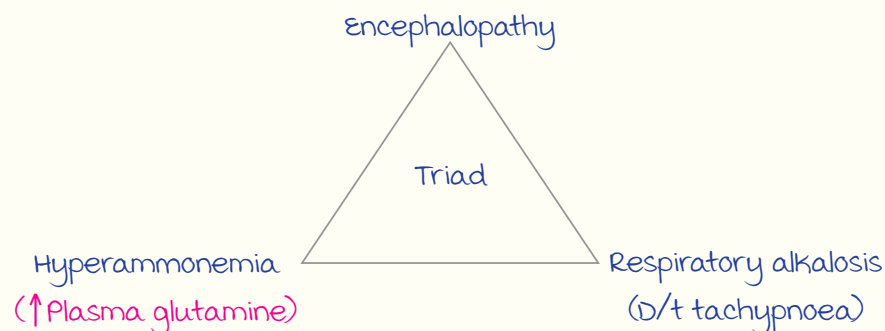


Argininosuccinic aciduria :

Trichorrhexis nodosa : Brittle hair with nodes.



Trichorrhexis nodosa.

General Features :

C/f in neonates :

- Feeding difficulties.
- Failure to thrive.
- Lethargy.
- Convulsions.
- Coma.

Investigations :

----- Active space -----

1. pH → ↓ : Organic aciduria.
 → ↑ / Normal : urea cycle disorder.

2. Plasma glutamine : ↑.

3. Tandem mass spectroscopy :

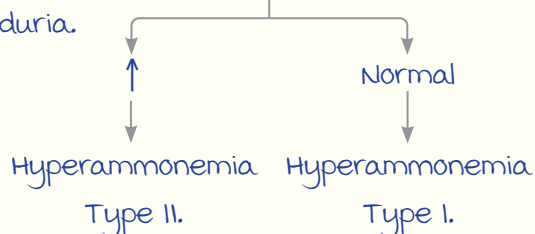
- Gold standard IOC for metabolic disorders.
- Screens for metabolites.

Specific amino acids ↑ :

- ↑ Ornithine : HHH syndrome.
- ↑ Citrulline : Citrullinemia.
- ↑ Argininosuccinate : Argininosuccinic aciduria.
- ↑ Arginine : Argininemia.

Non-specific amino acids ↑

Plasma orotic acid

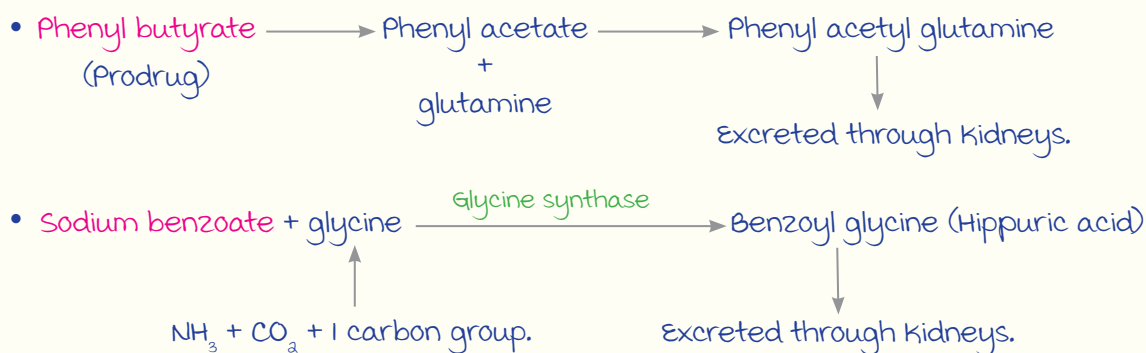
Treatment :

1. Supplement with arginine :

- Source of ornithine.
- Activator of NAG.
- Essential amino acid.
- C/I in arginase defect.

2. Acylation therapy :

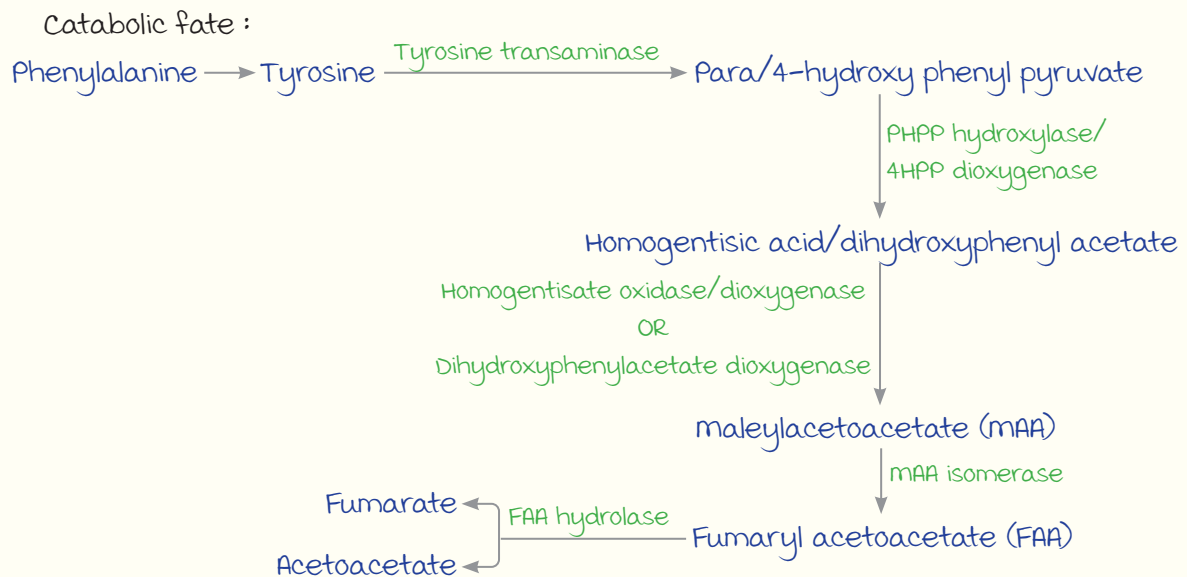
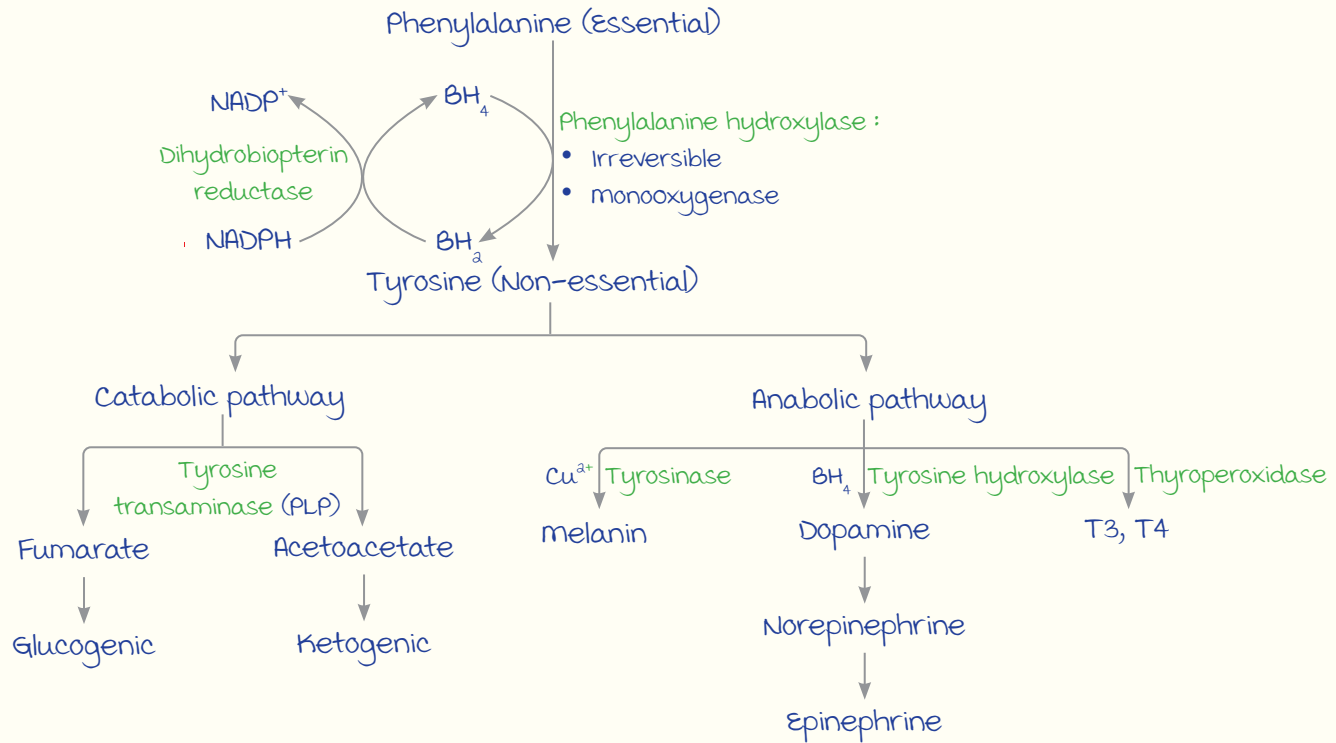
Phenyl butyrate & sodium benzoate : NH_3 scavengers.



AMINO ACIDS : PART 2

Phenylalanine & Tyrosine : Metabolism

00:00:25



Phenylalanine & Tyrosine : Disorders

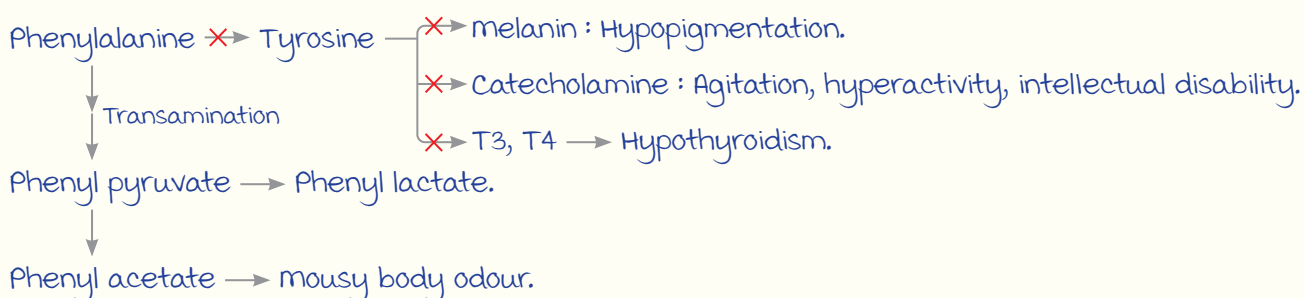
00:02:50

----- Active space -----

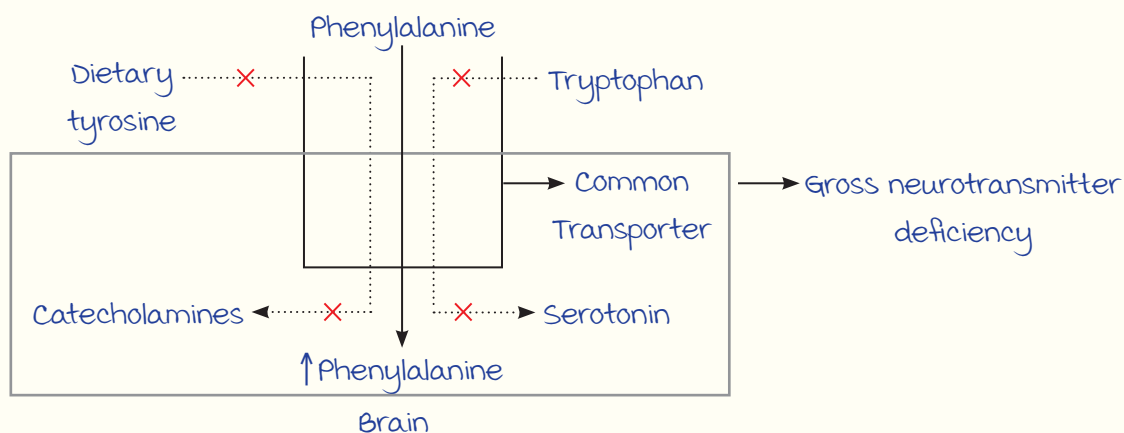
Enzyme defect	Disorder
Phenylalanine hydroxylase	Classical phenylketonuria (PKU)
- DHB reductase - GTP synthesis $\rightarrow$ BH₄ : - GTP cyclohydrolase - Pyruvoyl THB synthase - Sepiapterin reductase	Non-classical PKU
Homogentisate oxidase	Alkaptonuria
FAA hydrolase	Tyrosinemia type 1 : - m/c tyrosinemia - AKA Hereditary/hepatorenal tyrosinemia - Resembles porphyria - Rx : Nitisone
Tyrosine transaminase	Tyrosinemia type 2 : AKA Oculocutaneous tyrosinemia / Richner Hanhart syndrome
PHPP hydroxylase	Tyrosinemia type 3 / Neonatal tyrosinemia
Partial defect of PHPP hydroxylase	Hawkinsinuria $\rightarrow$ Swimming pool body odour

Classical Phenylketonuria :

Clinical features :



Neurological symptoms in PKU :



----- Active space -----

Lab diagnosis :

1. Guthrie's test (Bacterial inhibition test) :
Growth of *Bacillus subtilis*.
2. Ferric chloride test : Positive.
3. Blood phenylalanine levels.
4. Enzyme studies.

Treatment :

- Phenylalanine restricted diet.
- Large neutral amino acids (Tryptophan & tyrosine) : Competitive inhibition of phenylalanine.

Non-Classical PKU (DNA icon) :

Non-classic PKU → BH_4 deficiency → Defective functioning of : → Persistence of neurological symptoms after phenylalanine restriction.

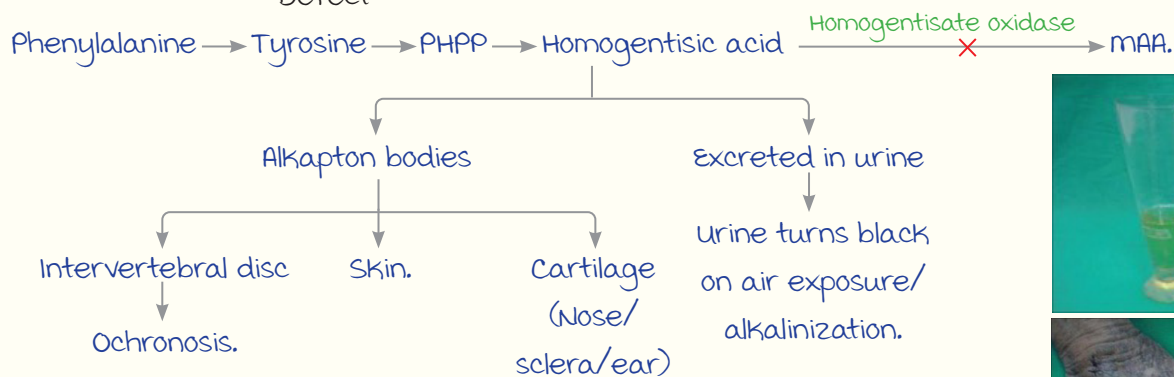
- Phenylalanine hydroxylase
- Tyrosine hydroxylase
- Tryptophan hydroxylase

Rx :

- Synthetic THB : Sapropterin dihydrochloride/Kuvan.
- Large neutral amino acids (Contains Trp and Tyr).

Alkaptonuria (DNA icon) :

Defect :

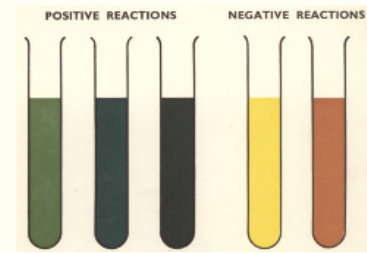


Clinical features :

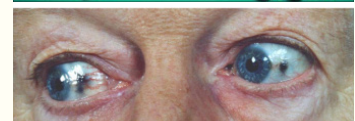
- Age of onset : middle age.
- Present with back pain.
- No intellectual disability.
- Black pigmentation of skin, intervertebral disc, cartilage.
- Blackening of urine on standing.



PKU : Hypopigmentation



Ferric chloride test



Alkaptonuria : Black pigmentation

Lab diagnosis :

- FeCl_3 test.
- AgNO_3 test.
- Alkalinize urine.
- Benedict's test : Positive.
- X-ray :
 - Bamboo spine.
 - Vacuum phenomenon.
(Aseptic gas collection within IV disc)
 - IV disc calcification.



Bamboo spine

----- Active space -----

Rx : Nitrofurantoin → Inhibits 4-hydroxyphenylpyruvate dioxygenase, HPPD.

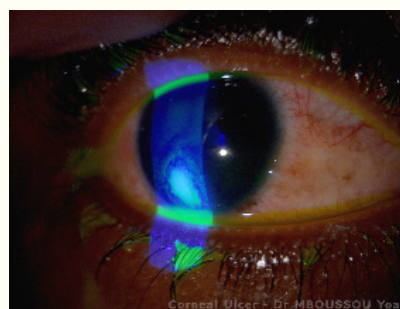
Type 2 Tyrosinemia (DNA icon) :

AKA Oculo-cutaneous tyrosinemia/Richner Hanhart syndrome.

Defect : Tyrosine transaminase.

Features :

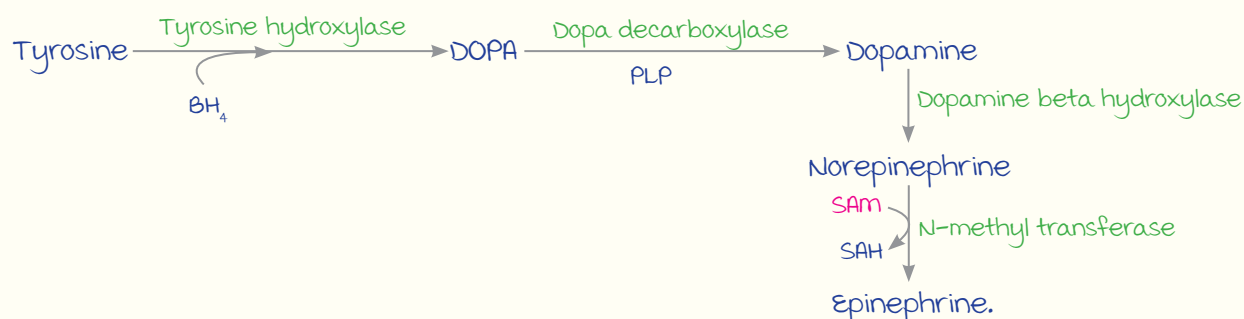
- Skin : Non-pruritic hyperkeratotic plaque on soles & palms.
- Corneal ulcers : Poorly stained with fluorescein.



Derivatives of Tyrosine

00:27:28

Catecholamines (DNA icon) :



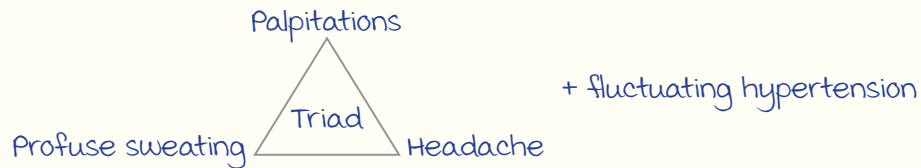
----- Active space -----

Degradation of catecholamines :

- Norepinephrine → Normetanephrine
 - Epinephrine → metanephrine
 - Dopamine → Homovanillic acid (HVA).
- Normetanephrine and metanephrine → Vanillyl mandelic acid (VMA).

Pheochromocytoma

- Tumor of adrenal medulla.
- c/f :



- Lab diagnosis :

a. 24 hr urinary tests for :

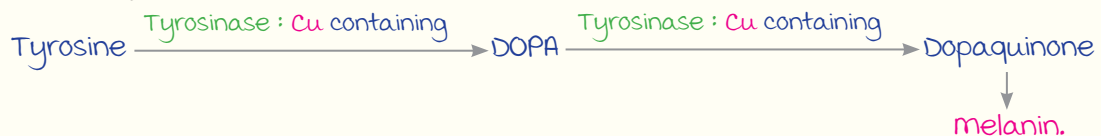
- VMA : Highest specificity.
- Catecholamines.
- Fractionated metanephrines : Highest sensitivity.
- Total metanephrines.

b. Plasma tests for :

- Catecholamines.
- Free metanephrines : Highest sensitivity.

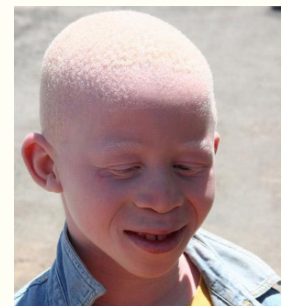
melanin

Site of synthesis : melanosomes (Stratum basale).



Albinism : Defect in tyrosinase

- milky white skin & hair.
- Photophobia.
- Lacrimation.



Albinism

Tryptophan : Metabolism & Derivatives

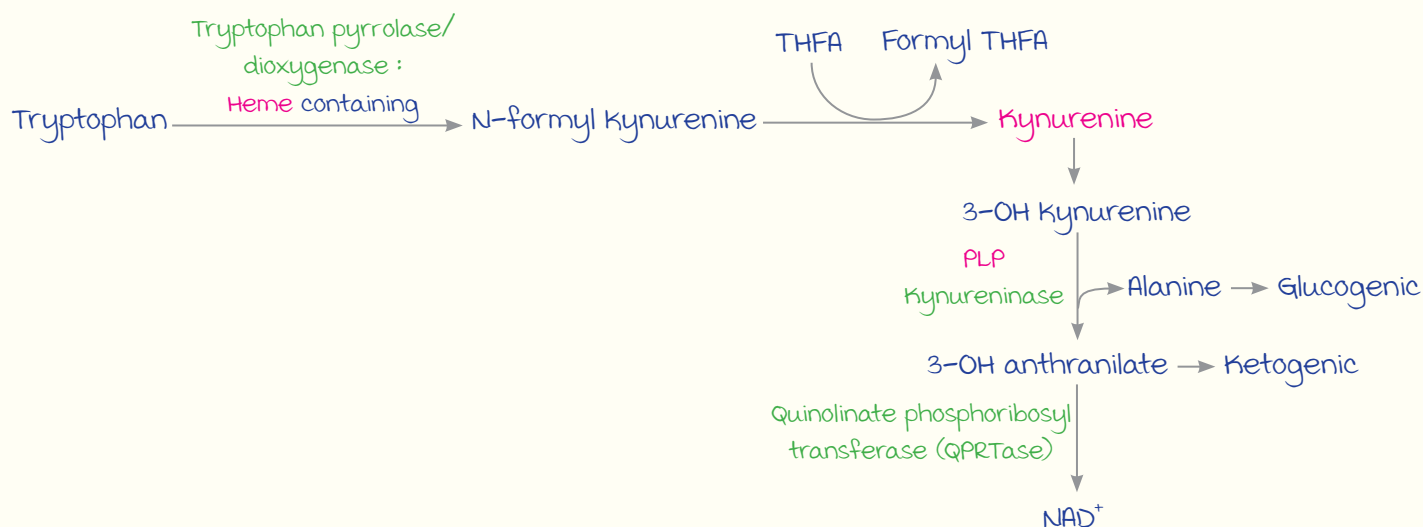
00:31:44

Properties :

- Aromatic AA.
- Non-polar.
- Essential.
- Ketogenic + glucogenic.

Catabolic Fate :

----- Active space -----



vitamin B ₆ deficiency
• $\downarrow$ Kynureninase activity. • $\uparrow$ 3-OH Kynurenine $\rightarrow$ Xanthurenic acid (Excreted in urine). • $\downarrow$ Synthesis of vit B₃ $\rightarrow$ Pellagra.

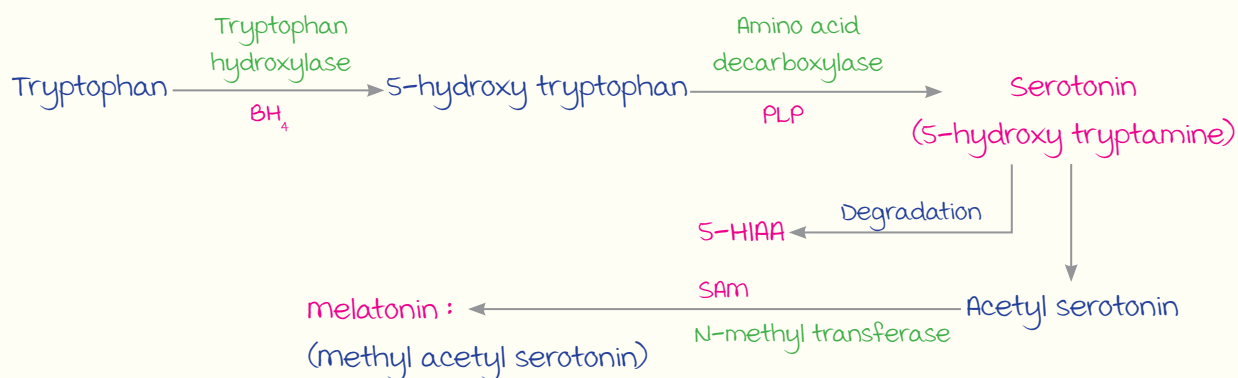
Niacin :

- Endogenously synthesized vitamin.
- 60 mg tryptophan $\rightarrow$ 1 mg niacin.

Derivatives (DNA icon):

Site : Argentaffin cells/Enterochromaffin cells of

1. GIT.
2. Brain.
3. Platelets.



- Site : Pineal gland.
- Neurotransmitter.
- Regulates circadian rhythm.

----- Active space -----

Tryptophan Metabolism : Associated Disorders

00:38:38

Carcinoid Syndrome :

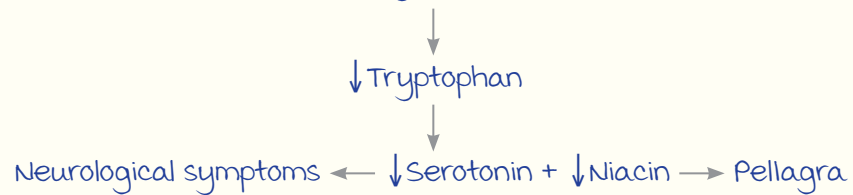
- Tumor of argentaffin cells $\rightarrow \uparrow \uparrow$ Serotonin $\rightarrow \uparrow$ 5-HIAA.
- $\downarrow$ Niacin.

c/f :

- Intermittent diarrhea.
- Flushing.
- Sweating.
- Features of pellagra.
- 24 hr 5-HIAA $\uparrow$.

Hartnup's Disease :

mutation : SLC 6A 19 mutation $\rightarrow$ Defect of tryptophan transporter (BOAT-I)



c/f :

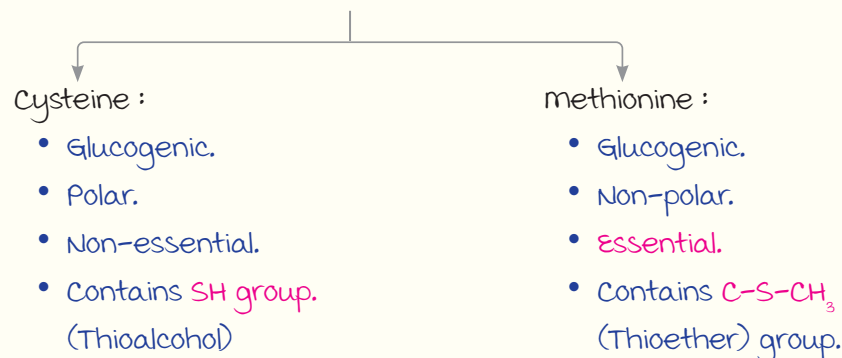
- mostly asymptomatic.
- Accumulation of tryptophan in intestine $\rightarrow$ compounds $\rightarrow$ Excreted in urine
(Indican & indigo blue) $\downarrow$ Bluish discoloration of diaper.
- Intermittent ataxia
- Wide based gait.
- Cutaneous photosensitivity : m/c symptom.

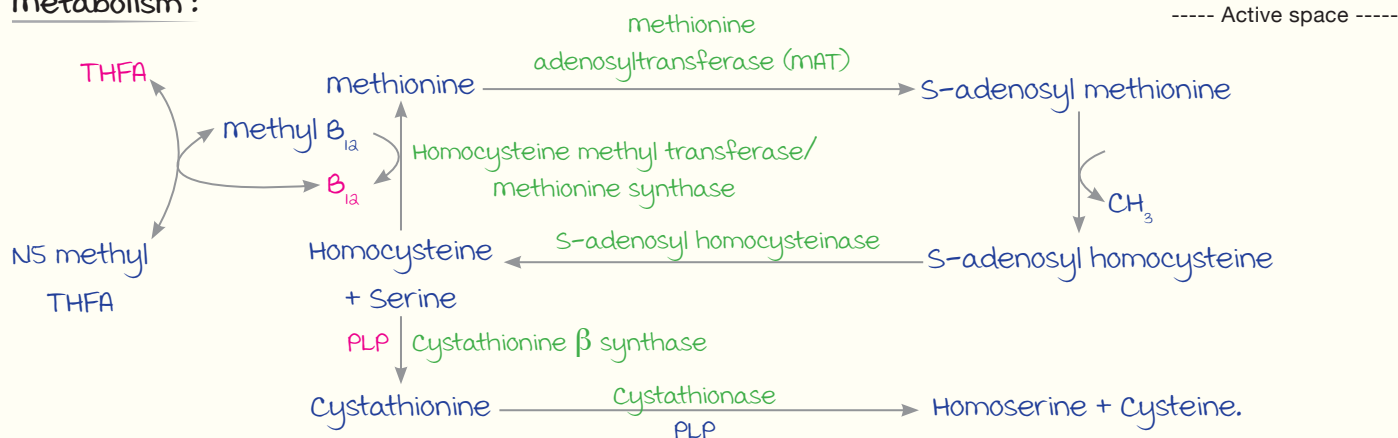
Test : Obermeyer test.

Rx : Lipid soluble esters of tryptophan + supplement niacin.

Sulphur Containing Amino Acids : Metabolism

00:44:02



metabolism :Sulphur Containing Amino Acids : Disorders & Derivatives

00:46:03

Defect	Disorders
cystathionine β synthase	Classical homocystinuria
- N5 methyl THFA - methyl B_{12} formation	Non-classical homocystinuria
MAT	Primary hypermethioninemia (Boiled cabbage odour)
Cystathionase	Cystathioninuria

Homocysteinemia
B_6, B_9 deficiency. B_{12} deficiency : N5 methyl THFA $\nrightarrow$ THFA. Functional deficiency of THFA $\rightarrow$ $\downarrow$ DNA synthesis $\rightarrow$ Nucleocytoplasmic asynchrony (Folate trap) $\rightarrow$ megaloblastic anemia + SACD + Homocysteinemia.

Classical Homocystinuria :

2 x Homocysteine $\rightarrow$ Homocystine $\rightarrow$ Excreted in urine.

Clinical features :

- Developmental delay.
- Skeletal deformities :
 - Tall stature.
 - Genu valgum/varum.
 - High arched palate.
 - Pes cavus : Charlie Chaplin gait.
 - Pectus carinatum/excavatum.

----- Active space -----

- Ectopia lentis : **Inferomedial**.
- ↑ Risk of thromboembolism.

Test : Cyanide nitroprusside test (+) (magenta color).



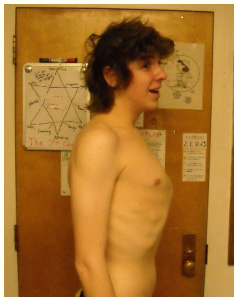
Arachnodactyly



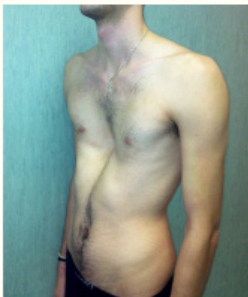
Tall stature



High arched palate



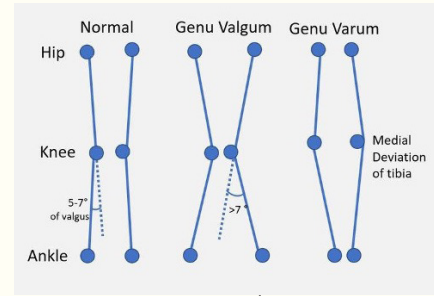
Pectus carinatum



Pectus excavatum



Pes cavus

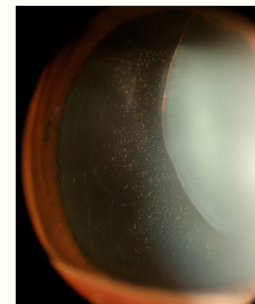


Genu valgum/varus

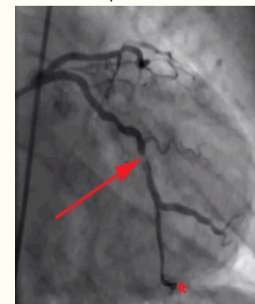
Skeletal deformities in homocystinuria

Classical vs. Non-classical homocystinuria :

	Classical homocystinuria	Non-classical homocystinuria
Enzyme	Cystathionine β synthase	methyl B_{12} formation ; methylene THF reductase
Defect in	Formation of cysteine	Remethylation of homocysteine to methionine
Homocysteine	↑	↑
Cysteine	↓	Normal
methionine	Normal	↓
Rx	• Cysteine supplementation • High dose vit B_6 • Betaine (Trimethyl glycine) • vit B_{12} & folic acid	• methionine supplementation • vit B_{12}, B_6, folic acid



Ectopia lentis



Thromboembolism

Cystinuria :

- Defect : Dibasic amino acid transporters in kidney.
- Excretion of : COLA.
 - Cystine.
 - Lysine.
 - Ornithine.
 - Arginine.
- Part of Garrod's tetrad.

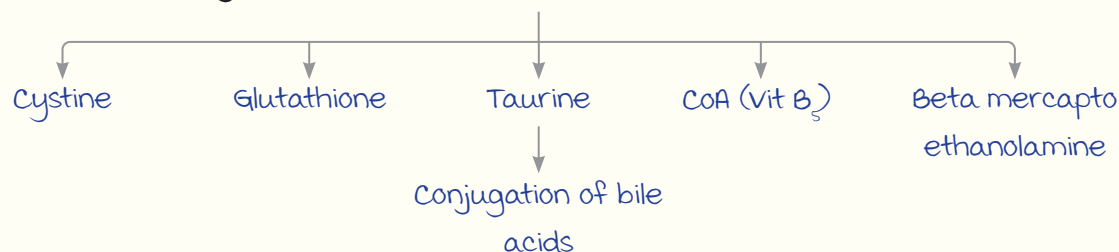


Cystine crystals : Hexagonal

----- Active space -----

Cystinosis :

- Defect : Cystine transporter in lysosomes.
- manifestation :
 - Renal failure.
 - Bone marrow suppression.
 - Corneal opacity.
 - Liver failure.

Derivative of Cysteine :Glutathione (GSH) :

- Tripeptide : Glutamic acid + cysteine + glycine.
- Active group : SH of cysteine.
- Pseudo peptide.

Functions :

1. Transport of amino acids : meister's cycle/Gamma glutamyl cycle.
2. Free radical scavengers : Glutathione peroxidase.
3. Conjugation.
4. Coenzyme.

Glycine & Serine

00:59:56

Glycine :

Required for conjugation.

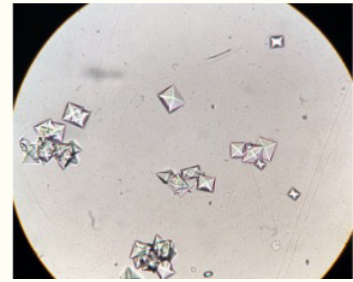
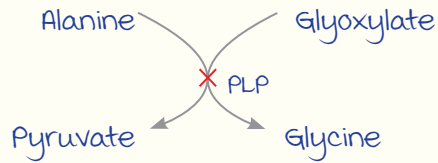
Derivatives :

1. Purine (C4, C5, N7).
2. Creatine : Glycine + arginine + methionine.
3. Heme.
4. Glutathione.
5. Collagen.

----- Active space ----- Hyperoxaluria :

- Primary :

Defect in glyoxylate alanine aminotransferase → ↑ Glyoxylate →



Oxalate stones : Envelope shaped

- Secondary :

- vit. B₆ deficiency.
- vit. C toxicity.

- Ethylene glycol poisoning.
- methoxyflurane.

Serine :

Functions  :

- Synthesis of :

a. Cysteine.

b. Phosphatidyl serine.

c. Choline.

d. Betaine.

2. Produces ethanolamine on decarboxylation.

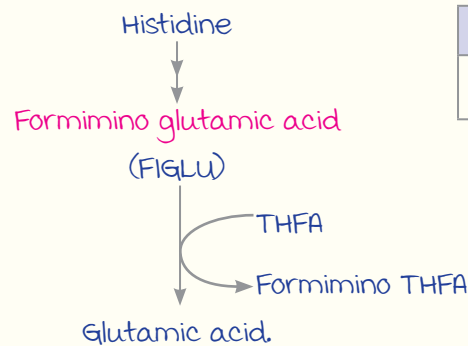
3. Precursor of selenocysteine.

4. Sphingosine : Serine + palmitic acid

Basic Amino Acids

01:02:30

Histidine :



Histidine load test

B₆ deficiency → FIGLU excreted in urine

Derivatives :

- FIGLU.
- Histamine : Decarboxylation of histidine.
- Carnosine : β alanine + histidine.
- Anserine : methyl carnosine.
- Homocarnosine : GABA + histidine.
- Ergothionine.

Arginine :

----- Active space -----

Derivatives :

1. Nitric oxide :

- Synthesized by NO synthase.
- Reactive free radical.
- Acts via cGMP.
- Endothelium derived relaxing factor (EDRF)

2. Ornithine & urea.

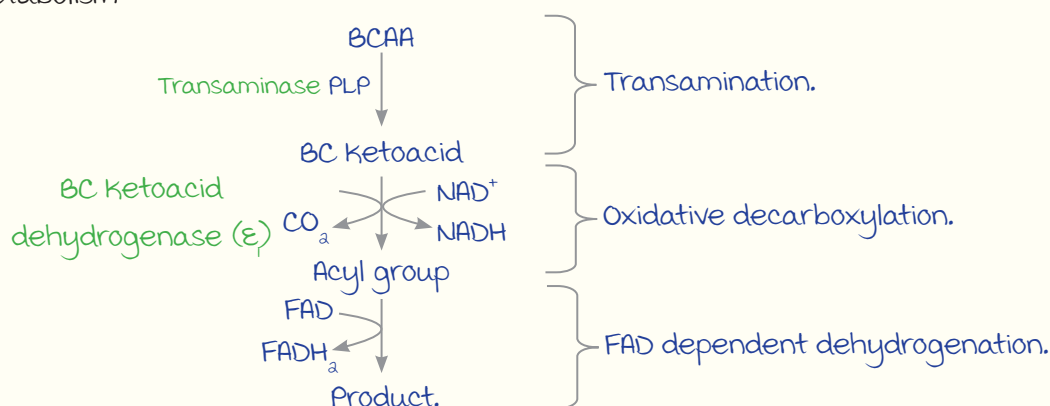
3. Creatine : Glycine + arginine + methionine.

Branched Chain Amino Acids

01:05:23



metabolism :

maple Syrup Urine Disease :

Defect : BCK DH

Features :

- Age of onset : Neonate.
- Feeding difficulty, convulsion, lethargy, coma.
- Boxing/kicking/Bicycling movement (Alternating hypo & hypertonia).
- Burnt sugar odour of urine (Caramel/maple syrup odour).

Lab diagnosis :

- $\uparrow$ BCAA & $\uparrow$ BCKA in blood & urine.
- Dinitrophenyl hydrazine (DNPH) test.
- Rothera's test : Positive.

Rx :

- Restrict BCAA.
- Supplement thiamine.

----- Active space -----

Overview

01:09:05

Enzyme Defects :

Disorder	Defective enzyme
msUD	Branched chain keto acid dehydrogenase
Isovaleric acidemia	Isovaleryl CoA dehydrogenase
Homocystinuria	Cystathionine β synthase
Phenylketonuria	Phenylalanine hydroxylase
Alkaptonuria	Homogentisate oxidase/Dihydroxyphenylacetate dioxygenase
Tyrosinemia type I	Fumaryl acetoacetate hydrolase
Tyrosinemia type II	Tyrosine transaminase
Tyrosinemia type III	PHPP hydroxylase

Laboratory Tests :

Test	Aminoaciduria
Ferric chloride test	PKU/Alkaptonuria
Dinitro phenyl hydrazine test	msUD
Guthrie test	PKU
Obermeyer test	Hartnup disease
Cyanide nitroprusside test	Homocystinuria
La Brosse VMA spot test	Pheochromocytoma
5-HIAA	Carcinoid syndrome

Peculiar Odours :

Disorder	Odour
Glutaric acidemia (Type 2)	Sweaty feet
Hawkinsinuria	Swimming pool
Isovaleric acidemia	Sweaty feet
msUD	maple syrup/Burnt sugar
Hypermethioninemia	Boiled cabbage
multiple carboxylase deficiency	Tomcat urine
Oasthouse urine disease	Hops-like
PKU	mousy/musty
Trimethylaminuria : • Defect in trimethylamine oxidase • Rx : Choline restriction	Fish odour
Tyrosinemia	Boiled cabbage/Rancid butter

METABOLISM OF LIPIDS

----- Active space -----

Classification of Fatty Acids (FA)

00:00:15

Lipid : Soluble in non-polar solvents only.

	Classification of lipids
Bloor's classification	- Simple lipids : - Fats - Oils - Waxes - Compound lipids : - Phospholipids - Glycolipids - miscellaneous : - Steroids - Lipid soluble vitamins - Derived lipids : - Fatty acid - Glycerol
Based on no. of carbon atoms	- Short chain (C2 to C6) - medium chain (C8 to C14) : Coconut oil (Least PUFA) - Long chain (> C16) - Very long chain (> C20)
Based on double bonds	- Saturated FA (Single bonds only) : Animal fat - Unsaturated FA (Double bonds)
Based on diet	- Essential FA : - Linoleic acid : Safflower oil, sunflower oil - α-linolenic acid - Semi essential FA : - Arachidonic acid - γ-linoleic acid
Unsaturated FA	
Based on no. of double bonds	- monounsaturated FA : mustard/rapeseed oil - Polyunsaturated FA : Safflower oil, sunflower oil
Polyunsaturated FA	
Based on position of final double bond	- Omega-3 FA : - α-linolenic acid : Flaxseed oil - Eicosapentaenoic acid - Cervonic acid : Breast milk, Fish & algal oil - Omega-6 FA : - Gamma linolenic acid - Linoleic acid - Arachidonic acid : PG, LT synthesis
Based on isomer	- Cis FA - Trans FA : Vanaspatti (Partially hydrogenated vegetable oil) - RDA : 2 to 7 g/d - WHO recommendation : < 1% total energy intake

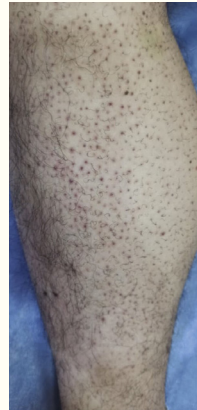
Applied biochemistry
Essential FA deficiency : • Acanthosis • Hyperkeratosis • Fatty liver • Scaly dermatitis • mitochondrial membrane damage
Docosahexaenoic acid (DHA)/Cervonic acid : • Highest concentration : Hippocampus • Function : - Infant brain development - Retina development • Can cross placenta • Deficiency : Retinitis pigmentosa
↑ Trans FA : • ↓ Fluidity of membrane • Insulin resistance • Dyslipidemia, CV risk (↑ TAG, ↑ LDL, ↓ HDL) • ↑ Inflammation
Significance of omega-3 FA : • Decreases : - Cardiovascular risk (↓ TG) - Platelet aggregation - mental illness, degenerative disease risk - Inflammation • Infant brain development • Benefits in ADHD, Non-alcoholic fatty liver disease

Yellow : Richest source.

----- Active space -----



Acanthosis



Hyperkeratosis



Scaly dermatitis

Essential FA deficiency

Phospholipids (PL) & Glycolipids

00:11:44

Glycerophospholipids :

Glycerol + 2 FA + PO_4 + nitrogenous/non-nitrogenous base.

Glycerophospholipid	Constituents	Present in
Phosphatidic acid (Simplest)	Diacyl glycerol (DAG) + PO_4	Cell membrane
Lecithin (Storehouse of choline)	DAG + PO_4 + choline (Phosphatidyl choline)	Lung surfactant, Cell membrane (most abundant)
Cephalin	DAG + PO_4 + ethanolamine	Cell membrane
Cardiolipin (Diphosphatidyl glycerol)	2 x Phosphatidic acid (PA) + glycerol	Inner mitochondrial membrane
Phosphatidyl serine	PA + serine	Apoptosis
Phosphatidyl inositol (2 nd messenger in hormonal pathways)	PA + inositol	Cell membrane

Cardiolipin

- A/w **Barth syndrome** (Cardioskeletal myopathy) :
 - Cardiomegaly + myopathy
 - mitochondrial disease
- Only antigenic PL : **False positive syphilis test**

Sphingophospholipids :

----- Active space -----

Sphingosine + FA + PO₄ + nitrogenous base.

Ceramide

Sphingophospholipids	Constituent	Present in
Sphingomyelin	Ceramide + PO ₄ + Choline	• myelin sheath • White matter of brain • Lung surfactant • Cell membrane

Glycolipids :

Ceramides + carbohydrate group.

Glycosphingolipids	Constituents	Present in
Cerebroside	Galactocerebroside	Ceramide + Galactose
	Glucocerebroside	Ceramide + Glucose
Globoside	Ceramide + oligosaccharide	-
Ganglioside (G _m)	Ceramide + oligosaccharide + N-acetyl neuraminic acid (NANA)	-

Sphingolipidoses

00:20:04

Sphingolipids $\xrightarrow[\text{X}]{\text{Degraded in}}$ Lysosomes.

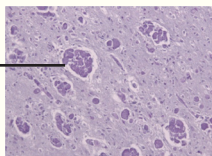
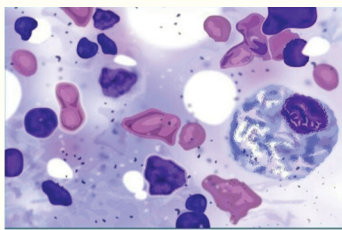
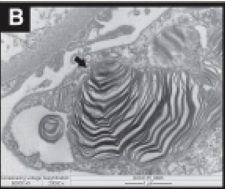
Sphingolipidoses : Type of lysosomal storage diseases.

Disorders	Enzyme defect	Features	Accumulating substance
G _m gangliosidosis (AR)	β -galactosidase	• Frontal bossing, depressed nasal bridge, long philtrum, low set ears • Cherry red spot (CRS) on macula • Angiokeratoma	G _m ganglioside
G _{m_a} gangliosidosis	Tay-Sachs disease (AR)	• Developmental delay • muscular weakness • Visual disturbances • Hepatosplenomegaly : - $\ominus$: Tay-Sach's disease - $\pm$: Sandhoff's disease • Hyperacusis • CRS on macula	G _{m_a} ganglioside
	Sandhoff's disease (AR)		

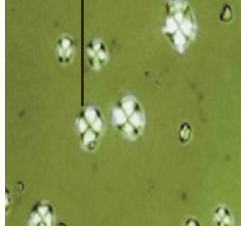


Cherry red spot

----- Active space -----

Disorders	Enzyme defect	Features	Accumulating substances
Krabbe's disease (AR)	β -galactocerebrosidase/ β -galactosidase	- CRS $\pm$ - Gross developmental delay - Opisthotonus posture with clenched fists - No visceromegaly - Severe neurological deficit  Globoid cell  Opisthotonus posture with clenched fists White matter of brain	Galactocerebroside
Gaucher's disease (AR) : m/c	β -glucocerebrosidase/ β -glucosidase	- Protruding abdomen - Crumpled tissue paper appearance - Massive hepatosplenomegaly - Thrombocytopenia : Bleeding manifestation - Erlenmeyer flask deformity - Pain & pathological fractures of long bones - No mental retardation (MR)/CR spot  Erlenmeyer flask deformity  Gaucher cell	Glucocerebroside in BM & liver
Niemann-Pick disease (AR)	Sphingomyelinase	- Developmental delay - CR spot in macula - Zebra body inclusions  	Sphingomyelin
Farber's disease (AR)	Ceramidase	Resembles rheumatoid arthritis	Ceramide

----- Active space -----

Disorders	Enzyme defect	Features	Accumulating substances
Fabry's disease (X-linked recessive)	α -galactosidase	- No visceromegaly - Angiokeratoma - No CR spot or MR - Lenticular opacities - maltese cross : urinary sediments  	Globotriosyl ceramide
metachromatic leukodystrophy (AR)	Aryl sulfatase A	No visceromegaly	-

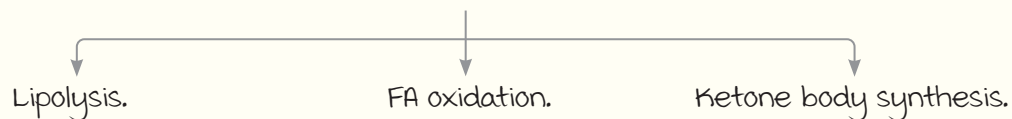
Note :

Wolman's disease : Acid lipase defect (Lysosomal storage disorder).

- Calcification of adrenal gland.
- Watery green diarrhoea.
- Hepatosplenomegaly.

Metabolism in Fasting State

00:33:27

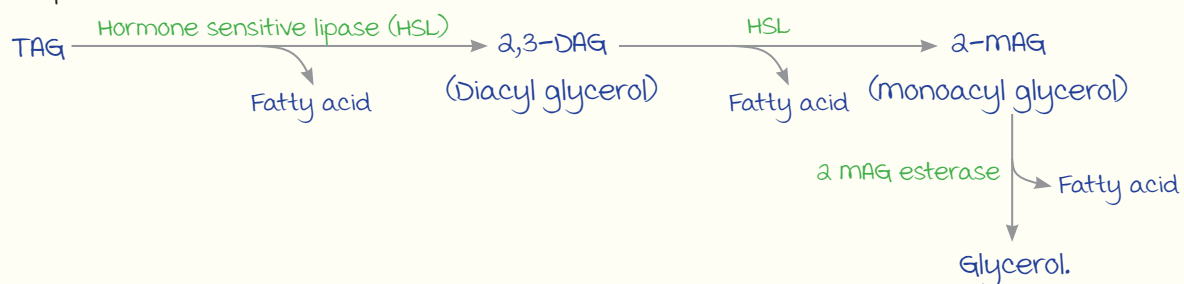


Lipolysis :

Hydrolytic cleavage of stored TAG.

Site : Adipose tissue.

Steps :



----- Active space -----

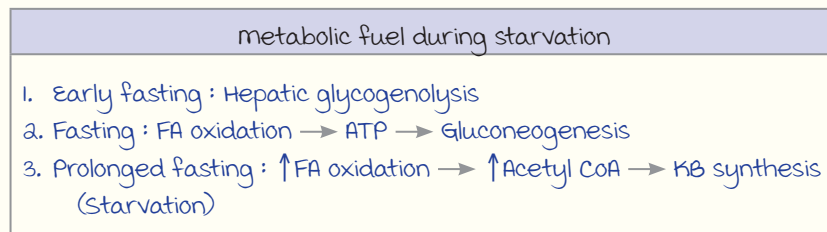
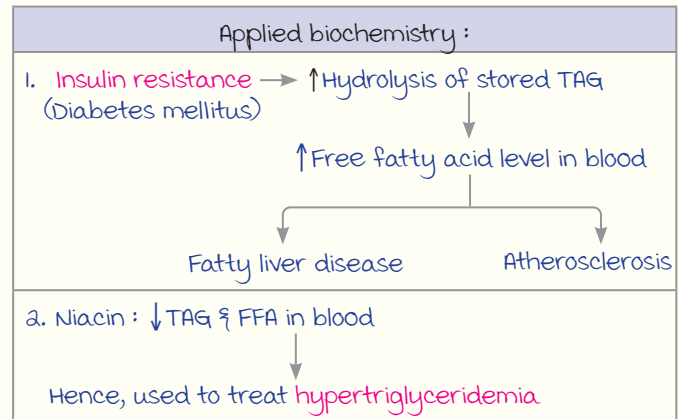
Activators :

- Glucagon.
- Epinephrine.
- ACTH.
- TSH.
- Thyroxine.
- MSH (melanocyte stimulating hormone).

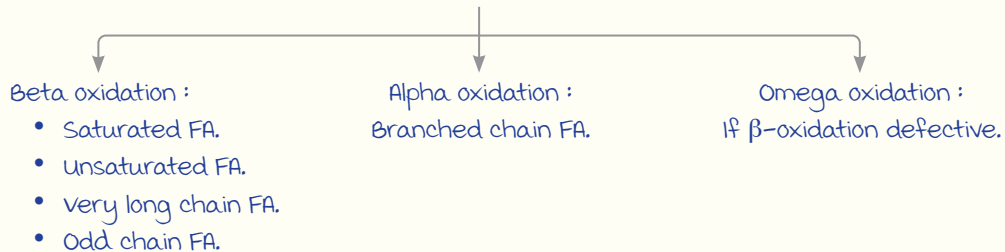
Inhibitors :

- Insulin.
- PGE_1 .
- Niacin

FA Oxidation :



Types :



Beta Oxidation

00:37:58

Sequential removal of 2 carbon acetyl CoA + oxidation of β carbon atoms.

Site :

- Liver
- Skeletal muscle
- Adipose tissue

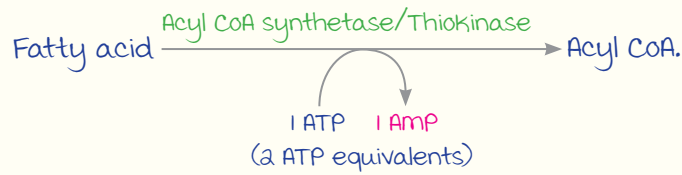
Organelle:

mitochondria.

Steps :

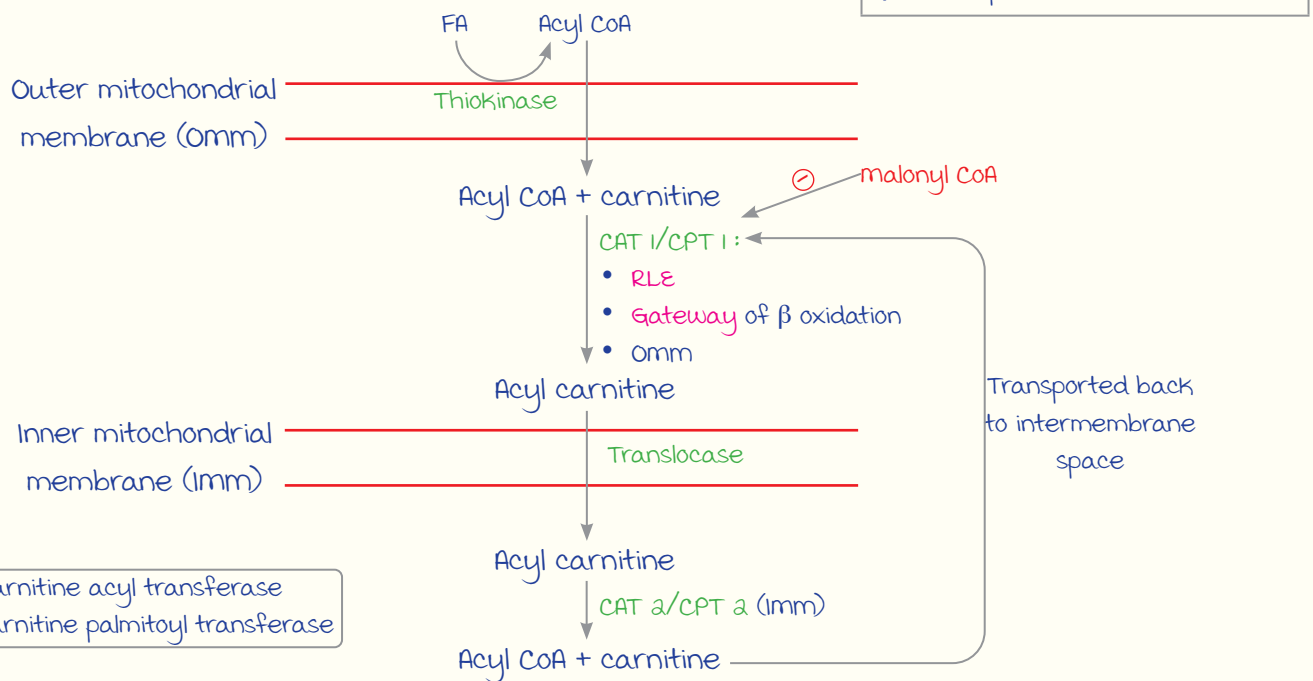
----- Active space -----

1. Activation of fatty acid : Occurs in the cytoplasm.

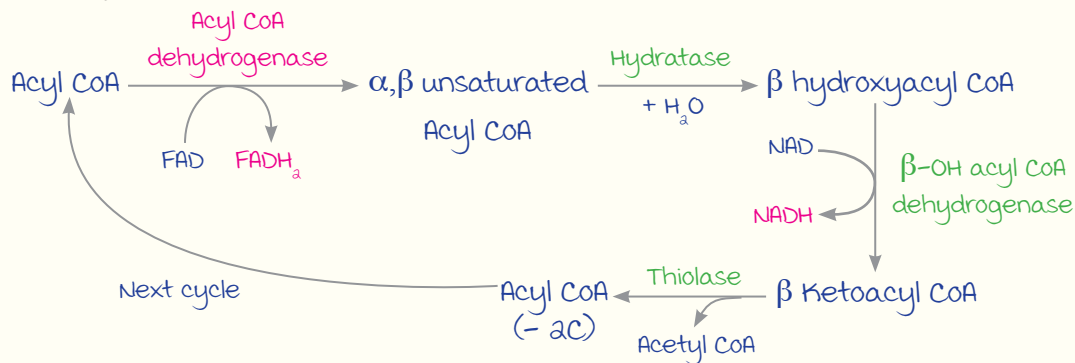


2. Carnitine transport :

- Transports Acyl CoA into mitochondria.
- Not required with medium & short chain FA (< 14C)



3. Fatty acid oxidation :



Energetics	
1 FADH _a :	1.5 ATP
1 NADH :	2.5 ATP
Total : 4 ATP	

----- Active space -----

Energetics of palmitic acid (C16) : m/c FA to undergo β oxidation.

C16	→	No. of acetyl CoA = $\frac{16}{2} = 8$ acetyl CoA → $8 \times 10 = 80$ ATP.
	→	β oxidation = $\frac{C16}{2} - 1 = 7$ β oxidation → $7 \times 4 = 28$ ATP.
	→	Activation : Utilizes 2 ATP equivalents → - 2 ATP.
Total : $108 - 2 = 106$ ATP		

Regulation :

CPT-I gateway : Inhibited by malonyl CoA.

- Fasting → Low I/G ratio $\xrightarrow{-}$ Acetyl CoA carboxylase → ↓ malonyl CoA $\xrightarrow{+}$ CPT - I
(RLE for FA synthesis)
- Fed state → High I/G ratio $\xrightarrow{+}$ Acetyl CoA carboxylase → ↑ malonyl CoA $\xrightarrow{-}$ CPT - I
(RLE for FA synthesis)

Other Oxidation Pathways & Disorders of Fatty Acid Oxidation

00:50:24

Other Oxidation Pathways :

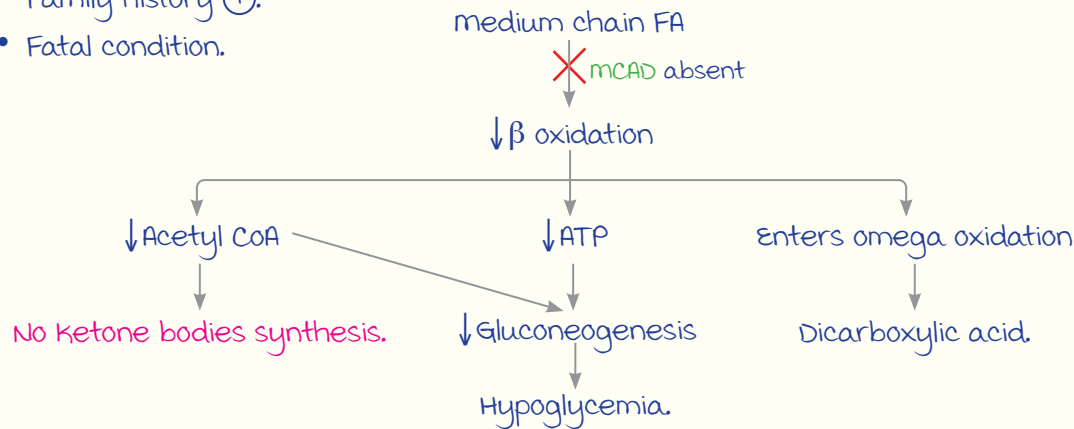
Oxidation pathway	Site	Important feature
β oxidation of unsaturated FA	mitochondria	Acyl CoA dehydrogenase step is bypassed ↓ 1.5 ATP less for every double bond
β oxidation of odd chain FA	mitochondria	Byproducts : Propionyl CoA (Glucogenic) + Acetyl CoA
Alpha oxidation : Branched chain FA (Phytanic acid)	• Peroxisome : major • SER : minor	• No acetyl CoA • No ATP produced • Phytanoyl CoA hydroxylase/oxidase
Omega oxidation	SER (microsome)	• Product : Dicarboxylic acid • No acetyl CoA • No ATP produced

Disorders of FA Oxidation :

----- Active space -----

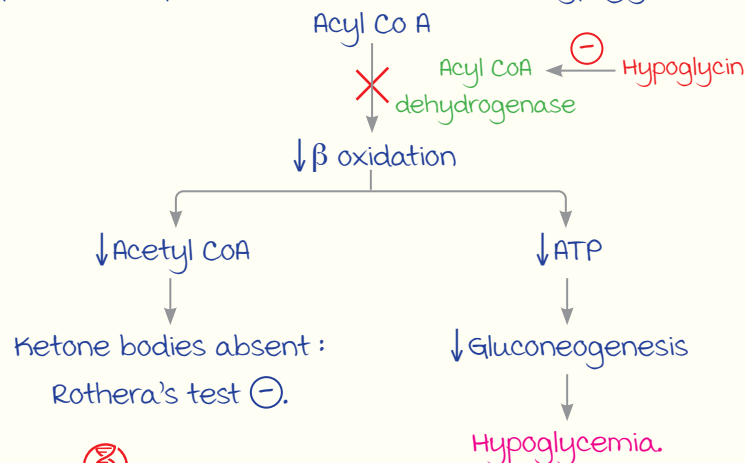
1. MCAD (medium chain acyl CoA dehydrogenase) deficiency :

- Causes seizures.
- Family history (+).
- Fatal condition.



2. Jamaican vomiting sickness :

D/t consumption of unripe ackee fruit → Contains hypoglycin.



3. Refsum's disease (🧬) :

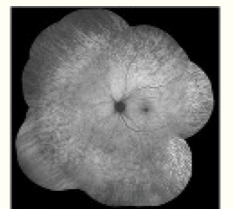
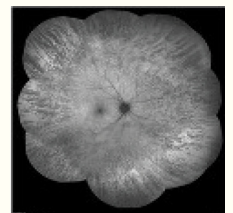
- Defect : Phytanoyl CoA oxidase (Hydroxylase) (Peroxisome) → X → Alpha oxidation.
- Features :
 - Asymptomatic > Symptomatic.
 - Anosmia.
 - Deafness.
 - Retinitis pigmentosa.
 - Ichthyosis.
 - Peripheral neuropathy.
 - Cardiac arrhythmias.

Rx :

Restrict dairy products & green leafy vegetables.



Ichthyosis (Scaly skin)



Retinitis pigmentosa

----- Active space -----

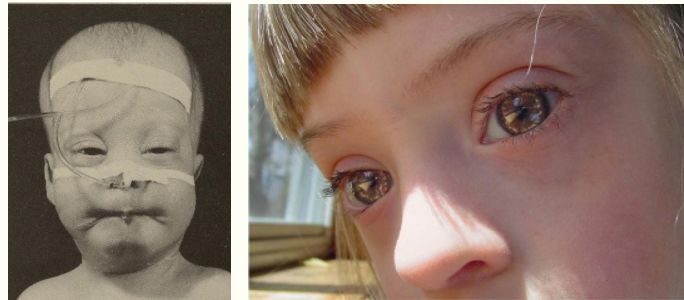
3. Zellweger syndrome/cerebrohepatorenal disease :

- Defect : Peroxisome targeting Sequence (PTS)

↓
Defective phytanic acid & very long chain FA oxidation (α -oxidation).

- C/f : Resembles Down's syndrome.

- | | |
|---------------------------------|-----------------------------|
| - mongoloid facies. | - High forehead. |
| - Hypertelorism. | - Brushfield spots in iris. |
| - upslanting palpebral fissure. | - Intellectual disability. |
| - Frontal bossing. | |



Zellweger syndrome : Resembles Down's syndrome

- Diagnosis : $\uparrow$ Plasma VLCFA & phytanic acid.

4. Acute fatty liver of pregnancy :

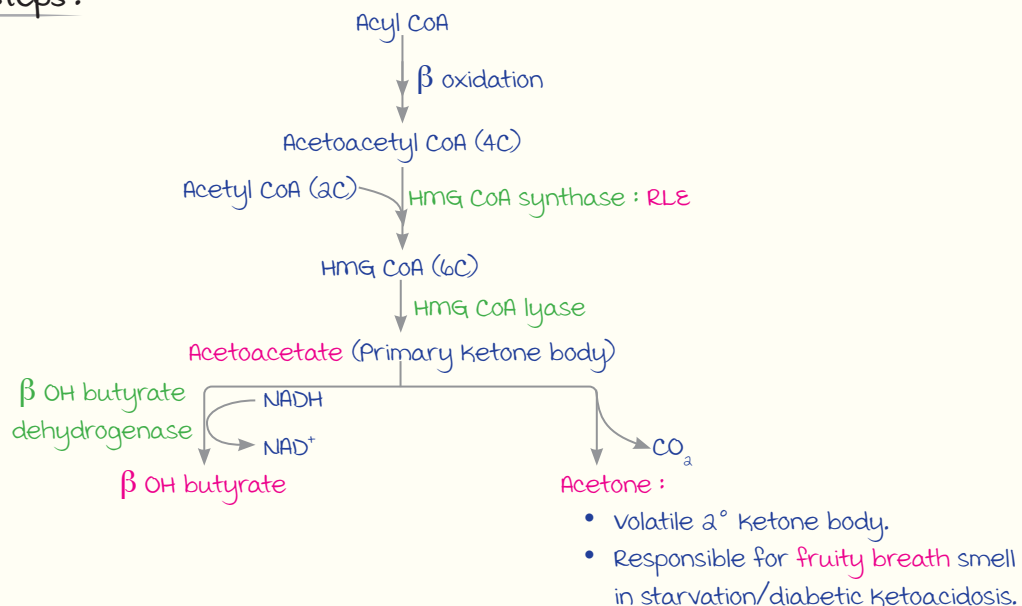
Long chain FA dehydrogenase deficiency in fetus $\rightarrow$ $\uparrow$ LCFA & derivatives in maternal blood.

Ketone Body Synthesis

01:01:54

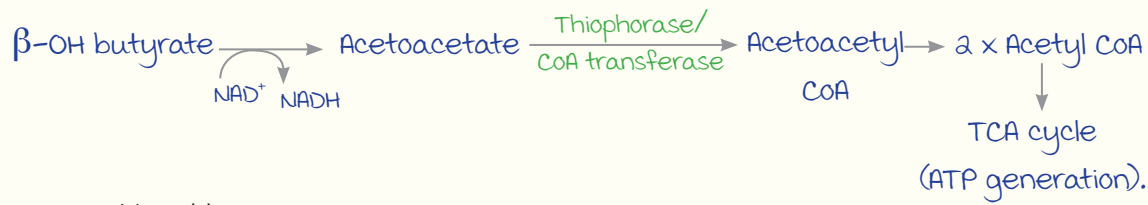
Site :

Exclusively liver mitochondria.

Steps :

Ketone Body utilization :

----- Active space -----

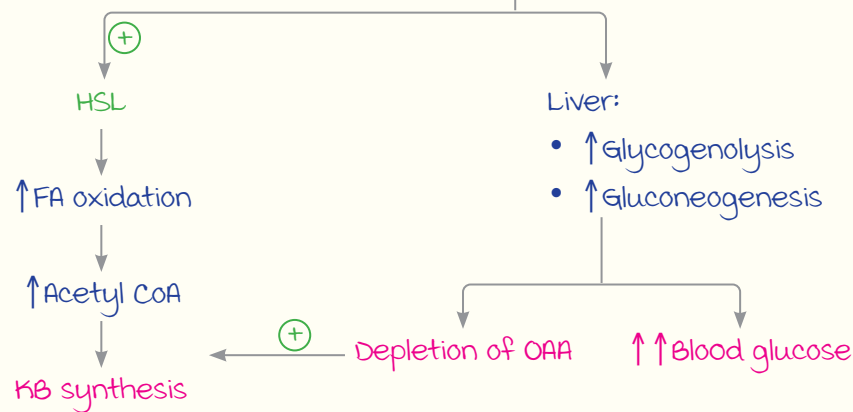


Never utilized by :

- Liver : D/t lack of thiophorase.
- RBC : D/t absence of mitochondria.

Diabetes ketoacidosis

- Diabetes : $\downarrow$ Insulin/Insulin resistance $\rightarrow$ Low I/G ratio (Simulates fasting state).



- Lab diagnosis :

- Rothera's test : $\oplus$.
- Ketostix.



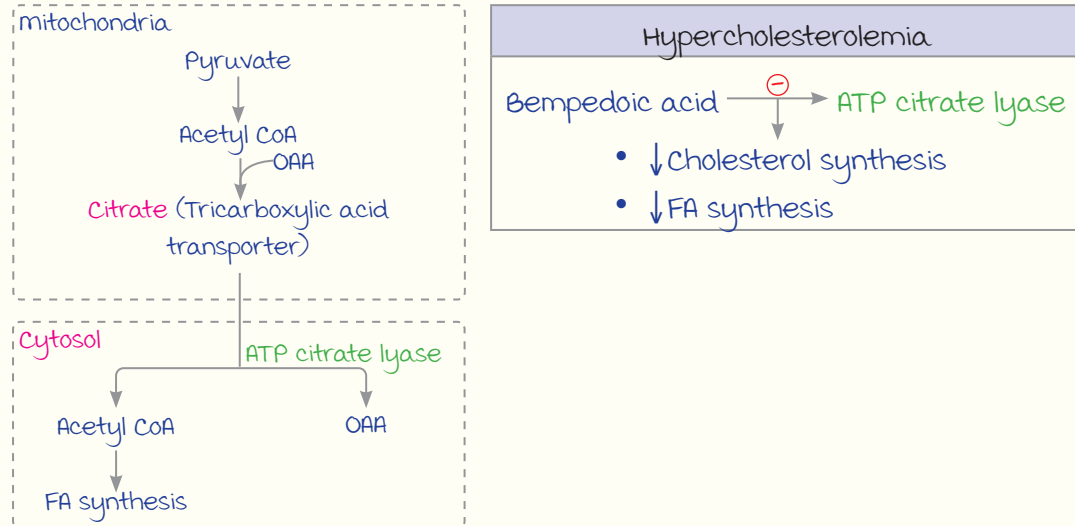
Ketostix



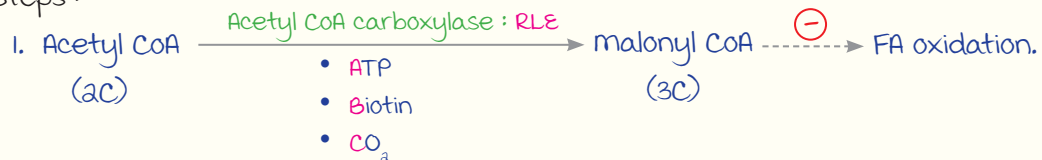
Rothera's test

Fatty Acid Synthesis :

Citrate transport :



Steps :



2. Fatty acid synthase complex :

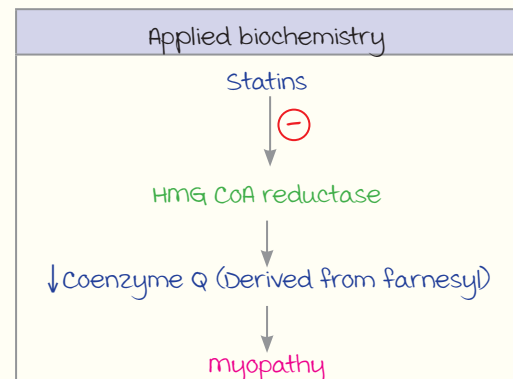
- Homodimer (X-shaped).
- Each monomer : 3 subunits
- Requires NADPH $\longrightarrow$ Reductase.
- Acyl carrier protein (ACP) : Contains pantothenic acid.
- Acetyl CoA carboxylase not included

Cholesterol Synthesis :

Exclusive animal steroid, not a metabolic fuel.

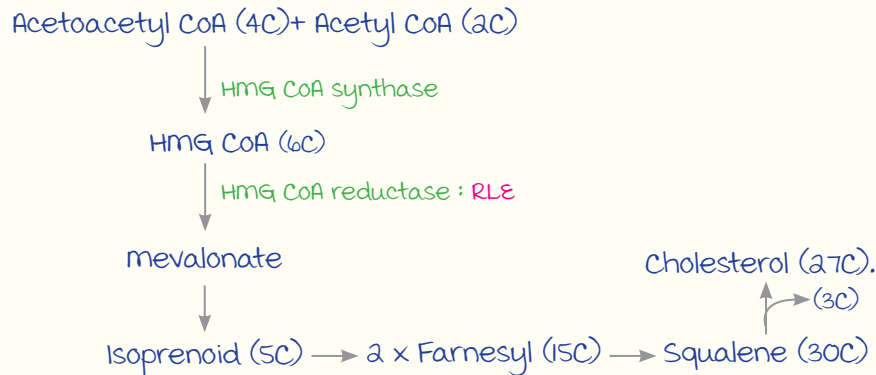
Site : Liver, adipose tissue, gonads, adrenal cortex.

Organelle : Cytoplasm + SER.



Steps :

----- Active space -----



Compounds derived from cholesterol :

1. Bile acids (Excretory product).
2. Steroid hormones.
3. Vitamin D.

Factors affecting HMG CoA reductase :

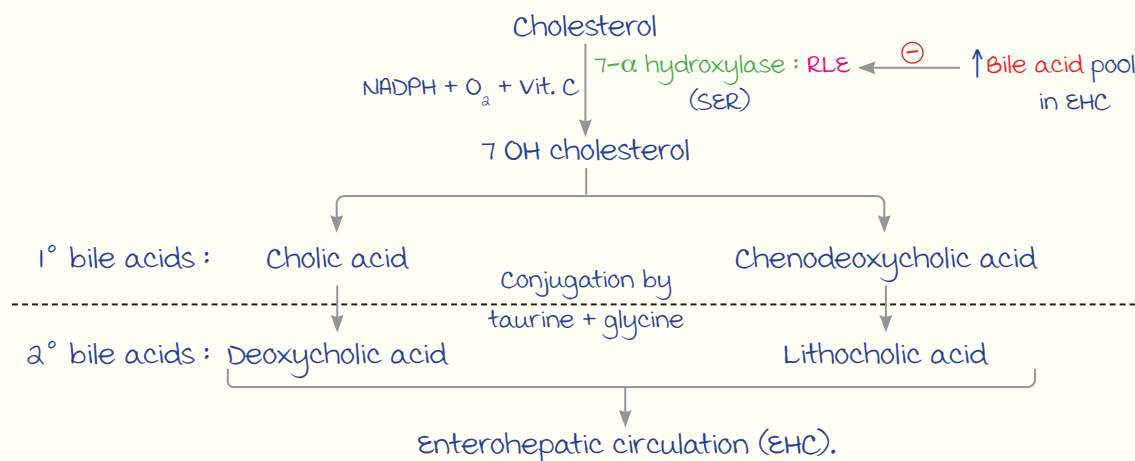
⊖	⊕
- Glucagon - Glucocorticoids - Dietary cholesterol - AMP	- Insulin - Thyroid hormone

Bile Acid Synthesis :

Site :

- Liver (Primary bile acid).
- Intestine (Secondary bile acid).

Steps :

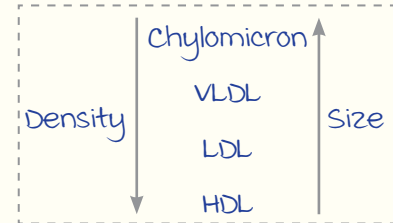


Lipoproteins

01:24:01

Characteristics :

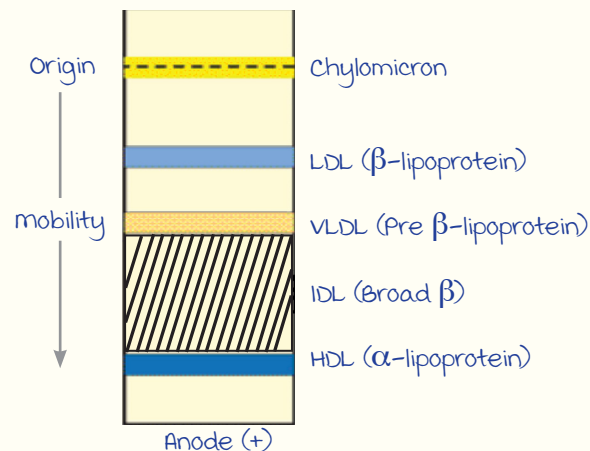
- maximum cholesterol : LDL.
- maximum TAG
- minimum density
- Remains at the point of application
- maximum size.
- maximum density
- minimum size
- Fastest electrophoretic mobility



Functions :

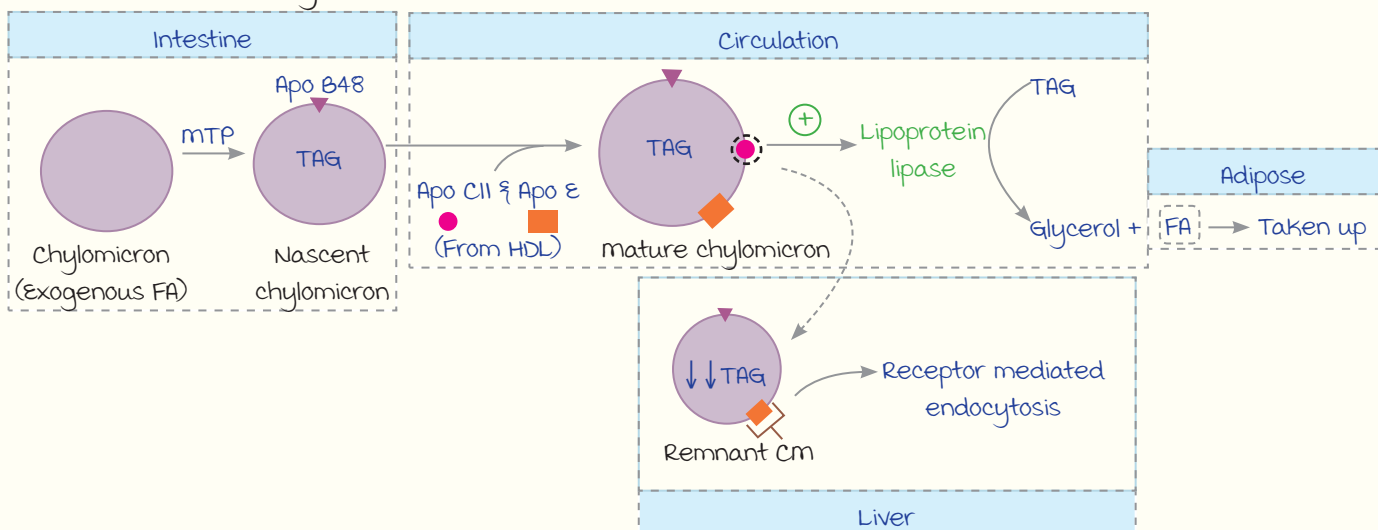
- Carry **exogenous TAG** to peripheral organs : Chylomicron.
- Carry **endogenous TAG** to peripheral organs : VLDL.
- Carry **cholesterol** from peripheral tissue to **adrenals** : HDL.

Electrophoretic pattern :



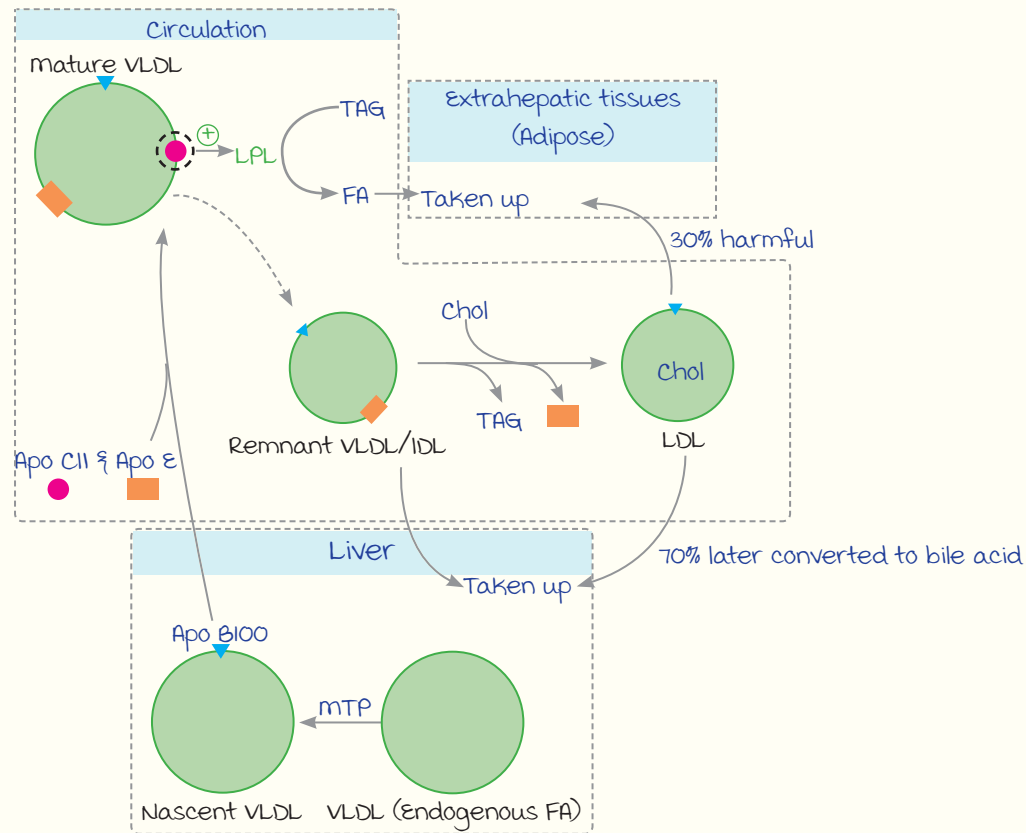
metabolism of Lipoproteins :

I. Chylomicron :

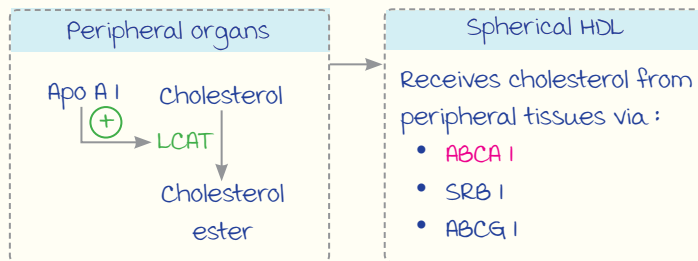


2. VLDL/LDL/IDL :

----- Active space -----



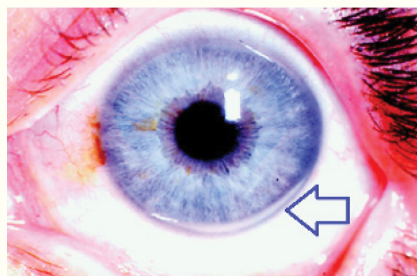
3. HDL : Reverse cholesterol transport.

Disorders of Lipoproteins

01:29:29

	Defect	Lipoprotein accumulated	Lipid levels
Familial chylomicronemia (Type I HL) Latest Rx modalities : • Lipogenic Ti parvovec • GOF LpL variant	LPL or Apo CII	Chylomicron > VLDL	• TAG ↑↑↑ • Cholesterol : Normal
	c/f		
	Pancreatitis can also be seen		
			
	Eruptive xanthoma	milky white plasma	Lipemia retinalis

----- Active space -----

	Defect		Lipoprotein accumulated	Lipid levels
Familial hypercholesterolemia (Type II HL) : m/c Latest Rx modalities : • Lomitapide (MTTP ⊖) • mipomersen • VERVE 101 : Genome editing on CRISPR cas9.	II A	LDL receptor or Apo B100 mutation or PCSK9 defect	LDL	• Cholesterol : ↑↑↑ (Risk of CAD) • TAG : Normal
	II B	Polygenic		• Cholesterol : ↑ • TAG : ↑
	c/f			
	Family h/o CAD (Coronary artery disease), chest pain			
				
	Corneal arcus		Tendon xanthoma	
	Defect		Lipoprotein accumulated	Lipid levels
Familial dysbetalipoproteinemia (Type III HL)	Apo E		Remnant chylomicron & VLDL	• TAG : ↑ • Cholesterol : ↑
	c/f			
				
	Tubero-eruptive xanthoma		Palmar xanthoma	

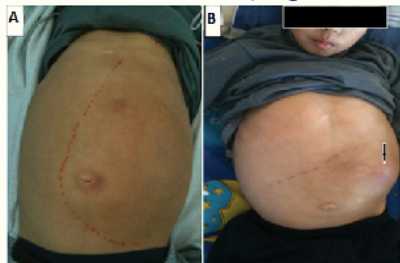
Tangier's Disease :

Defect : ABC AI

Findings : Cholesterol gets accumulated in peripheral tissues.

Features :

1. Hepatosplenomegaly :



2. Orange tonsils :



3. Peripheral neuropathy.

Abetalipoproteinemia :

----- Active space -----

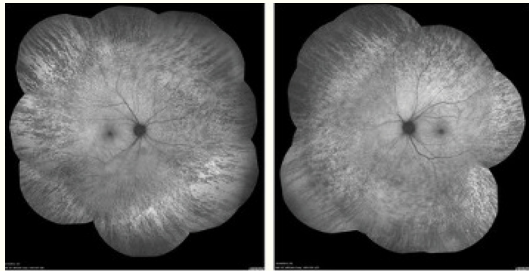
Defect : mTTP/mTP (microsomal triglyceride transfer protein).

Findings :

- ↓ Chylomicron → Bleeding manifestations (Transports fat soluble vitamins such as vitamin K).
- ↓ VLDL. → Steatorrhea.
- ↓ IDL.
- ↓ LDL.

Features :

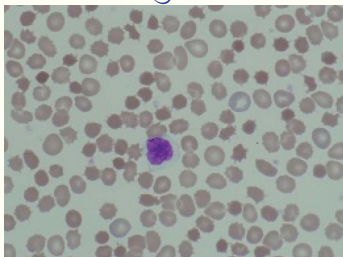
1. Pigmentary retinopathy :



2. Petechial rash :



3. Acanthocytes :



4. Neurological deficits.

Sitosterolemia :

Features similar to type IIA HL.

Defect :

- ABC G5
- ABC G8

Finding :

- ↑ Cholesterol.
- ↑ LDL.

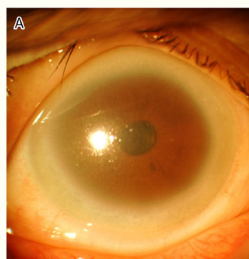
Fish Eye Disease :

Defect : Partial LCAT deficiency.

Features :

- Hepatosplenomegaly.
- Corneal opacity.

Finding : ↓ HDL.



Fish eye disease

MOLECULAR BIOLOGY

DNA

00:00:10

DNA Replication :

Salient features :

- Both strands act as templates.
- Bidirectional : Always 5' → 3'.
- Semidiscontinuous.
- Semiconservative.
- Requires primer.

Enzymes & proteins involved :

Enzymes/proteins	Function
Helicase	Unwind DNA
Primase	Synthesize RNA primer
Single strand binding protein (SSB)	Stabilises single stranded DNA
Topoisomerase	Relieves torsional strain in unwound DNA
DNA ligase	Joins the nick

DNA polymerase :

Enzyme	Function
Prokaryotic DNAP	
DNAP I (AKA Kornberg's enzyme)	• Removal of primers & gap filling • DNA repair (major) • DNA proof reading
DNAP II	• DNA repair • DNA proof reading
DNAP III (most processive)	• Leading strand synthesis • Synthesis of Okazaki fragments • DNA proofreading

Enzyme	Function
Eukaryotic DNAP	
ε	Leading strand synthesis
δ	Lagging strand synthesis
γ	mitochondrial DNA synthesis
β	DNA repair activity
α	Primase activity

Klenow fragment :

- DNAP I without 5' → 3' exonuclease activity.
- Use : Recombinant DNA technology.

DNA Repair Defects :

----- Active space -----

Defects in DNA	Repair mechanisms	Associated disorders
Bulky adducts/ Pyrimidine dimers	Nucleotide Excision Repair (NER)	• Xeroderma pigmentosa - Cutaneous photosensitivity - ↑ Risk of skin cancer • Cockayne syndrome • Trichothiodystrophy
Abasic sites	Base Excision Repair (BER)	MUTYH-associated polyposis
Base mismatch	mismatch Repair (MMR)	Hereditary Non-Polyposis Colorectal Cancer (HNPCC) : - Genes defect : MSH-2, MLH-1 - microsatellite instability
• Double strand breaks • Single strand breaks • Intrastrand cross-links	Non-Homologous End Joining (NHEJ)	• SCID • Radiosensitive SCID
	Homologous Recombination (HR)	• Ataxia telangiectasia : - Features of cerebellar ataxia - Dilated capillaries • Nijmegen break syndrome • Bloom's syndrome • Werner syndrome • Rothmund Thomson syndrome • Breast cancer susceptibility (BRCA 1 & 2)

Telomere :

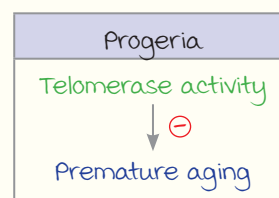
Ends of the chromosomes.

Telomerase :

- Properties :
 - Contains an intrinsic RNA primer.
 - Reverse transcriptase activity (+).
- Absent in : Somatic cells.
- Present in : Hematopoietic cells, germ line cells, cancer cells, lymphocytes, stem cells.
- mechanism :
 - Extends 3' end of DNA.
 - No critical shortening of DNA.
 - Provide immense replicating potential to DNA.
 - No hayflick limit ;

Hayflick limit (Theory of ageing) :

Cell division $\rightarrow \rightarrow \rightarrow \rightarrow$ Critical shortening Stops after certain number.



----- Active space -----

Transcription

00:10:48

- DNA $\rightarrow$ RNA.
- Only one strand transcribed : Template strand.
- Other strand : Coding strand.
 - Strand not involved in transcription.
 - Same sequence as that of RNA with T replaced by U.

Enzymes :

RNA polymerase (RNAP) :

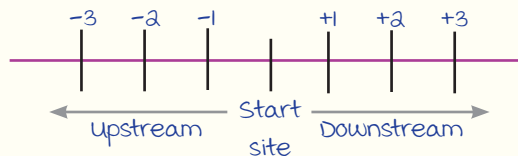
- Prokaryotic :
 - σ subunit : Binds to promoter.
- Eukaryotic RNAP.

	RNAP-I	RNAP-II	RNAP-III
Sensitivity to α -amanitin	Least	Highest	Intermediate
major products	rRNA (most abundant)	• mRNA • miRNA • snRNA • lncRNA	• tRNA • 5S rRNA

Regulation :

Promoters of transcription :

- Coding region sequences that specify the start site of transcription.



- E.g. :
 - Pribnow box : -10 bp (Prokaryotes).
 - TATA box : -25 bp (Eukaryotes).

Enhancers/silencers/repressors :

- Present upstream/downstream.
- Non-specific.
- Located far away from start site.

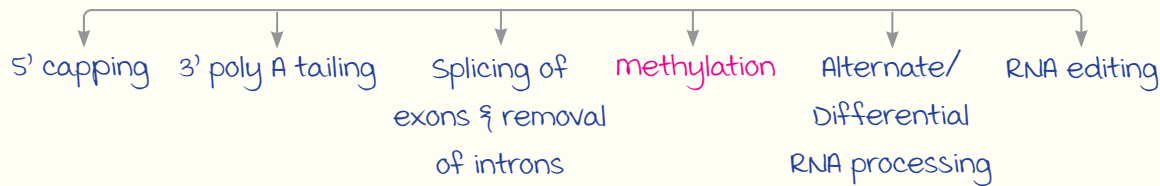
Termination :

Termination of transcription requires ρ factor.

Post Transcriptional modification :

----- Active space -----

Steps :



5' capping :

7 methyl guanosine cap requires :

- Guanylyl transferase.
- methyl transferase.
- Functions :
 - Stabilizes mRNA : Prevents the attack of 5' → 3' exonuclease.
 - Facilitates initiation of translation.
 - Binding of mRNA to initiation complex.

3' Poly A Tailing :

- Addition of 40-200 adenosine residues at 3' end.
- Enzyme : Polyadenylate polymerase.
- Functions :
 - Stabilizes mRNA : Prevents the attack of 3' → 5' exonuclease.
 - Facilitates exit of mRNA from nucleus → Cytoplasm for translation.
 - Recruitment of 40S ribosome : Poly A tail & poly A tail binding complex (PAB).

Splicing of exons & removal of introns :

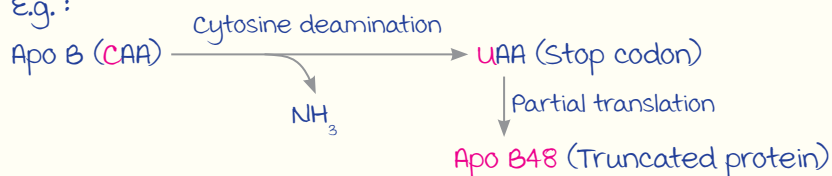
- Carried out by spliceosomes
 - snRNA
 - Proteins
 - Primary transcript : Heteronuclear RNA
 } snRNP/ Snurps
- Introns removed → Exons joined together (Lariat).

Alternate RNA processing :

Splicing in different sequences → Different proteins from same gene.

RNA editing :

E.g. :



----- Active space -----

RNA

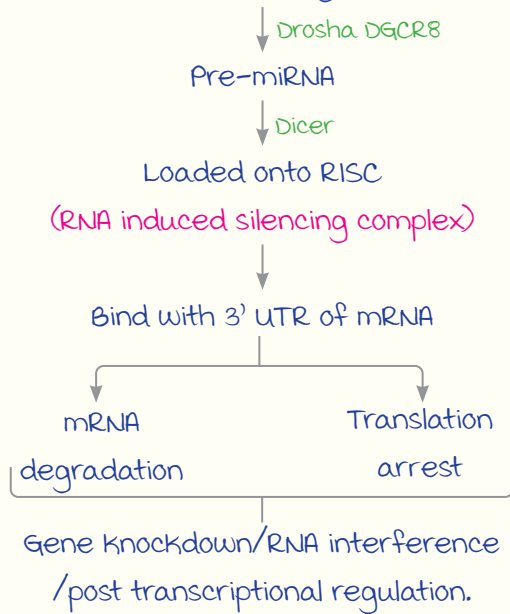
00:19:45

miRNA :

Small non-coding ssRNA of 21-22 nucleotide length.

Function : **Post-transcriptional regulation** of gene expression.

Formation : Pri-miRNA (Endogenous)



.....
Note :
.....

siRNA :

- Same mechanism as miRNA.
- usually **exogenous**.

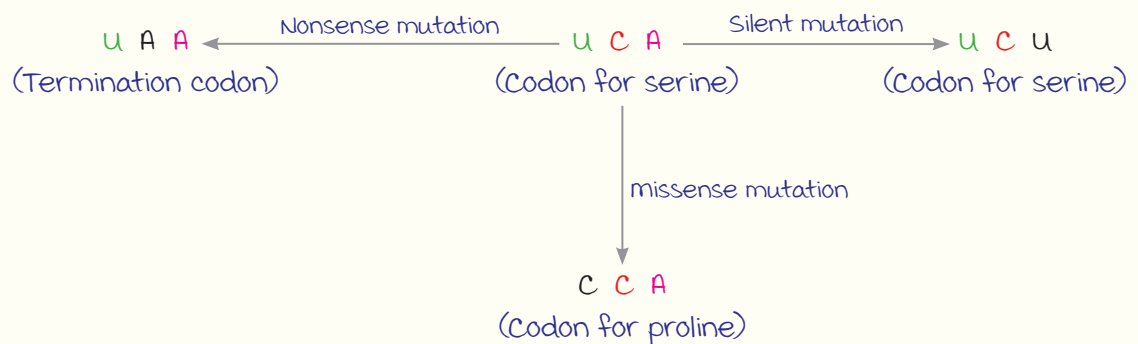
Mutations and Epigenetics

00:22:31

mutations :

Change in DNA sequence

Point mutation (Base substitution) : **m/c**.



Frame shift mutation :

Insertion/deletion of nucleotide → Distorted reading frame → Loss of 1° protein structure
↓
Loss of biological activity.

----- Active space -----

Epigenetics :

- Heritable, reversible chemical modification of DNA or chromatin.
- **No change** in DNA sequence.

DNA methylation :

- Enzyme : **DNA methyl transferase**.
- Site : Cytosine residues.
- Effect : Gene silencing.

Histone modifications :

- Histone acetylation → Euchromatin formation → ↑ Gene expression (Enzyme : **Histone acetyl transferase**).
- Histone deacetylation → Heterochromatin formation → ↓ Gene expression (Enzyme : **Histone deacetylase**).

Note :

Salient features of genetic code :

- **Triplet codon** : 3 nucleotides.
- **universal** : Each codon codes for same protein in all species. (Except : mitochondrial DNA)
- **unambiguous** : 1 codon codes for only 1 amino acid.
- **Degenerate** : 1 amino acid can have > 1 codon (In 3rd base).
- Non-overlapping.
- Start codon : **AUG**.
- Stop codons : **UAA, UGA, UAG**.

Genomic Analysis

00:26:50

Blotting techniques :

Technique	Target molecule
Southern blot	DNA
Northern blot	Specific RNA
Western blot	Specific antigen/protein
South western blot	DNA protein interaction

----- Active space -----

microarray :

	Chip composition	Addition	Results	Applications
DNA microarray	Known oligonucleotide	Fluorescent-labelled unknown DNA	Fluorescence detected	- DNA sequencing - Point mutation detection - Study of genomics
c-DNA microarray	Known c-DNA	Fluorescent-labelled unknown RNA		- Study of gene expression - Study of transcriptome
Protein microarray	Known Agl Ab	Fluorescent-labelled unknown Agl Ab		- Study of antigen expression - Study of proteomics
Array CGH	c-DNA from normal & cancer cells	Fluorescent-labelled : - Red : Cancer cell - Green : Normal cell	Fluorescence colour based result	- Comparison of Normal vs diseased genome - Study deletions, insertion, amplification, aneuploidy - Cannot detect balanced translocation

Karyotyping :Steps  :

1. Collect cells from peripheral vein using heparin syringe.
2. Culture in **phytohemagglutinin**.
3. Incubation at **37°C** for 3 days.
4. Harvest with colchicine.
5. Routine staining with **Giemsa**.

Banding techniques :

Banding technique	use
Giemsa (G)	Light & dark banding
Quinacrine (Q)	Similar to Giemsa, but examination with UV fluorescence microscope
Reverse (R)	Denatured chromosome → Light & dark bands in reverse pattern
Centromeric (C)	Heterochromatin (Centromere) stained preferentially

Human Genome Project :

- 1996 → 1st human gene map.
- 2003 → Sequencing of human genome complete.

Results :

- Exons : 1.5%
- No. of protein coding genes : 30000 – 40000.
- Haploid set of chromosome : **3.2 billion bp**.

Fluorescent In-situ Hybridisation (FISH) :

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AKA Spectral Karyotyping

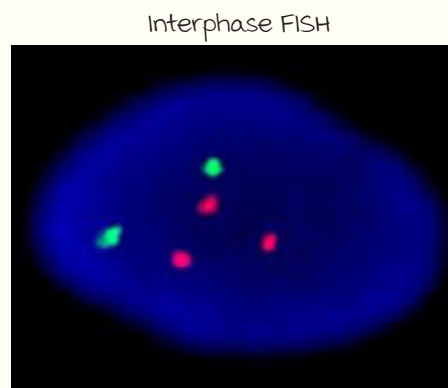
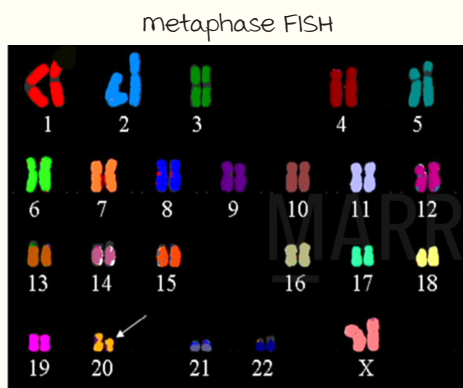
Detected by : Fluorescent probes.

Uses :

To detect :

- microdeletions
 - Amplification
 - Newly isolated gene to chromosomal location.
 - **Fusion genes**
 - Structural abnormalities
 - **Translocations**
 - Aneuploidy
 - Copy number variations
- } Numerical abnormality

Types :



- Growing cells not required
- Rapid results
- Greater detection limit

Recombinant DNA Technology

00:35:25

Restriction Endonuclease (RE) :

- Present in bacteria : Restrict phage entry (Defense mechanism).
- molecular scissors : Cut at specific **palindromic sequence** (Type II RE).
- Create sticky/staggered ends.

Applications :

- Creation of restriction maps.
- Restriction fragment length polymorphism (RFLP).
- Synthesis of recombinant proteins.

vector :

Carriers of foreign DNA to host cell (**Transformation**).

I. Plasmid : Extrachromosomal bacterial dsDNA → Confer **antibiotic resistance**.

Insert size : 0.1 to 10 Kbp.

----- Active space -----

2. Phage DNA : Linear DNA in phages.

Insert size : 10 to 20 Kbp.

3. Artificial chromosome (AC) :

- Bacterial (BAC) } 50 - 250 Kbp.
- Phage (PAC) }
- Yeast (YAC) : 500 to 1000 Kbp.

4. Cosmid : Plasmid with cohesive end site.

- Insert size : 30 - 50 Kbp.

Restriction map & RFLP :

- Unique band pattern formed when DNA is treated with specific restriction enzyme.
- Same restriction enzyme → unique pattern in each individual.
- Mutation of gene →

Abolish restriction site.	} Detected
Create restriction site.	

DNA Fingerprinting :

- Band pattern of unknown DNA $\xrightarrow{\text{matched with}}$ Band pattern of known DNA.
- Used in medico-legal cases (Identifying suspect etc) & disputed paternity.

DNA Footprinting :

- Study of DNA-protein interaction.
- method :

DNA is mixed with protein

↓ DNase

Breakdown of DNA areas not bound to protein.

↓

Fragments bound to proteins not visualized (Footprint regions).

PCR : Genome Editing and DNA Sequencing

00:42:04

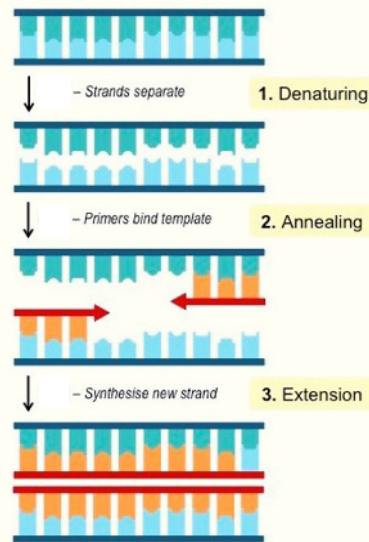
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Steps  :

- No. of DNA after n cycles = 2^n .
- Half the number of amplicons to nth cycle : $(n-1)^{\text{th}}$ cycle.

master mix :

- DNA sample.
- Primers.
- Nucleotides.
- Taq polymerase.
- MgCl_2 .



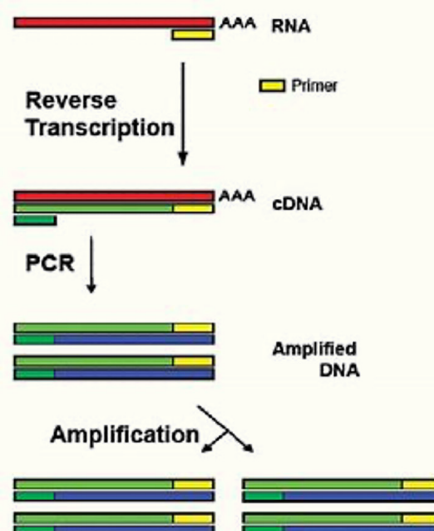
Steps of PCR

Real time PCR (Quantitative PCR) :

- Simultaneous amplification & detection.
- Signals on target :
 - Taqman probe.
 - SYBR green.

Reverse Transcriptase PCR (RT-PCR) :

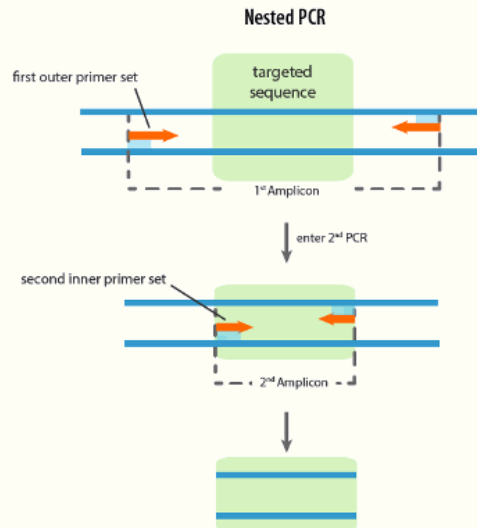
Amplification of RNA study of gene expression.



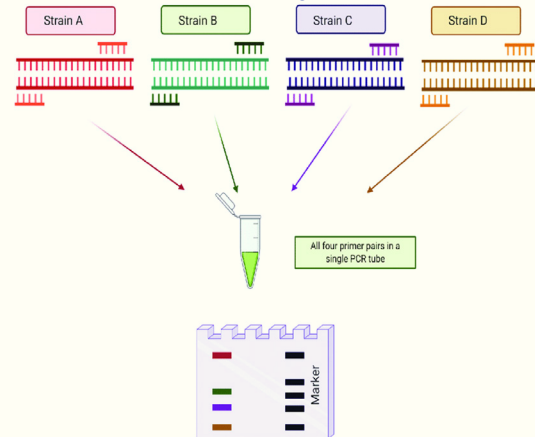
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Nested PCR :

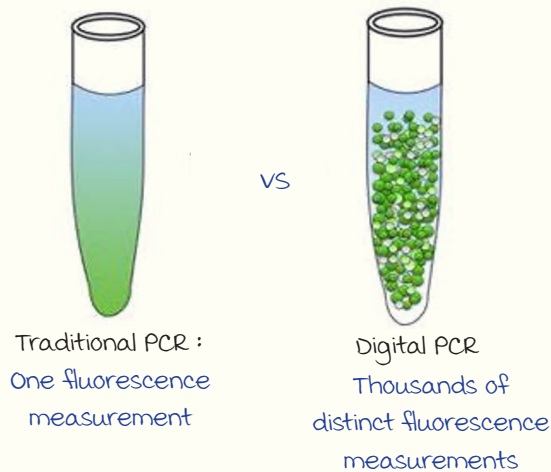
↑ Specificity.

multiplex PCR :

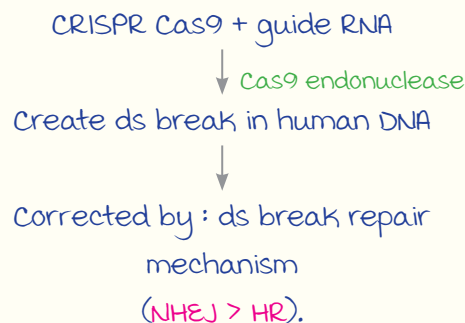
- multiple random primers added.
- Syndromic approach : Detect infective strains for management.

Droplet Digital PCR :

- multiple PCR in same reaction mixture in multiple droplets.
- Detects mutations with low allele frequency.

Genome Editing :

CRISPR Cas9 : Present in Bacteria.



DNA sequencing :

----- Active space -----

Sanger's sequencing : uses labelled **dideoxynucleotides**.

Dideoxy NT : No functional 3'-OH group → No 3'-5' phosphodiester bond → **Controlled chain termination** → Bands detected by fluorescent label

Next generation sequencing :

- used in population based studies: Heterogenous DNA samples
- Steps :
 - Spatial separation of DNA samples.
 - Amplification.
 - massive **parallel** sequencing.

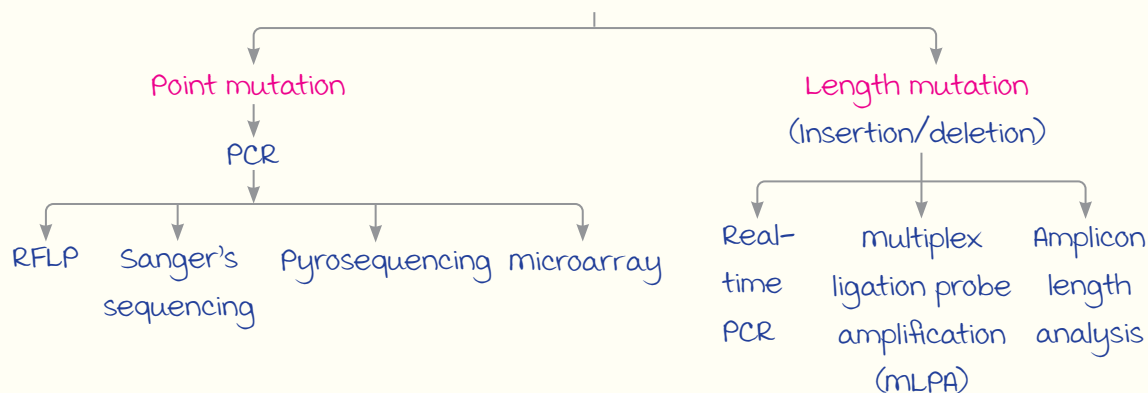
Overview of Techniques

00:51:55

DNA protein interaction detection :

1. Southwestern blotting.
2. DNA footprinting.
3. **Chromatin immunoprecipitation (ChIP)**.

Mutation detection :



Aneuploidy detection :

1. Karyotyping.
2. FISH.
3. Quantitative fluorescence PCR/Real-Time PCR.
4. Array CGH.
5. MLPA.

----- Active space -----

Study of gene expression :

1. Northern blot.
2. cDNA microarray (Array CGH).
3. RT-PCR ($\pm$ real time PCR).

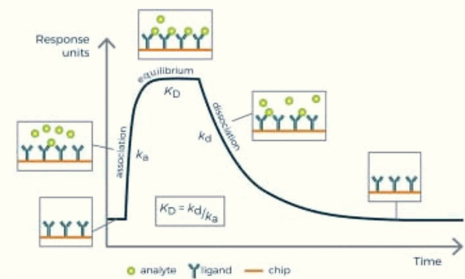
Strain typing of bacteria :

Chromosome mutation analysis of strains of bacteria.

1. RFLP.
2. Spoligotyping.
3. multiple interspersing repeat unit (MIRU) typing.
4. Next generation sequencing.

Surface plasmon resonance :

Real-time technique : Ab affinity towards antigen.



Nucleotides

00:55:32

- Nitrogenous base + pentose sugar + phosphate group.
- Nucleotides absorb UV light : 260 nm.
(mutagenic effect of UV light)

Types of nitrogenous bases :

Double ring : Purine	Single ring : Pyrimidine
Adenine Guanine	Cytosine Deamination → Uracil Methylation → Thymine

Important Linkages :

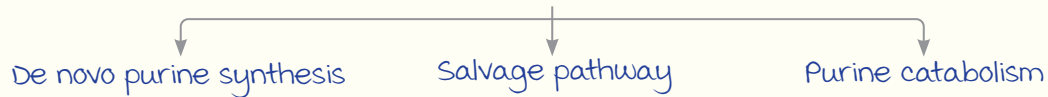
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Nucleic acid :

Nucleotide

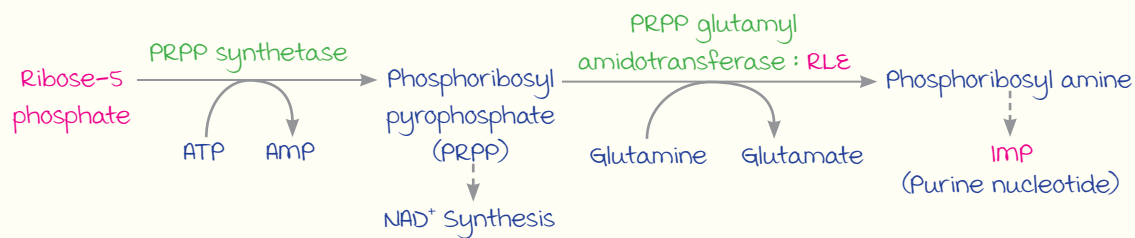
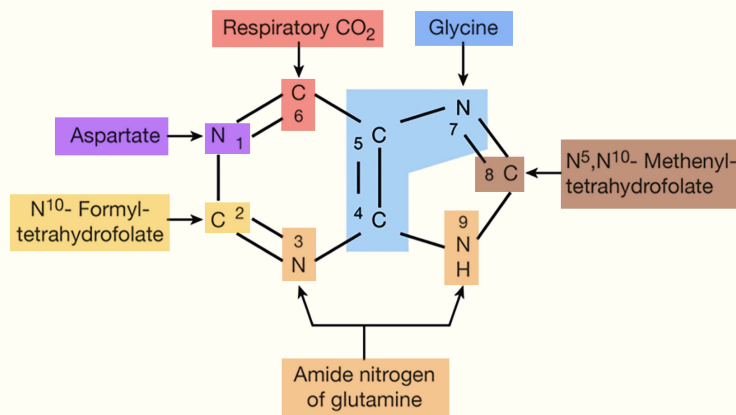


Purine metabolism :



De novo purine synthesis :

- Site :
 - major : liver.
 - Occurs in all organs except brain, RBC & WBC.
- Source of purine ring :



----- Active space -----

Salvage pathway :

- Recycles purine nucleosides & bases to purine nucleotides.
- Conserves energy in organs without de novo synthesis.
- Phosphoribosylation of purines :

Substrate	Donor	Enzyme	End product
Adenine	PRPP	Adenine phosphoribosyl transferase (APRTase)	AMP
Hypoxanthine		Hypoxanthine guanine phosphoribosyltransferase (HGPRTase)	IMP
Guanine			GMP
Adenosine	ATP	Kinase	AMP
Guanosine			GMP

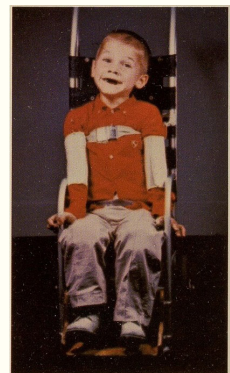
Purine catabolism :

- Site : Liver
- Organelle : Cytoplasm
- End product : uric acid.

Enzyme deficiency	Features
Adenosine deaminase	SCID : Both T cells & B cells affected
Purine nucleoside phosphorylase	Immunodeficiency affecting only T cells
Xanthine oxidase	• Xanthinuria • ↓ Uric acid • Inhibitors : - Allopurinol - Febuxostat

Disorders of Purine Synthesis :Lesch Nyhan Syndrome :

- Defect : HGPRTase.
- Hypoxanthine $\not\rightarrow$ IMP $\rightarrow$ ↑ Purine catabolism $\rightarrow$ ↑ Uric acid.
- c/f :
 - Compulsive self-mutilation.
 - Behavioural & cognitive disturbances.



Lesch Nyhan Syndrome

Kelley Seegmiller syndrome :
Partial defect of **HGPRTase**.

Gout :

- c/f :
 - Acute inflammatory arthritis
 - Hyperuricemia.
 - uric acid nephrolithiasis
- Definitive diagnosis :
Needle shaped **negatively birefringent** msu crystals on polarized light microscopy.
- Etiology :
 - $\uparrow\uparrow$ activity of
 - **PRPP synthetase**
 - **PRPP glutamyl amidotransferase**
 - Von Gierke's disease.
 - Lesch Nyhan Syndrome.
 - Alcohol consumption $\rightarrow \uparrow \text{NADH} \rightarrow \uparrow \text{Lactate} \rightarrow \downarrow \text{excretion of uric acid}$
($\uparrow$ Renal threshold)
 - Secondary gout d/t
 - Renal failure
 - malignancy
 - Chemotherapy



Tophi : In SC tissue



msu (monosodium urate) crystals

$\downarrow$ excretion of uric acid
($\uparrow$ Renal threshold)

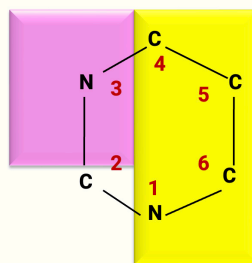
$\downarrow$
Precipitates gout.

Pyrimidine metabolism :

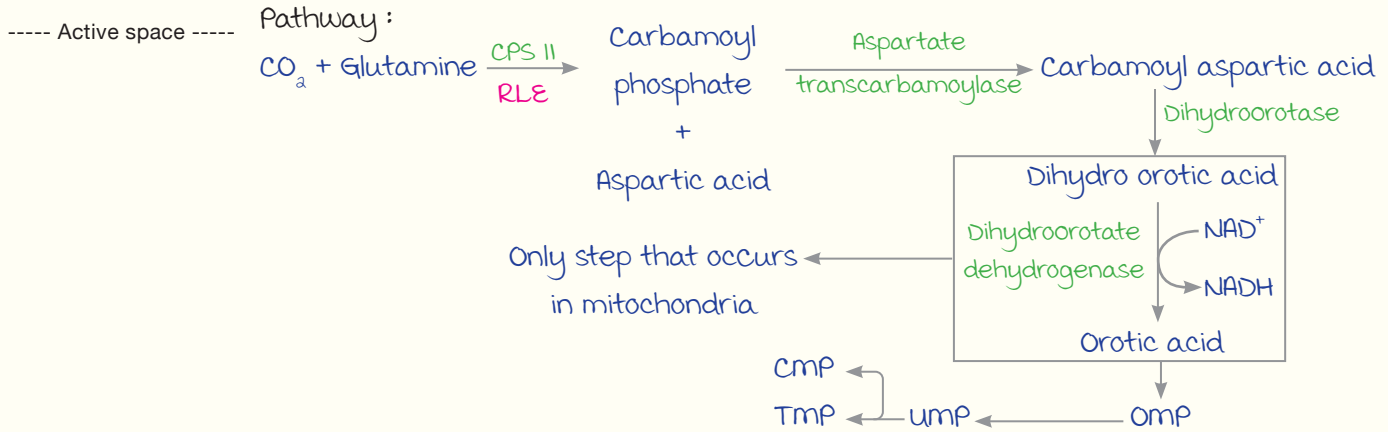
Site : Liver

Organelle : Cytoplasm & mitochondria.

Sources of pyrimidine ring :



■ Aspartic acid
■ Glutamine



End products of pyrimidine catabolism :

- β -alanine (From cytosine & uracil)
 - β -amino isobutyrate (From thymine)
 - Pseudouridine : Excreted unchanged.
- } Water soluble (No symptoms of excess catabolism)

Hereditary orotic aciduria :

- Defect in : UMP synthase
 - Orotate phosphoribosyl transferase
 - Orotidine monophosphate decarboxylase

- c/f :

macrocytic, megaloblastic anemia (Does not respond to vit B_{12} / folic acid / Fe supplementation).

VITAMINS AND MINERALS

----- Active space -----

Endogenous vitamins :

- Synthesised by the body :
 - Niacin : From tryptophan.
 - Vitamin D : From cholesterol.
- Produced in the body : By microbiome
 - Vitamin K.
 - Pantothenic acid.
 - Biotin.

Fat soluble vitamins : A, D, E & K.

Water soluble vitamins : B complex & C

Vitamin A vs Vitamin D

00:00:53

	Vitamin A		Vitamin D	
Forms	• Retinal • Retinoic acid • Retinol		• Ergocalciferol (D_2) : Plant sources • Cholecalciferol (D_3) : Animal sources	
metabolism	Intestine	β -carotene / Retinoids (Diet) $\rightarrow$ Retinol $\downarrow$ + Fatty acids	Skin	Diet $\rightarrow$ Cholecalciferol + D-binding protein 7-dehydrocholesterol $\xrightarrow{\text{UVB rays (290-315 nm)}}$ Cholecalciferol $\downarrow$ 25-hydroxylase
	Liver : Stored in HSCs	Carried in chylomicron $\leftarrow$ Retinyl ester $\downarrow$	Liver	25-hydroxycholecalciferol $\downarrow$ PTH ($\uparrow Ca^{2+}$, $\downarrow PO_4^{3-}$) 24-hydroxylase $\downarrow$ 1- α -hydroxylase (RLE)
	Plasma	Transported by retinol binding protein (RBP) & transthyretin	Kidney	1,25-dihydroxycholecalciferol (Calcitriol) (Biologically active form) $\downarrow$ Excess excretion 24, 25 - DHCC (Calcitroic acid)
Functions	• Vision : Retinal • Regulation of gene expression • Normal reproduction : Retinol • Maintenance of normal epithelium of skin & mucosa • Anti-oxidant : β-carotene • Growth & differentiation : Retinoic acid • Therapeutic use : - β-carotene cutaneous photosensitivity (Porphyria) - All-trans retinoic acid : Promyelocytic leukemia (Differentiation therapy) - 13-cis-retinoic acid (Isotretinoin) : Cystic acne & childhood neuroblastoma		• Regulation of Ca^{2+}, PO_4^{3-} ($\uparrow Ca^{2+}$, $\uparrow PO_4^{3-}$) : - Bone : Bone remodelling - Kidney : Reabsorb Ca^{2+} & PO_4^{3-} - Intestine : Absorb Ca^{2+} • Immunomodulatory : Prevent TB (Cathelicidin) • Anti-proliferative : Prevent Ca (Colon, breast, prostate)	

----- Active space -----

	Vitamin A	Vitamin D
Deficiency	• Eye manifestations : - Loss of vision to green light (earliest) - Night blindness - Conjunctival → Corneal xerosis - Bitot's spots - Corneal ulcer → Keratomalacia • Skin manifestations : - Follicular hyperkeratosis : Phrynoderma - Squamous metaplasia of mucus secreting epithelium	• ↑ unmineralised matrix : - Before closure of epiphysis : Rickets. - After closure : Osteomalacia. • Genu valgum & genu varum • Windswept deformity • Rachitic rosary (Non tender)
Toxicity	• Acute toxicity : - Pseudotumor cerebri (In arctic explorers) - Exfoliative dermatitis - Hepatomegaly - Lysosome affected • Chronic toxicity (> 50,000 IU/d) : - Bony exostoses - Hepatomegaly, cirrhosis • Pregnancy : Teratogenic (13-cis retinoic acid : C/I in pregnancy)	Calcinosis : Ca^{2+} deposited in blood vessels
Sources	• Halibut liver oil (Richest) • Plant sources : Carrots (Richest), green leafy vegetables • Animal sources : Egg, milk, cheese	Sunlight >> Diet
Assessment	• ↑ Dark adaptation time • Carr & Price reaction : Direct assay	• Vit. D assay (25-OH-CC) • S. osteocalcin
RDA	Retinol : • Children (1 - 6 y) : 400 µg • men & women : 600 µg • Pregnancy : 800 µg • Lactation : 950 µg	• Children : 400 IU • Adults : 200 IU • Pregnancy : 400 IU



Conjunctival & corneal dryness



Bitot's spot



Corneal ulcer



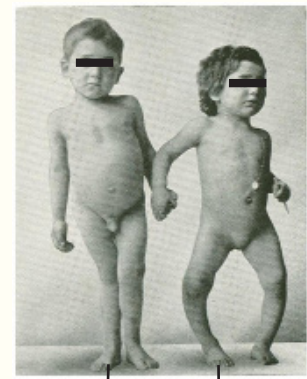
Windswept deformity



Rachitic rosary



Phrynoderma/Toad skin



Genu valgum & Genu varum

Vitamin E vs. Vitamin K

00:14:55

----- Active space -----

	Vitamin E	Vitamin K
Forms	Alpha tocopherol	K_1 : Phylloquinone (Dietary Source) K_2 : menaquinone (Bacterial flora) K_3 : menadione (Synthetic; Water soluble)
Functions	most potent natural anti-oxidant (Chain-breaking) - Prevents LDL oxidation. - Protects PUFA in membranes Therapeutic uses : - Retrolental fibroplasia - Intermittent claudication - Bronchopulmonary dysplasia - Intraventricular hemorrhage - Slow aging	Post-translational gamma carboxylation : - Prothrombin (Factor II) - Factor VII - Factor IX - Product of gene gas-6 - matrix gla protein - Factor X - Nephrocalcin - Protein C - Protein S - Osteocalcin Pro-coagulant : <pre> graph LR A[Reduced vit. K (Hydroxyquinone)] -- "Clotting factors Gamma carboxylated" --> B[vit. K epoxide] B -- "vit. K epoxide reductase" --> C[vit. K quinone] C -- "Clotting factors Gamma carboxylated" --> A </pre>
Deficiency	- Axonal degeneration - Peripheral neuropathy - Spinocerebellar ataxia Hemolytic anemia - Eye manifestations : - Pigmentary retinopathy - Ophthalmoplegia - Nystagmus	Bleeding manifestations : ↑CT, PT - Common in premature babies - D/t prolonged antibiotics
Toxicity	- Interferes with platelet aggregation - Interferes with vit. K function	Hemolysis → Hyperbilirubinemia ↓ Kernicterus (In babies)
RDA	- males : 10 mg/d - Females : 8 mg/d - Pregnancy : 10 mg/d - Lactation : 12 mg/d	-

Anticoagulants

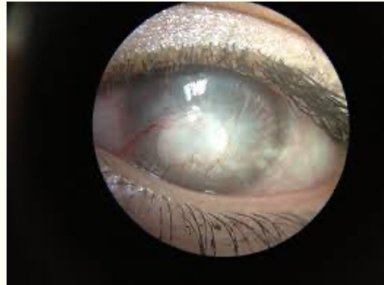
warfarin & dicoumarol :

⊖ vit K epoxide reductase

----- Active space -----

Vitamin B₁ vs. B₂

00:21:18

	Vitamin B ₁ (Thiamine)	Vitamin B ₂ (Riboflavin)
Features	- Sources : - Aleurone layer of grains - Animal food : Egg, meat, yeast - Functions : Coenzyme role (TPP) - Oxidative decarboxylation : PDH, BCKDH, AKGDH Transketolase - Causes : - Polishing of rice - Chronic alcoholism	- AKA Warburg yellow enzyme - Heat stable - Present in flavoproteins. - Active form : - FAD - FMN
Deficiency	manifestations : Dry beri-beri (PNS affected) : - Symmetrical motor & sensory neuropathy - Loss of reflexes, muscle cramps - muscle atrophy (if severe) Wet beri-beri : - Peripheral edema - Dyspnea - Cardiomegaly - Pulmonary edema - High output cardiac failure Wernicke's encephalopathy (WE) : - Global confusion - Ophthalmoplegia - Truncal ataxia - Horizontal nystagmus Wernicke Korsakoff's : - Features of WE + dementia + confabulatory psychosis	- magenta coloured tongue  - Angular stomatitis & cheilosis - Fissures in lips  - Eye : Keratitis, conjunctivitis & corneal vascularization  - Seborrheic dermatitis
Assessment	Erythrocyte transketolase level - urinary thiamine excretion	Erythrocyte glutathione reductase - urinary riboflavin

Vitamin B₃, B₅, B₆, B₇

00:24:41

----- Active space -----

vitamin B₃ (Niacin) :

Functions :

1. Coenzyme :

- NAD + All dehydrogenases.
- NADPH generating :
 - HMP oxidative.
 - Cyt. isocitrate DH.
 - malic enzyme.
- NADPH utilising : All reductases.

2. Therapeutic use : Lipid modifying drug (Hypertriglyceridemia).

Deficiency of Vit. B₃ : Pellagra

1. Dermatitis (Photosensitivity) : Casal's necklace.
2. Diarrhoea
3. Dementia.
4. Death.

Niacin toxicity (DNA icon) :

1. PG mediated flushing.
2. Hyperuricemia.
3. Glucose intolerance.
4. Cystoid macular edema.
5. Gastric irritation.
6. Fulminant hepatitis.



Flushing

Note :

Pellagra-like symptoms :

- Hartnup's disease : ↓ Absorption of tryptophan → ↓ Niacin.
- Carcinoid syndrome : ↑ Tryptophan Shunted → Serotonin (↓ Niacin production).
- maize/corn diet : Poor source of tryptophan.
- Jowar/sorghum staple diet : ↑ Leucine $\xrightarrow{\ominus}$ QPRTase → ↓ Niacin production.

vitamin B₅ (Pantothenic Acid) (DNA icon) :

Function : Present in CoA (Acyl CoA & acetyl CoA) & Acyl carrier protein (FA synthase complex).

Deficiency : Gopalan's burning foot syndrome/nutritional meralgia.

mRI : eye of tiger sign (Pantothenate kinase-associated neurodegeneration).

----- Active space -----

Vitamin B₆ :

Active form : Pyridoxal phosphate.

Function : Coenzyme role.

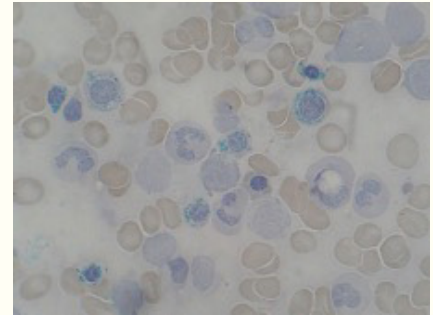
- Transamination
- Simple decarboxylation.
- Transsulfuration
- Tryptophan metabolism :
Kynureninase.
- Heme synthesis : ALA synthase
- Glycogenolysis : Glycogen phosphorylase.

Deficiency manifestations :

- Pyridoxine dependent seizures.
- Sideroblastic anemia : ↓ ALA synthase.
- Pellagra-like symptoms.

urinary metabolites in B₆ deficiency :

- Homocysteine.
- Oxalate.
- Xanthurenic acid.



Ringed sideroblast

Assessment :

- Enzyme activity : Erythrocyte transaminase.
- Load test : Tryptophan load test → Excretion of xanthurenic acid.
- Direct measurement : Estimation of B₆.

Toxicity of B₆ : Sensory neuropathy.

RDA : Depends on protein intake.

Biotin (Vitamin H/B₇) :

Function : Coenzyme role.

- Carboxylation reaction :
 - Pyruvate carboxylase.
 - Acetyl CoA carboxylase.
 - Propionyl CoA carboxylase.

Deficiency :

- Cause :
 - Leiner's disease : Linked to complement 5a deficiency.
 - Raw egg consumption : Avidin $\xrightarrow{\ominus}$ Biotin.
- Symptoms :
 - Depression, hallucinations.
 - Scaling, seborrheic dermatitis & erythematous rash.
 - Alopecia.

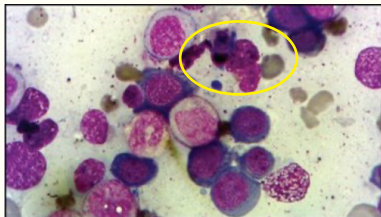
Biochemical tests :

----- Active space -----

- urinary biotin concentration.
- Serum & urine propionic acid level.
- ↓ Activity of biotin dependent enzymes in lymphocyte.

Folic Acid vs. Vitamin B₁₂

00:32:49

	Folic acid (vit. B ₉)	Vit B ₁₂
Form & function	Tetrahydrofolic acid (Active) : Carrier of 1-carbon groups	• Adenosyl B₁₂ (Active) : methyl malonyl CoA mutase • methyl B₁₂ (Active) : Homocysteine methyl transferase/ methionine synthase
	Homocysteine $\xrightarrow{B_{12}, B_9}$ methionine	
Sources	Green leafy vegetables (Plant sources only)	mainly animal sources
Deficiency	manifestations : • megaloblastic anemia • Neural tube defects : - Spina bifida - Anencephaly • Homocysteinemia	• Causes : 1. Nutritional : Strict vegans 2. Gastric : ↓ Intrinsic factor (IF) - Autoimmune pernicious anemia - Gastrectomy 3. Intestinal : - Crohn's disease - Fish tapeworm - Stagnant loop syndrome • manifestations : 1. megaloblastic anemia 2. Subacute combined degeneration : - Numbness - Loss of reflex - Peripheral neuropathy
Assessment	• Serum folate/red cell folate • Histidine load test : FIGLU excretion in urine • Serum homocysteine • Serum amino imidazole carboxamide ribose-5-phosphate (AICAR)	• Serum cobalamine • Serum homocysteine • Urine homocystine • Serum methyl malonate (Specific) • Schilling test • Peripheral smear
Bone marrow & peripheral smear	 megaloblast in bone marrow	 macrocytes in peripheral smear

----- Active space -----

Vitamin C (Ascorbic Acid)

00:37:37

Synthesis : Humans can't synthesise vit. C → Lack L-gulonolactone oxidase.

Functions :

1. Collagen synthesis : Prolyl & lysyl hydroxylase.
2. Iron absorption : Ferrireductase.
3. Tyrosine metabolism : PHPP hydroxylase.
4. Bile acid synthesis : 7- α -hydroxylase.

Deficiency :

- Scurvy :
 - a. Bleeding manifestations.
 - b. Iron deficiency anemia.
 - c. Scorbutic rosary (Angulated, tender).



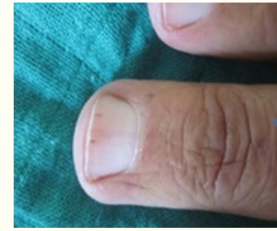
Bleeding gums



Hemarthrosis



Petechial rashes



Splinter hemorrhages

- Barlow's disease (Infantile scurvy) : D/t deficiency during weaning (6 - 12 months).
- Treatment : Supplement vit. C.

Minerals

00:39:55

Copper :

Copper deficiency :

	Wilson's disease	menke's disease/kinky or steely hair syndrome
Etiology	mutation in ATP7B gene : Defective Cu transport in liver	mutation in ATP7A : • X-linked recessive • Defect in Cu transporter in intestine
Features	Kayser Fleischer rings seen	Enzymes affected : • Xanthine oxidase • Lysyl oxidase (Collagen affected)
Dx	• ↑ S. Cu • ↓ S. Ceruloplasmin • Liver copper assay (Gold standard) • ↓ 3-methyl histidine excretion in urine • ↑ 24h urine copper	• ↓ S. Ceruloplasmin • ↓ S. Cu

Selenium, Zinc & Chromium :

----- Active space -----

	Selenium	Zinc
Functions	• Anti-oxidant • Selenocysteine containing enzymes : - Glutathione peroxidase - Thioredoxin reductase - Deiodinase - Selenoprotein P	• metalloenzymes (Carboxypeptidase, carbonic anhydrase) • Stabilise human insulin • Spermatogenesis • maintenance of immunity
Deficiency	• Keshan disease (Seen in China) : Endemic cardiomyopathy (d/t dietary deficiency) • Kashin-Beck disease : Chronic joint disorder (Also d/t ↓ iodine)	• Hypogeusia : ↓ Taste. • Acrodermatitis enteropathica : Diarrhoea + perioral & perineal rashes.



Acrodermatitis enteropathica

Miscellaneous Topics on Nutrition

00:44:25

Energy available from macronutrients (in kcal/g) :

- Carbohydrate : 4.
- Protein : 4.
- Fat : 9.
- Alcohol : 7.

Specific dynamic action (Thermic effect of food) :

Energy used for digestion & absorption of food :

- Carbohydrate : 5.
- Protein : 30 (max.).
- Lipids : 15.
- mixed food : 10.

Respiratory quotient :

 $RQ = \text{CO}_2 \text{ exhaled} / \text{O}_2 \text{ consumed}$:

- Carb : 1.
- Protein : 0.81
- Lipids : 0.71.
- Alcohol : 0.66.

MISCELLANEOUS TOPICS IN BIOCHEMISTRY

TCA Cycle

00:00:06

Site : mitochondria.

Features :

- Amphibolic pathway
 - Catabolic
 - Citrate $\xrightarrow{\text{use}}$ FA synthesis.
 - Anabolic
 - α KG $\xrightarrow{\text{use}}$ Glutamate $\rightarrow$ GABA.
 - Succinyl CoA $\xrightarrow{\text{use}}$ Heme synthesis.
- Final common oxidative pathway of lipids, carbohydrates & proteins.
- Acetyl CoA : Completely oxidised.
- Unidirectional steps :
 - Citrate synthase.
 - α ketoglutarate dehydrogenase.
- Succinate dehydrogenase :
 - Linked to complex II of ETC.
 - Produces FADH_a .
- SLP : Succinate thiokinase.
- Anaplerotic reaction : Filling up of intermediates of TCA cycle.

Pyruvate carboxylase $\xleftarrow{\text{Allosteric activator Acetyl CoA}}$

Energetics
3 NADH 1 FADH_a 1 ATP
Total : 10 ATP

Vitamins Involved :

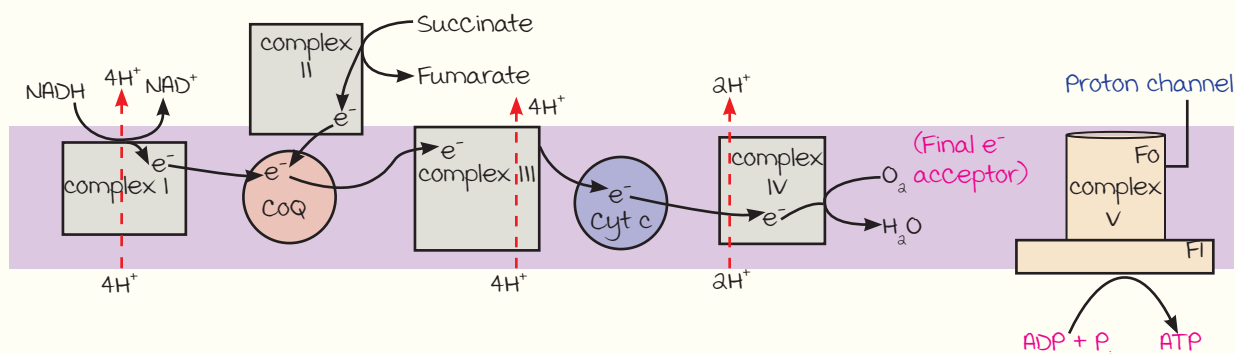
- Pantothenic acid (CoA).
- Thiamine (α -KGDH).
- Riboflavin (FAD).
- Niacin (NAD^+).

Inhibitors :

Inhibitor	Enzyme
Fluoroacetate	Aconitase
Arsenite	α ketoglutarate dehydrogenase
malonate	Succinate dehydrogenase

Electron Transport Chain (ETC)

00:02:48



Complexes :

----- Active space -----

Complex	Name	Redox couples	Protons pumped	Inhibitors
Complex I	NADH Q oxido-reductase	• FMN • Fe-S complex	4 H ⁺	• Rotenone • Amobarbital • Piericidin A
Complex II	Succinate Q oxido-reductase	• FAD • Fe-S complex	None	• malonate • Carboxin • TTFA (Thenoyl trifluoro acetone)
Complex III	Q cytochrome C oxido-reductase	• Cyt b, cyt C₁ • Fe-S complex	4 H ⁺	• Antimycin A • BAL (Dimercaprol)
Complex IV	Cytochrome C oxidase	• Cyt a-a₃ • Cu A - Cu B centre	2 H ⁺	• CO • Cyanide • H₂S
Complex V	ATP synthase	• F_o : H⁺ channel • F₁ : ATP synthesis	None	Inhibitors of oxidative phosphorylation : • Atractyloside : ADP/ATP transporter • Oligomycin : F_o complex • Venturicidin

Uncouplers of Oxidative Phosphorylation :

Chemical :

1. 2,4-Dinitrophenol.
2. Dinitroresol.
3. FCCP.
4. High dose aspirin.

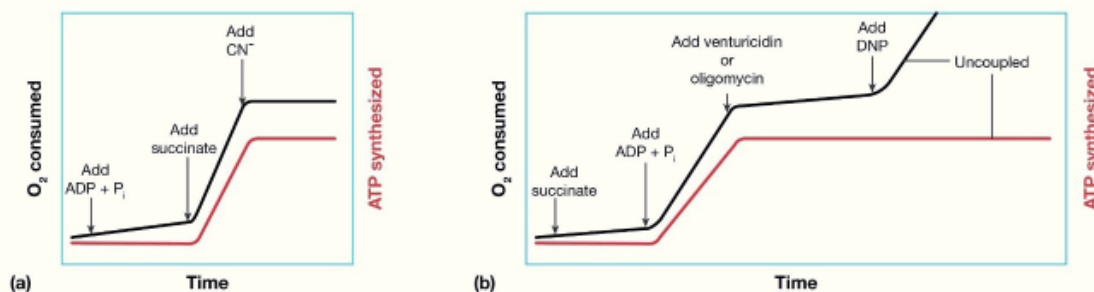
Physiological :

1. Thermogenin (uncoupling protein I).
2. Thyroxine.
3. Long chain fatty acid.
4. Ionophores : valinomycin, gramicidin.

Effects of Inhibitors of ETC :

Inhibitor	O ₂ consumption (Respiration)
At complexes	↓
Oxidative phosphorylation	↓ (↑ H ⁺ conc. in lms)
Uncouplers	↑

ATP production is stopped.



Effects of inhibitors & uncouplers on ETC

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Heme Metabolism

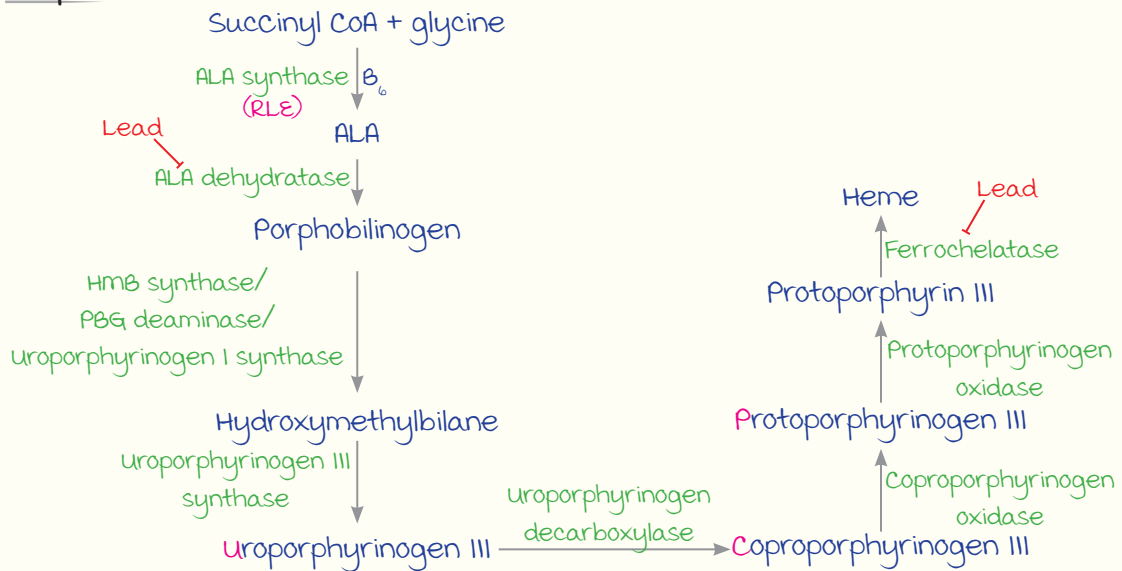
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Site :

Liver, erythrocyte precursors.

Organelle :

Partly cytoplasmic & partly mitochondrial.

Steps :Porphyrias :

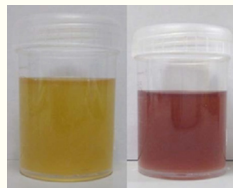
mode of inheritance :

m/c is autosomal dominant except :

- Congenital erythropoietic porphyria (CEP).
- ALAD enzyme deficiency (ADP).
- Erythropoietic protoporphyria (EPP).
- X-linked protoporphyria (XLP).

Types :

----- Active space -----

Porphyria	Defective enzyme	Clinical features
ADP	ALA dehydratase	-
Acute intermittent porphyria (m/c acute)	- uPBG synthase/ PBG deaminase/ HMB synthase - Accumulation of : - Porphobilinogen - ALA	Neuro-visceral manifestations : - No photosensitivity - m/c : Abdominal pain - H/o drug intake 
Congenital erythropoietic porphyria (CEP)	- uroporphyrinogen III synthase - Accumulation of : Porphyrins - Late in pathway	Cutaneous photosensitivity ⊕  Erythrodontia Non-immune hydrops fetalis
Porphyria cutanea tarda (m/c)	uroporphyrinogen decarboxylase	Cutaneous photosensitivity : Blisters ⊕ 
Sideroblastic anemia (X-linked)	ALA synthase	Ringed sideroblasts
Erythropoietic protoporphyria (EPP) (m/c in children)	Ferrochelatase	Non-blistering photosensitivity

Diagnosis :

- Ehrlich test : Non-specific
 - Pink : urobilinogen (UBG).
 - Red : Porphobilinogen (PBG).
- Hoesch test.
- Watson Schwartz test : Differentiates b/w UBG & PBG.
- Soret band at 400 nm.
- Wood's lamp : Red fluorescence.

Lead Poisoning :

- Inhibits : ALA dehydratase, Ferrochelatase.
- H/o : Occupational exposure (Paints), children playing with painted toys.
- C/f : Abdominal pain.
- Biomarkers : ↑ urinary ALA.

